Beyond Social Preferences: Mental Representations in Games*

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Abstract

Behavior in economic games moves with contextual features that leave rules and payoffs unchanged—framing effects that existing theories record but do not explain. We propose a model in which players represent a game by matching it to familiar social situations: the retrieved representation directs attention to some payoff consequences over others and supplies beliefs about the counterpart. We test the model in large pre-registered experiments on dictator, ultimatum, and trust games ($N \approx$ 12,750), measuring representations directly from open-ended explanations classified at scale. Representations vary systematically across games and predict actions and beliefs within them. Randomized stories and a market frame shift representations, actions, and beliefs coherently but not uniformly: the same market frame reduces dictator transfers and ultimatum offers yet raises trust-game sending. Measured representation shifts account for treatment effects for both players, and for who changes behavior when the same person faces a redescribed game. Beliefs move with representations rather than with the counterpart's actual behavior, so context also produces systematic forecast errors, with allocative consequences: the market frame destroys surplus in the ultimatum game while creating it in the trust game. Overall, our findings suggest that context changes behavior by changing the situation players perceive themselves to be in.

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1 Introduction

How much weight people give to their own interests against the interests of others is not a fixed trait: it varies, sometimes dramatically, with the situation they are in. Markets are said to make people more selfish—exposure to market institutions erodes moral concern for third parties, and monetary incentives can crowd out prosocial motivation rather than reinforce it (Falk and Szech, 2013; Bartling et al., 2015; Gneezy and Rustichini, 2000; Bowles and Polanía-Reyes, 2012). War is said to make people more cooperative—individuals exposed to conflict subsequently participate more, trust more, and share more within their communities (Bauer et al., 2016; Voors et al., 2012). Social proximity is said to make people more moral—shrinking the distance to a counterpart, by naming or identifying her, raises giving and the force of fairness norms (Hoffman et al., 1996; Leider et al., 2009; Small and Loewenstein, 2003). The standard reading of this variation runs through social preferences and social norms that respond to the incentives and institutions each context supplies: fairness concerns of the kind that vary across cultures and bargaining environments (Roth et al., 1991; Henrich et al., 2001), disciplined by reciprocity and norm enforcement (Fehr and Gächter, 2000).

Yet the same instability appears where incentives and institutions have no room to operate. Game theory describes a strategic situation by its rules: the players, the actions available to them, and the payoffs that follow. Given the rules, behavior is pinned down by two additional objects, beliefs about what others will do and preferences over outcomes. This apparatus embeds a strong assumption: that players represent the game as the analyst writes it down. A large experimental literature suggests they do not. Behavior in the same game—same players, same actions, same payoff consequences—moves with contextual details that the standard description treats as irrelevant: how the endowment is labeled, what the interaction is called, which experiences the situation happens to evoke (Camerer, 2003; Levitt and List, 2007). The trader who looks prosocial in the laboratory reverts to self-interest in his own marketplace (List, 2006); the prisoner's dilemma relabeled the Community Game doubles cooperation, and does so through what players expect of each other

(Liberman et al., 2004; Ellingsen et al., 2012). If contexts reshape behavior through norms and incentives, what exactly does a payoff-irrelevant word carry into an abstract game? What is the mechanism that makes the same person selfish in one description and generous in another?

Figure 1 illustrates the puzzle with the simplest possible game. Participants in our control condition play one of two dictator games over a \$12 budget. In the first, following Krupka and Weber (2013), the dictator is provisionally allocated \$12 and the other participant \$0, and the dictator can transfer any amount (DG-KW). In the second, adapting the take frame of List (2007), the provisional allocation is \$8 and \$4, and the dictator can transfer up to \$8 or *take* up to \$4 (DG-LT). The two games have identical feasible final allocations and identical payoff consequences for every final allocation; a theory of stable distributional preferences predicts identical choices (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000). Behavior differs sharply: the two allocation distributions in Figure 1 do not line up, and the average recipient ends up with \$4.20 in DG-KW but \$4.97 in DG-LT.[1][2][3]

The leading account of such instability appeals to social norms.[4] Krupka and Weber (2013) show that payoff-identical actions carry different degrees of social appropriateness in the two frames, and that elicited norms—combined with a stable preference for norm compliance—predict behavioral differences across dictator-game variants. This account is powerful but incomplete, in two ways. First, it measures the norm in each frame without explaining why minor, payoff-irrelevant changes move collectively shared judgments in the

[1]Equality of the two distributions is rejected by Kolmogorov–Smirnov ($D = 0.178$), Mann–Whitney, and Welch tests, all $p < 0.001$. In DG-KW, 18.9% of dictators give nothing; in DG-LT, only 10.5% take everything, 19.0% leave the default allocation untouched, and 9.2% give the recipient more than half.

[2]This comparison is distinct from that in List (2007), where a take option is added to a give-only game holding the initial allocation fixed, and giving falls. DG-KW and DG-LT instead differ in both the default allocation and the action set, so the higher final transfers in DG-LT are not in tension with List's finding; a fifth of DG-LT participants simply leave the \$8/\$4 default in place.

[3]Roughly half of participants choose the equal split in both games (51.4% and 51.7%), consistent with the special salience of the 50–50 norm (Andreoni and Bernheim, 2009); the instability is concentrated in the other half of the population—a point we return to below.

[4]The phenomenon is general: payoff-preserving expansions of the action set (Bardsley, 2008), hidden-information options (Dana et al., 2007), and costless exit (Lazear et al., 2012) all move behavior over unchanged alternatives.

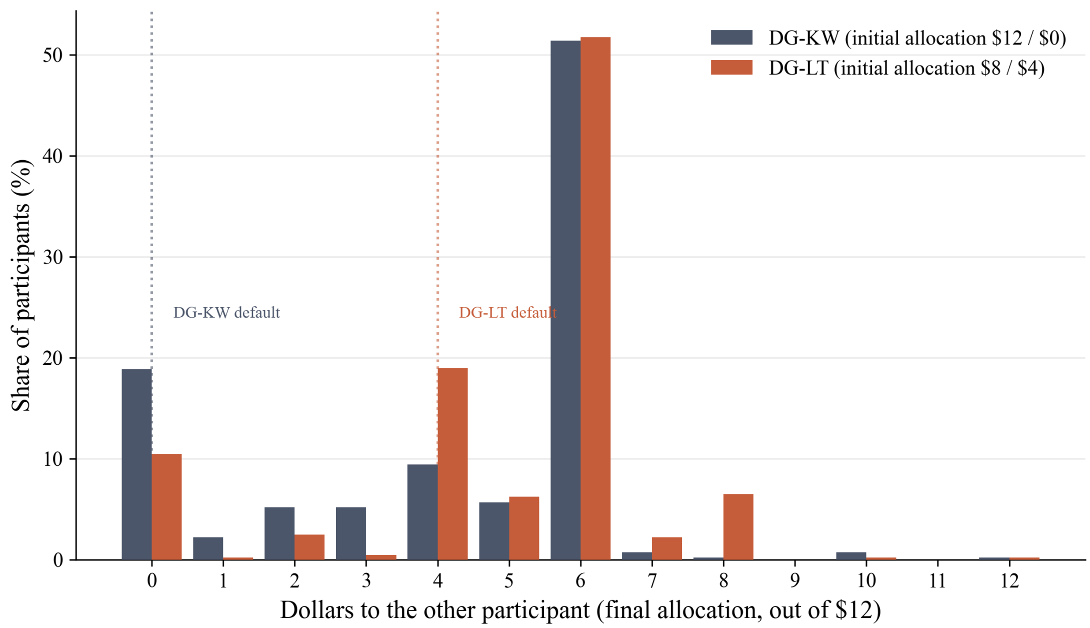

Figure 1: Final allocations to the other participant in two payoff-equivalent dictator games, control condition. In DG-KW the dictator is provisionally allocated \$12 and can transfer between \$0 and \$12; in DG-LT the provisional allocation is \$8/\$4 and the dictator can transfer between $-$\$4 and \$8. Dotted lines mark the default (no-transfer) allocation in each game. $N = 403$ (DG-KW) and $N = 400$ (DG-LT).

first place: the context-to-norm mapping remains a black box. Second, the instability lives *within* the person, not only across frames. When the same participant plays both games of Figure 1, nearly half change their choice (Appendix E): the same individual who shares generously under one description keeps under another, and any account built on stable individual traits—a taste for fairness, a fixed sensitivity to norms—records this instability without explaining it. Fehr and Charness (2025) list the first gap among the central open questions in the social preferences literature; Camerer (2003) flagged it two decades ago: framing effects are real, but "the key step is figuring out what general principles (or theory of framing) can be abstracted" from them.

This paper proposes and tests such a theory, building on the cognitive foundations developed by Bordalo et al. (2026b). Before choosing, a player asks: *what kind of situation is this?* She answers by matching the game, including its payoff-irrelevant contextual features, to similar situations she has experienced or can imagine—splitting a windfall with a friend, negotiating a price with a middleman, investing in a joint project (Tversky, 1977; Gilboa and Schmeidler, 1995; Bordalo et al., 2012, 2016, 2020). The retrieved situation is the player's *representation* of the game, and it directs attention to some payoff consequences over others. We formalize a representation as two objects retrieved together: attention weights over payoff elements—own earnings, total surplus, and a sharing norm—and beliefs about the other player's behavior. A representation says both what the game is about and whom one is facing. Given a representation, the player maximizes utility, so behavior inherits the instability of representations: contextual changes that shift which situation the game evokes shift norms, beliefs, and actions together, even though rules and payoffs are unchanged.

We test this account in a large pre-registered experiment on Prolific ($N \approx$ 12,750 across roles and games) built around three classic games: the dictator game, the ultimatum game, and the trust game. The design has three ingredients. First, we *measure representations directly*. After each decision, participants explain in open-ended text what drove their choice; a large language model, validated by human coders, classifies each answer into a Moral category (fairness, generosity, equality), a Self-interest category (own payoff), or a Mutual Benefit/Cooperation category (joint gains), alongside the social proximity of the counterpart the answer evokes. Second, we *measure beliefs*: first movers in the ultimatum and trust games report the behavior they expect from the second mover, both in response to their chosen action and in response to a common reference action, which allows belief comparisons across participants who chose different actions. Third, we *manipulate context while holding rules fixed*: relative to an abstract control, participants either play the same game framed as a market interaction between a data producer and a broker, or read and summarize a story about a workplace bonus or about charitable aid before playing the abstract game.

Four results emerge. First, absent any manipulation, representations vary systematically across games and predict behavior within each game. Moral representations dominate the dictator and ultimatum games but drop sharply in the trust game, where cooperative representations surge from near-absence to over a third of responses: the same subject pool spontaneously represents payoff-equivalent transfers in morally different ways depending on the strategic setting.[5] Within games, categories predict actions as the attention weights imply—with a twist that only strategic structure explains: self-interested dictators transfer 10% of the budget against 46% for moral dictators, yet self-interested ultimatum proposers offer 42%, close to the moral proposers' 50%, because rejection risk disciplines low offers. In the trust game, cooperative senders send the most and hold the most optimistic beliefs about repayment. The two components of the representation are also linked as attention implies: moral proposers act on the norm regardless of what they expect, while self-interested proposers and senders track their beliefs.

Second, contextual manipulations shift representations and behavior coherently, and in ways that are hard to reconcile with context simply priming generosity or selfishness. The market frame crowds out moral representations in every game while raising both Self-interest and Mutual Benefit/Cooperation, and behavior follows the attention weights rather than a single "market makes selfish" direction: dictator giving and ultimatum offers fall, but trust-game sending *rises*, together with markedly more optimistic beliefs about second movers.[6] Nor is this a shift in means alone: under the market frame the equal-split spike collapses while zero offers and full trust-game sends surge—mass moving between qualitatively different actions, the signature of a changed mixture of representations rather than of a single preference parameter sliding. The aid story, despite its charitable content, *reduces* moral representations and increases self-interested ones relative to the bonus story in the dictator and ultimatum games, and it reduces giving; it leaves the trust game unaffected.

Third, representations quantitatively account for treatment effects. For each treatment comparison we estimate simple benchmark regressions of actions and beliefs on representation categories (and, where relevant, beliefs), and combine the estimated coefficients with

[5] Moral shares are 68%, 77%, and 44% of classified responses in the dictator, ultimatum, and trust games; Mutual Benefit/Cooperation rises from below 4% in the first two to 37% in the trust game.

[6] Moral representations fall by 27, 43, and 24 percentage points in the dictator, ultimatum, and trust games; dictator giving falls by 11 percentage points and ultimatum offers by 22, while trust-game sending rises by 15.

the treatment-induced shifts in representations to *predict* treatment effects. Predicted effects line up with actual effects across 18 outcome-by-comparison cells spanning both players and all games; predicted magnitudes fall short of actual ones, consistent with attenuation from coarse category measurement.[7] The same representation categories thus organize heterogeneity *within* a context and instability *across* contexts. Within-subject designs in the dictator experiment confirm the link at the individual level: participants whose representation changes across the two payoff-equivalent dictator games change their choice at more than twice the rate of those whose representation is stable (80% versus 36%).

Fourth, beliefs do not behave as rational expectations. Ultimatum proposers systematically underestimate acceptance in the abstract and story conditions (by roughly 16–17 percentage points) but swing to overestimation under the market frame. Under rational expectations, context could move beliefs only insofar as it moves behavior; instead, context moves beliefs through representations, and errors follow—with real allocative consequences, as the market frame's destruction of ultimatum surplus shows.

Our contribution is threefold. To the social preferences literature (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000; Charness and Rabin, 2002; Fehr and Charness, 2025), we offer direct evidence on a mechanism that reconciles stable preferences with unstable behavior: what varies across contexts is not the taste for fairness but the representation that determines which motives receive attention. To the norms literature (Krupka and Weber, 2013; Kimbrough and Vostroknutov, 2016; Bicchieri, 2006; Fehr and Schurtenberger, 2018), we supply the missing upstream step: norms shift across payoff-equivalent frames because frames change the situations games evoke, and we show the same mechanism operates in strategic games through beliefs, not only through injunctive judgments. The belief evidence also speaks to the literature on stated beliefs in games (Costa-Gomes and Weizsäcker, 2008; Rey-Biel, 2009), where actions and elicited beliefs often fail to line up and the leading diagnosis is that subjects perceive the game differently across the two tasks: we measure

[7] Regressing actual on predicted effects across the 18 cells yields $R^2 = 0.87$; predicted-to-actual ratios are near one half for the largest first-mover effects.

that perception directly, and beliefs retrieved with the representation explain both when the two align and when they part ways.

To the literature on framing and context effects (Hoffman et al., 1994; Liberman et al., 2004; Dufwenberg et al., 2011; Eckel and Grossman, 1996; Cherry et al., 2002), we offer a unifying framework and a measurement strategy—open-ended reasoning classified at scale—that turns framing from a catalog of effects into a testable theory, in the spirit of Bordalo et al. (2026b), who develop the cognitive machinery for individual decisions and explicitly point to games as the natural next application.[8] The representational lens also carries welfare content: attention weights map directly into surplus and inequality (Section 2.1), so the same contextual shifts that move behavior have signed allocative consequences—under the market frame, destroying surplus in the ultimatum game while creating it in the trust game (Section 5).

The paper proceeds as follows. Section 2 presents the model: how a representation—attention together with beliefs—maps into behavior in the three games, and how context and experience determine which representation a game evokes. Section 3 describes the experimental design, establishes the within-context results in the control sample, and measures the similarity structure that links our treatments to the theory. Section 4 presents treatment effects on representations, actions, beliefs, and forecast errors, organized by the lever each treatment moves. Section 5 shows that representation shifts account for treatment effects, heterogeneity, and surplus, and returns to the motivating puzzle. Section 6 concludes.

2 Model

This section develops the framework in two steps. Section 2.1 defines a representation—attention to payoff elements together with beliefs about the other player—and derives be-

[8] Related formal approaches include case-based decision theory (Gilboa and Schmeidler, 1995) and analogy-based expectations (Jehiel, 2005), in which agents extrapolate from similar past cases or bundle games into analogy classes. Our contribution is empirical as much as theoretical: we measure the representation, not only its behavioral footprint.

havior taking the representation as given. Because the moral motive attaches to the sender's *action* rather than to realized outcomes, the ultimatum and trust games reduce to *modified dictator games*. Section 2.2 models where representations come from: a game description is matched, by similarity and by frequency of experience, to a category of situations, and the retrieved category slants the representation. This second step turns our two manipulations into theory-driven levers—the stories move the frequency with which a category is retrieved, the market frame moves the description itself—and it delivers the paper's organizing predictions: heterogeneity within a context, coherent shifts across contexts, and systematic forecast errors. Proofs of all propositions are in Appendix A.

2.1 How Representations Shape Strategic Behavior

Two players interact once. Player S is the first mover—the dictator, the proposer, the investor; generically, the *sender*—and player R is the second mover or *receiver*. The sender's initial wealth is normalized to one, and her action is the share $t \in [0, 1]$ she sends to the receiver. Total final wealth is $W(t)$, which may depend on the amount sent, π_S denotes the sender's material payoff, and π_R denotes the receiver's material payoff.

A *representation* of the game is a pair $\mathbf{r} = (\boldsymbol{\alpha}, \mathbf{x})$: an attention profile $\boldsymbol{\alpha} = (\rho, \sigma, \mu)$ over payoff elements and, in the strategic games, a believed behavior $\mathbf{x}$ of the receiver. Given her representation, the sender maximizes

$$u_S(t) \;=\; \rho\, W(t) \;-\; \sigma\, \pi_R \;-\; \frac{\mu}{2}\, (t - t^e)^2\,,$$

where the three payoff elements are total wealth, the receiver's material payoff, and a sharing norm anchored at the *equal-payoff action* t^e—the action that equalizes the two players' material payoffs under the sender's representation: the equal split $t^e = \frac{1}{2}$ in the dictator and ultimatum games and, in the trust game, where the multiplier breaks that coincidence, the send that equalizes expected payoffs at the believed return (Proposition 3)—and the Greek letters are attention weights. Because $\pi_S = W(t) - \pi_R$, the objective can be rewritten as

$$u_S \;=\; \sigma\,\pi_S \;+\; (\rho - \sigma)\,W(t) \;-\; \frac{\mu}{2}\,(t - t^e)^2\,, \tag{1}$$

which is the form we use throughout and makes the mapping to motives transparent: σ is attention to own earnings, $\rho - \sigma$ is attention to the total surplus over and above one's own share, and μ is attention to the norm. The attention weights and beliefs are properties of the *situation the game evokes*, not stable traits of the player: the same person can be a moral dictator and a self-interested price-setter. The undistorted case $\rho = \sigma = \mu = 1$, in which no motive is slanted by the representation, gives inequity-averse utility in the spirit of Bolton and Ockenfels (2000); standard models prescribe *stability*—an attention profile that stays fixed while context varies—and, in the strategic games, *rational expectations*—beliefs that track actual behavior. The functional form thus nests the motives of standard distributional preference models (Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000; Charness and Rabin, 2002), with two differences. First, the moral term is *idealistic*: it attaches to what the sender does, not to what materializes afterwards. Whether the receiver accepts, rejects, or reciprocates, the sender has still sent the same t; only material payoffs are contingent on the other player's move. This assumption embodies a simple moral psychology—one feels right or wrong about one's own conduct—and it is what makes the strategic games tractable; in the trust game the anchor is evaluated at the sender's *believed* return, so the norm still judges her own conduct within her own representation, not the realized outcome. Second, and central to everything that follows, beliefs are part of the representation rather than equilibrium objects: the process that determines what the game is *about* also determines whom one imagines playing against.

Dictator game. Here $W(t) = 1$ and $\pi_R = t$; there is no second mover to forecast, so $\mathbf{x}$ is empty.

Proposition 1 (Dictator transfer)**.** *The optimal transfer is* $t^{DG} = \max\left\{0,\ \frac{1}{2} - \frac{\sigma}{\mu}\right\}$. *It never* *exceeds one half, is decreasing in the selfishness ratio σ/μ, and is independent of ρ.*

A moral representation (high μ) anchors the transfer at the equal split; a self-interested one (high σ, low μ) drives it to zero. A cooperative representation loads on ρ, which is idle in a fixed-pie game with no strategic margin: cooperative dictators act on their residual σ/μ, which places them between the two poles when those residual weights are balanced. The first testable ranking follows: dictator transfers ordered Moral > Mutual Benefit/Cooperation > Self-interest, with a wide gap between the extremes.

Ultimatum game. If R accepts, payoffs are as in the dictator game; if R rejects, both players earn zero. The sender's representation now includes a believed behavior of the receiver, which we embed in an *acceptance schedule*: the sender believes an offer t is accepted with probability

$$p(t) \;=\; a + b\,t, \qquad a, b > 0,$$

so the belief component of her representation is the pair $\mathbf{x} = (a, b)$. Because the moral payoff is idealistic, only the material part of utility is scaled by acceptance, and the sender solves

$$\max_{t} \;\; p(t)\,\big(\rho - \sigma t\big) \;-\; \frac{\mu}{2}\,\big(t - \tfrac{1}{2}\big)^2 . \tag{2}$$

Proposition 2 (Ultimatum offer)**.** *The objective in* (2) *is strictly concave, and the optimal offer is*

$$t^{UG} \;=\; \frac{1}{2} \;+\; \frac{b\rho - (a+b)\,\sigma}{2b\sigma + \mu}, \tag{3}$$

censored to $[0,1]$*; the offer is interior if and only if* $\big|b\rho - (a+b)\,\sigma\big| < b\sigma + \mu/2$*. At an interior optimum: (i) if* $p(1) = a + b \le 1$*, then* $t^{UG} > t^{DG}$ *for every attention profile; (ii) the offer is increasing in* ρ*, decreasing in* σ*, and moved toward the equal split by* μ *(raised when* $t^{UG} < \frac{1}{2}$*, lowered when* $t^{UG} > \frac{1}{2}$*); (iii) the offer is decreasing in believed acceptance* a*, with sensitivity* $\big|\partial t^{UG}/\partial a\big| = \sigma/(2b\sigma + \mu)$*, which is increasing in* σ *and decreasing in* μ*.*

Part (i) is the strategic discipline of the ultimatum game: because low offers risk rejection, *every* representation offers more than it would transfer as dictator, so the enormous category gaps the model produces in the dictator game must compress in the ultimatum game—the acceptance schedule disciplines precisely those senders whose own attention profile would otherwise push offers down. Part (iii) delivers two sharp empirical predictions. First, holding the representation fixed, offers should *fall* with believed acceptance—a proposer who thinks low offers will be accepted makes them. Second, the belief sensitivity itself depends on attention: a moral (high-μ) proposer acts on the norm regardless of what she expects, while a self-interested (high-σ) proposer tracks her beliefs. Attention to total wealth also acquires a role it lacks in the dictator game: $\partial t^{UG}/\partial \rho = b/(2b\sigma + \mu)$, which is strictly positive yet vanishes when acceptance is insensitive to the offer ($b = 0$). The pie is fixed, but rejection destroys it, so weight on total wealth raises the offer purely through the fear of that destruction.

The acceptance schedule need not be taken as a primitive. Appendix B founds it in the receiver's own moral psychology—the offer's inequity is felt whether or not it is accepted, and rejection carries a psychological benefit of *protest*—so the believed pair (a, b) is a representation of the *receiver's motives*, of how the sender perceives the receiver's representation: imagining a punitive receiver (our *Moral bad* receiver category of Section 3.1) scales acceptance down across the board, while imagining a self-interested receiver makes acceptance sensitive to money. Under that foundation the condition $a + b \leq 1$ in Proposition 2 also acquires a natural reading: the imagined receiver's protest weight is large enough that even a full-split offer is not certain to be accepted.

Trust game. The amount sent is multiplied by $1 + r$, with $r > 0$ the net return, so total wealth is $W(t) = 1 + rt$.[9] The sender believes the receiver returns a share s, with $0 \leq s < 1$, of the multiplied amount $(1 + r)t$, so the belief component of her representation is $\mathbf{x} = s$ and the receiver's believed material payoff is $\pi_R = (1 - s)(1 + r)t$.[10] Expected payoffs equalize

[9] In the experiment the multiplier is $m = 1 + r = 3$ on a \$6 endowment, so $r = 2$.

[10] We take the believed returned share to be constant for simplicity; like the acceptance schedule, it can be endogenized: Appendix B founds the return schedule in the receiver's moral psychology—a norm of returning

at the send $t^e(s) = \big[1 + (1+r)(1-2s)\big]^{-1}$, which is increasing in the believed return and equals one half exactly at $s = r/\big(2(1+r)\big)$—one third of the output at $r = 2$—so here the equal-payoff anchor moves with the belief. The sender solves

$$\max_{t} \;\; \rho\,(1 + rt) \;-\; \sigma\,(1+r)\,t\,(1-s) \;-\; \frac{\mu}{2}\,(t - t^e(s))^2 \,. \tag{4}$$

Proposition 3 (Trust send)**.** *The optimal send is*

$$t^{TG} \;=\; t^e(s) \;+\; \frac{\rho r - \sigma(1+r)(1-s)}{\mu}, \tag{5}$$

censored to $[0,1]$*; the send is interior if and only if* $-\mu\, t^e(s) \;<\; \rho r - \sigma(1+r)(1-s) \;<\; \mu\big(1 - t^e(s)\big)$*. At an interior optimum, the send is increasing in* ρ*, decreasing in* σ*, and moved toward the equal-payoff send* $t^e(s)$ *by* μ*; it is increasing in the believed return share* s *with sensitivity* $\mathrm{d}t^e/\mathrm{d}s + \sigma(1+r)/\mu$*, where* $\mathrm{d}t^e/\mathrm{d}s = 2(1+r)\big[1 + (1+r)(1-2s)\big]^{-2} > 0$*: beliefs move the send through the anchor itself as well as through the material term, so the sensitivity is strictly positive even under norm domination. As* $\mu \to 0$ *the problem becomes linear and the solution goes to a corner: a purely material sender sends everything if she believes the returned share exceeds the break-even level,* $s(1+r) > 1$*, and nothing if it falls short; the anchor is likewise censored—*$t^e(s) \geq 1$ *for* $s \geq \frac{1}{2}$*—so a norm-dominated sender who expects at least half of the output back is pulled to the full send. Finally, at interior optima,* $t^{TG} \geq t^{DG}$ *if and only if* $r\big[\rho - \sigma(1-s)\big] + \sigma s \geq \mu\big(\frac{1}{2} - t^e(s)\big)$*; for* $s \geq r/\big(2(1+r)\big)$ *the right side is non-positive and* $\rho \geq \sigma(1-s)$ *suffices.*

The trust game is where both components of the representation bite at once. Attention to total wealth is no longer idle: ρ multiplies the return r, so cooperative senders—those who attend to the multiplied pie itself—send the most. Moral senders are anchored near the equal-payoff send. Self-interested senders send the least unless they are optimistic, and

half of a target pie—and shows that the comparative statics survive when the believed share declines in the send; the receiver's *actual* return margin, anchored at the realized equal-payoff return, instead rises with the send (same appendix).

their sends are the most belief-sensitive: as in the ultimatum game, the material channel of the sensitivity, $\sigma(1+r)/\mu$, rises with own-payoff attention and falls with moral attention—but unlike in the ultimatum game, the anchor itself moves with the belief, so even norm-dominated senders track the expected return through $\mathrm{d}t^e/\mathrm{d}s$. Whether the send sits above or below the anchor—the sign of the second term in (5)—is ambiguous: it is positive if and only if $\rho/\sigma > (1+r)(1-s)/r$, and for a purely material sender ($\rho = \sigma$) this reduces to the break-even condition $s(1+r) > 1$, so the term is positive exactly when she believes trusting pays. The belief sensitivity $\mathrm{d}t^e/\mathrm{d}s + \sigma(1+r)/\mu$ holds at any interior optimum regardless of this sign (at a censored send it is locally zero). And in contrast to the ultimatum game, beliefs now enter with a *positive* sign—optimism about the return raises the send, while optimism about acceptance lowers the offer. This sign flip across games is a distinctive joint prediction of the model. Table 1 collects the comparative statics.

Table 1: **Summary of Comparative Statics**

	Attention and beliefs				Belief sensitivity
Transfer	ρ	σ	μ	belief	$\lvert\partial t/\partial \text{belief}\rvert$
Dictator t^{DG}	0	$-$	$\to \frac{1}{2}$	NA	NA
Ultimatum t^{UG}	$+$	$-$	$\to \frac{1}{2}$	$-$ (acceptance a)	$\sigma/(2b\sigma+\mu)$
Trust t^{TG}	$+$	$-$	$\to t^e(s)$	$+$ (return share s)	$\mathrm{d}t^e/\mathrm{d}s + \sigma(1+r)/\mu$

Notes: Signs at interior optima; solutions are censored to $[0,1]$. "$\to \frac{1}{2}$" ("$\to t^e(s)$") means the moral weight moves the transfer toward the norm anchor—the equal split in the fixed-pie games, the equal-payoff send in the trust game—so over the empirically relevant range its effect is positive. The material channel of the belief sensitivity is increasing in σ and decreasing in μ: self-interested representations track beliefs most. In the trust game the anchor itself moves with the believed return, adding the weight-free term $\mathrm{d}t^e/\mathrm{d}s$, so even moral representations—which act on the norm—track beliefs there.

Welfare and inequality. The comparative statics about the sender's action immediately deliver comparative statics for total surplus and for the division of payoffs, which is where representations acquire social content. Let V_g denote expected total material wealth at an interior optimum—$V_{DG} = 1$, $V_{UG} = p(t^{UG})$ (the pie survives only if the offer is accepted), $V_{TG} = 1 + r\,t^{TG}$—and let $D_g = \pi_S - \pi_R$ denote the payoff gap conditional on trade (in the ultimatum game, given acceptance; the other games have no rejection margin).

Proposition 4 (Welfare). *At interior optima: (i) surplus attention raises welfare where the pie responds to the action, with strength quadratic in that response:* $\partial V_{DG}/\partial\rho = 0$, $\partial V_{UG}/\partial\rho = b^2/(2b\sigma + \mu)$, $\partial V_{TG}/\partial\rho = r^2/\mu$; *(ii) own-payoff attention destroys welfare in both strategic games,* $\partial V_{UG}/\partial\sigma = -b\big[(a + b)\mu + 2b^2\rho\big]/(2b\sigma + \mu)^2 < 0$ *and* $\partial V_{TG}/\partial\sigma = -r(1 + r)(1 - s)/\mu < 0$, *and unambiguously lowers the counterpart's expected payoff; the sender's own expected payoff can fall as well, precisely when* $b(1-2t^{UG}) > a$*—a steep believed schedule with a low intercept; (iii) in the strategic games, norm attention raises welfare exactly when the material pull is selfish—*$\partial V_g/\partial\mu$ *has the sign of* $t^e - t^g$, *with* $t^e = \frac{1}{2}$ *in the fixed-pie games—while dictator-game welfare is unaffected by any weight; (iv) optimism raises welfare:* $\partial V_{UG}/\partial a = (b\sigma+\mu)/(2b\sigma+\mu) > 0$ *and* $\partial V_{TG}/\partial s = r\big[\mathrm{d}t^e/\mathrm{d}s+\sigma(1+r)/\mu\big] > 0$.

Part (i) formalizes cooperation through attention: a sender who attends to total wealth gives more precisely where giving protects (ultimatum) or produces (trust) the pie, and welfare rises with the *square* of the pie's sensitivity to the action, because attention moves the action by that sensitivity and the action moves the pie by it again. Part (ii) is the reverse face: own-payoff attention is socially destructive in every strategic game, and it is not even reliably self-serving—when rejection risk is steep, the offer cut costs more in expected acceptance than it saves in shares. Part (iii) says the norm is a social technology against selfish material pulls, not a complement to cooperative ones: when the pull is generous (an optimistic cooperator in the trust game), the anchor drags the send back toward the equal-payoff point and destroys surplus—though the drag is milder for optimists, whose anchor itself sits above one half.

These statements evaluate the ultimatum pie at the sender's own beliefs: $V_{UG} = p(t^{UG})$ is expected surplus under her model of the world. This is by design—a rational-expectations welfare measure would impose exactly the belief–behavior alignment the data reject—and the ultimatum game is the only place where the choice matters: dictator surplus is constant, and trust surplus $1 + r\,t^{TG}$ is belief-free given the send, with the division evaluated at the actual share below. *Realized* welfare, by contrast, requires the receiver: the pie survives, and the division is settled, only through what the receiver actually does, which depends on her own representation. Appendix B closes the model on this side—actual acceptance and return schedules are retrieval mixtures over the receiver's categories, cued by the very action she faces (equation (9))—and the next remark shows which welfare statements survive the substitution.

Remark 1 (Believed versus Realized Welfare). Let actual acceptance follow any increasing, differentiable schedule $\hat{p}$, held fixed as the sender's parameters vary, and let $\hat{V}_{UG} = \hat{p}(t^{UG})$ denote realized expected surplus at the offer chosen under the believed schedule. Parts (i)–(iii) of Proposition 4 survive with the same signs: attention weights enter neither schedule, so they move realized surplus only through the action, $\partial\hat{V}_{UG}/\partial\theta = \hat{p}'(t^{UG}) \cdot \partial t^{UG}/\partial\theta$ for $\theta \in \{\rho, \sigma, \mu\}$, and only magnitudes change. Part (iv) reverses in the ultimatum game: optimism no longer raises acceptance directly—it only lowers the offer, and actual acceptance falls with it—so $\partial\hat{V}_{UG}/\partial a < 0$; its trust-game half is unchanged.

Believed and realized welfare thus part ways exactly where beliefs and behavior do—an optimistic proposer expects more surplus and produces less—which is the forecast-error mechanism of Section 4.3 behind the ultimatum-game surplus accounting of Section 5.2.

Proposition 5 (Inequality). *Conditional on trade,* $D_{DG} = 2\sigma/\mu$ *and* $D_{UG} = 2\big[(a+b)\sigma - b\rho\big]/(2b\sigma + \mu)$*: own-payoff attention widens the gap, surplus attention narrows it (strictly in the ultimatum game), and norm attention compresses it toward zero,* $\partial D_{UG}/\partial\mu = -D_{UG}/(2b\sigma + \mu)$ *and* $\partial D_{DG}/\partial\mu = -D_{DG}/\mu$*. In the trust game, for an actual returned share* s_a*,* $D_{TG} = 1 - t\big[1 + (1+r)(1-2s_a)\big]$*: a larger send narrows the gap if and only if* $s_a < (2+r)/\big(2(1+r)\big)$*, and norm attention compresses the gap toward its value at the anchor,* $1 - t^e(s)\big[1 + (1+r)(1-2s_a)\big]$*, which is zero precisely when the belief is correct* $(s = s_a)$*: the norm eliminates conditional inequality exactly when the sender forecasts the return correctly, and the residual gap at the anchor measures her belief error.*

The equal-payoff anchor coincides with the equal split in the fixed-pie games but not in the trust game: with $r = 2$—the experiment's tripling of the send—the equal-split send equalizes payoffs only if the receiver returns exactly one third of her output, and the anchor $t^e(s)$ instead moves with the believed return. The trust game is thus where a payoff-based sharing norm separates from an action-based one, and the belief sensitivities of Section 3.2 take the payoff-based side. The population reading connects these statics to the treatments: a treatment that reweights retrieved categories toward high-σ, low-μ representations in the ultimatum game destroys expected surplus and widens conditional inequality, while one that reweights toward high-ρ representations and optimistic beliefs in the trust game creates surplus—the pattern of the surplus accounting in Section 5.2.

From measured categories to model parameters. Our empirical categories map onto the representation as follows. A **Moral** representation (fairness, generosity, equality) corresponds to high norm attention μ. A **Self-interest** representation corresponds to high own-earnings attention—σ high with $\rho \approx \sigma$, since own earnings are $\sigma\pi_S$ when the surplus weight $\rho - \sigma$ is nil—and low μ. A **Mutual Benefit/Cooperation** representation (joint gains, teamwork) corresponds to surplus attention, $\rho > \sigma$. For second movers, the protest foundation of Appendix B distinguishes **Moral good** (fairness, reciprocity: $\hat{\rho} > \hat{\sigma}$ with moderate protest weight) from **Moral bad** (condemnation of the sender: a high protest weight, hence rejection), with **Self-interest** making acceptance money-sensitive and returns low. The belief components (a, b) and s are measured directly by the belief and hypothetical-belief questions. The model does not restrict how $\mathbf{x}$ relates to $\boldsymbol{\alpha}$ across people: beliefs may be anchored on a commonly held standard, or may covary with own attention, as under *projection*—a cooperative sender imagining a cooperative receiver. Which regime prevails is an empirical question that Section 3.2 answers game by game.

Three testable rankings and two belief-channel signs follow. Transfers should rank Moral > Self-interest in the dictator game with a wide gap; the gap should compress in the ultimatum game; sends should rank Mutual Benefit/Cooperation > Moral > Self-interest in the trust game. Offers should fall with believed acceptance in the ultimatum game; sends should rise with the believed return in the trust game. And the action–belief sensitivity should be strongest for Self-interest representations in both strategic games, since it is increasing in σ and decreasing in μ.

2.2 How Representations Are Determined

The sender does not see a game form; she sees a *description* $\mathbf{d} = (\mathbf{r}_d, \boldsymbol{\kappa}_d)$, where $\mathbf{r}_d$ is the representation implied by a literal reading of the rules and $\boldsymbol{\kappa}_d$ is the context: the labels, the cover story, the setting in which the interaction is embedded. Her memory contains *categories* of experience $c \in \mathbf{C}$—classes of situations she has lived or can imagine, such as sharing a windfall, haggling over a price, investing in a joint project—each with a typical representation $\mathbf{r}_c$, a typical context $\boldsymbol{\kappa}_c$, and a time-weighted frequency F_c (Bordalo et al., 2020, 2026b). Facing the description, she selects a category and a representation $\mathbf{r} = (\boldsymbol{\alpha}, \mathbf{x})$ to solve

$$\max_{c \in \mathbf{C},\, \mathbf{r} \in \mathbf{R}} \; F_c \cdot S\big[(\mathbf{r}, \boldsymbol{\kappa}_d), (\mathbf{r}_c, \boldsymbol{\kappa}_c)\big] \;+\; S\big[(\mathbf{r}, \boldsymbol{\kappa}_d), (\mathbf{r}_d, \boldsymbol{\kappa}_d)\big] \;+\; \epsilon_c, \tag{6}$$

where S is a similarity function (Tversky, 1977; Gilboa and Schmeidler, 1995) and ϵ_c is an i.i.d. extreme-value shock. The chosen representation trades off fidelity to the description (the second term) against assimilation to the best-fitting category of experience (the first term).

Equation (6) has two forces, and each of our manipulations targets one of them. First, a higher frequency F_c fosters a category's retrieval, increasing the likelihood that the sender slants her representation toward $\mathbf{r}_c$, *for a given description.* This is the **story lever**: reading and summarizing a story about a workplace bonus or about charitable aid is a recent, vivid experience that raises the retrieval weight of its category, while the description of the game itself is unchanged. Second, a change in the context $\boldsymbol{\kappa}_d$ that makes the description more similar to the contexts $\boldsymbol{\kappa}_c$ of experiences in some category also fosters that category's retrieval, *for given frequencies*. This is the **frame lever**: describing the identical game as a transaction between a data provider and a broker moves the description itself into the neighborhood of market experiences.

The frequency weight has a natural recency structure. Let each experience e in category c have age τ_e and let $F_c = \sum_{e \in c} \delta^{\tau_e}$ with $\delta \in (0, 1)$: frequency is experience discounted by recency. The description just read is then itself the most recent trace—age zero, weight $\delta^0 = 1$—which is why fidelity carries a fixed unit weight in equation (6): both terms are recency-weighted similarities. This reading disciplines the trade-off between memory and description. A story is an age-zero experience added to its category, so a single story can rival a lifetime of discounted experience—which is why the story lever works at all; and when discounting is strong every F_c is small, so the description dominates and memory cannot drown it unless a category is both frequent and recent. A natural refinement scales fidelity by a cue-strength parameter increasing in the description's distinctiveness—its dissimilarity from the stored contexts—so that a description unlike anything in memory resists assimilation.

Equation (6) is not bespoke. It is, structurally, the description-augmented recognition problem of Bordalo et al. (2026b) (their equation 20): the chosen representation plays the role of their tuned attention vector; category prototypes and their recency-weighted frequencies F_c are defined exactly as theirs; fidelity to the description enters without a frequency multiplier in both frameworks (the present description is the age-zero trace); and the extreme-value shock delivers their logit retrieval probabilities. Their results transfer: attention tunes to the retrieved category and neglects discrepant context, which makes categorization sticky, and a change in a feature's bottom-up salience can re-categorize the problem—their theory of framing is our frame lever. Two departures are deliberate, and they are the games contribution. First, the attended features here are the payoff elements of a strategic problem rather than those of a known lottery: their bound $\alpha \in [0, 1]$ shades perceived values relative to true ones, while here no veridical attention profile exists—behavior identifies only relative weights, and the undistorted case $\rho = \sigma = \mu = 1$ is a statement about ratios, not levels.

Second, beliefs: in their framework perceived probabilities are attention-distortions of an objective distribution, but in a one-shot game there is no objective distribution over the counterpart to distort—the category supplies the belief itself, the imagined receiver (a, b) or s as prototype content. This is the model's one genuinely new retrieval object, and it is what makes systematic forecast errors possible: the third implication below. They close by listing strategic behavior as a natural application of problem recognition; the model of this section is that application.

Three implications organize the empirical work.

1. *Heterogeneity within a context.* Within a condition, people hold different representations (they have different histories F_c and draw different shocks ϵ_c), so behavior is heterogeneous, and its heterogeneity is structured by the mixture of retrieved categories, not merely by noise—measured below by the between-category share of outcome variance, equation (7) in Section 5.1.

2. *Instability across contexts.* Across treatments, the distribution of retrieved categories shifts—through F_c for the stories, through $\boldsymbol{\kappa}_d$ for the market frame—and the action moves in the direction that *combines* the attention differences (across $\boldsymbol{\alpha}$) and the belief differences (across $\mathbf{x}$) of the categories gaining and losing weight. Holding category-typical actions fixed, the mean effect of a manipulation is the composition change $\Delta E[t^g] = \sum_c \Delta q_c \, \bar{t}_c^g$, where q_c is the retrieval probability of category c and $\bar{t}_c^g$ its typical action in game g: the product of a retrieval shift, whose size is gated by the similarity between the game's description and the manipulated content, and the spread of category-typical actions in that game. Both factors are game-specific and measurable, so the same manipulation can move one game strongly and another barely at all—not because any payoff element drops out of the representation, but because retrieval shifts less where similarity is low and matters less where the retrieved categories act alike. Heterogeneity and instability are two faces of the same object.

3. *Forecast errors and learning.* Nothing in (6) guarantees that the two players' representations are shared or mutually consistent: the sender's imagined receiver is retrieved from the sender's memory, not from the receiver's behavior. Systematic, context-dependent forecast errors are therefore an equilibrium object of the theory, not a pathology—and learning in games under a stable context can be read as convergence toward representations that no longer generate salient surprises, which is what renders equilibria stable.[11]

The theory's inputs are, in principle, measurable: the similarity of our contexts to broad categories of experience—cooperation problems, conflict problems, moral problems—and to receiver types who accept or reject, return or keep. Section 3.3 reports two rounds of such measurement—the structural similarity of the stories to the games, and the similarity of every context in the design to vignette exemplars of the categories and receiver types themselves—and uses them to discipline the interpretation of the treatments.

3 The Experiment

This section proceeds in three steps. It first describes the experimental design and measurement (Section 3.1). It then uses the control sample to establish the empirical content of the representation measure: how stated categories vary across games and relate to actions and beliefs (Section 3.2). Finally, it examines the similarity structure that links the treatments to the retrieval model (Section 3.3).

3.1 The Protocol

Games and roles. Participants, recruited on Prolific, are randomly assigned to one of three canonical games (Forsythe et al., 1994; Güth et al., 1982; Berg et al., 1995)—DG, UG,

[11] Appendix B formalizes this extension: the receiver faces a richer description that includes the sender's action, so her retrieval is cued by the offer itself; aggregate acceptance and return schedules are retrieval mixtures of category schedules, and a treatment that moves the two players' descriptions differently moves mean forecast errors.

or TG—to one role, and to one of the four conditions below. In the DG and UG the budget is \$12; in the TG the sender's endowment is \$6 and the multiplier is $m = 3$. To make decisions consequential while keeping the sample large, the chosen allocation is implemented for one participant (pair) in ten. In the main text we focus on the KW version of the dictator game (provisional allocation \$12/\$0) and on first movers (Player 1); the LT version (\$8/\$4 with a take option) appears in the Introduction, enters the similarity gradient of Section 4.1 and the dispersion accounting of Section 5.1, and returns in Section 5.3; second movers (Player 2) return in Section 5, with full detail in Appendix C. Deferring second movers is a matter of identification, not of importance: their clean comparisons require the hypothetical responses at fixed actions described below, and Section 5 reports them and folds both players into the accounting exercise.

Conditions. We compare four conditions that hold rules, action sets, and payoff consequences fixed:

- **Control**: the game in abstract framing ("you and your match are provisionally allocated a certain amount of dollars...").

- **Market**: the same game framed as a market interaction. The first mover is a *data provider* who sets the price of her data; the second mover is a *broker* who resells it to the research team (and, in the UG, can reject the price; in the TG, can transfer revenue back). Roles, action sets, and payoffs are identical to Control.

- **Bonus**: before playing the abstract game, participants read and summarize a story about a team leader deciding whether to share a performance bonus with a teammate.

- **Aid**: before playing the abstract game, participants read and summarize a story about a welfare recipient deciding whether to share unexpected financial aid with a struggling neighbor.

The stories manipulate which real-world situation the game evokes without mentioning the game itself; the market frame manipulates the description of the interaction directly. The two manipulations are designed to move different ingredients of the evoked situation. The stories share the same sharing structure—one person controls a windfall, another person has a plausible claim on part of it—and vary what the situation is about. The Bonus story foregrounds joint contribution inside an ongoing work relationship: the counterpart is a teammate, socially proximate, with a claim rooted in his contribution. The Aid story foregrounds the protagonist's own need and scarcity: the counterpart is a mere acquaintance, social proximity recedes, and one's own material stakes come to the fore. The market frame is designed to evoke anonymous, arm's-length exchange between parties defined by their roles rather than by any personal relationship—a produced and priced endowment, an intermediary with a value-added function—the situation of firms transacting rather than of persons sharing. Full instructions and stories are reproduced in Appendix I. Our two pre-registered comparisons are **Market vs Control** and **Aid vs Bonus**; the latter holds constant the presence of a story and varies only its content.[12]

Measuring representations. We measure representations in the most direct way available: by asking participants to describe them. After choosing, each participant answers an open-ended question about what drove her decision (the "reasons" question). Free text does not force participants into categories chosen by the analyst, and it can be collected at scale.[13] Following the pre-registered procedure, a large language model (GPT-4.1) classi-

[12]The experiments were pre-registered on AsPredicted: #261370 (dictator games, including the within-subject designs analyzed in Appendix E and the hypothetical-allocation representation measure analyzed in Appendix F) and #286187 (ultimatum and trust games, including belief elicitation and its accuracy). Pre-registered sample sizes are 400 participants per between-subject dictator cell plus 600 per within-subject design, and 600 participants per role in each ultimatum/trust cell; participants failing an attention check or twice failing comprehension questions are screened out. The final Player 1 sample comprises 1,604 (DG-KW), 1,603 (DG-LT), 2,400 (UG), and 2,400 (TG) observations; the Player 2 sample comprises 2,402 (UG) and 2,346 (TG). Trust-game receivers are recruited only for senders who send a positive amount, which is why fewer than 2,400 were fielded.

[13]Open-ended elicitation and automated text classification are an emerging measurement strategy in economics (Ferrario and Stantcheva, 2022; Haaland et al., 2024); here they are deployed inside incentivized games and tied to a model of retrieval.

fies each answer—with classification rules validated by human coders[14]—into one of three substantive categories for Player 1 (*Moral*, *Self-interest*, *Mutual Benefit/Cooperation*) and four for Player 2 (*Moral good*, *Moral bad*, *Self-interest*, *Mutual Benefit/Cooperation*), plus a residual *No clear justification* (henceforth "unclassified").[15] These *strategic* categories are the main categorization: they map onto the attention weights of Section 2.1. Alongside them, the classification codes the *social proximity* of the counterpart the answer evokes (no mention, abstract stranger, anonymous peer, teammate/coworker, friend): whether the player thinks of the other party as a concrete person or as an abstraction. This second dimension carries independent content, because reduced social distance is known to raise giving and the weight of fairness norms in these very games: identification and proximity increase dictator giving (Hoffman et al., 1996; Bohnet and Frey, 1999; Charness and Gneezy, 2008), altruism declines systematically with social distance (Leider et al., 2009; Goeree et al., 2010), and an identified counterpart carries more moral weight than a statistical one (Small and Loewenstein, 2003). Table 2 reproduces representative answers. The categories are audible in the raw text: moral answers reason from fairness and equal division; self-interested answers reason from own need; cooperative answers reason from the joint total and from what the other player will do with what they receive. The social-proximity coding is audible too: high-proximity answers construe the counterpart as a peer ("we're both in this together"), low-proximity answers mention no one at all.

We treat the assigned category as a (noisy, discrete) measure of the participant's representation—the attention profile $\boldsymbol{\alpha}$ of Section 2.1—and three features of the data support this reading. First, categories are not relabeled actions. If participants merely described their choice in

[14] Human coders classified 3,990 responses in six validation subsamples—for each classification batch, two disjoint subsamples drawn stratified on the LLM-assigned category, each coded throughout by one research assistant blind to the condition and to the game. Human–LLM agreement is 76% overall (Cohen's κ = 0.64): 79–80% for Player 1's scheme and 71% for Player 2's finer five-category scheme, where disagreements concentrate at the *Moral good*/*Mutual Benefit* boundary. Appendix G details the protocol and reports rates by subsample and condition.

[15] The residual category accounts for 1–3.5% of control responses; it rises under the market frame, and a footnote in Section 4.2 shows that excluding unclassified answers is innocuous for the treatment effects. Appendix F reports a second, pre-registered representation measure based on participants' descriptions of real-world situations similar to hypothetical final allocations; the two measures deliver consistent conclusions.

Table 2: **Representative Open-Ended Answers, by Category (Control Condition)**

Category	Game	Response
Panel A: Player 1		
Moral	DG-KW	"I am not a greedy person. it just seems fair to split this unexpected money evenly"
Moral	UG	"Fair is fair and I could only hope they did the same, type of thing."
Moral	TG	"Splitting it equally between us so we both get the same seemed the fairest way, and also even if they decide not to send anything back I still get something"
Self-interest	DG-KW	"I don't know these other people. I don't know their story, I don't know who they are, all I know is that I need that $12 so I'm going to choose myself"
Self-interest	UG	"It's strategic...I'm trying to keep a bit more for myself, many responders will still accept this because some money is better than no money..."
Self-interest	TG	"It felt like a safe amount as if they sent back $0 I'd still be able to keep some money."
Mutual Benefit/Coop.	UG	"Actually i love team work. I love it when two or more people work together with honesty and transparency to achieve a common goal which benefits both party."
Mutual Benefit/Coop.	TG	"Because I hope for the best in people and that they will want to share the bonus. (It is in all our best interests.)"
Panel B: Social proximity of the evoked counterpart		
High proximity	UG	"It's split evenly and is most fair. I don't get more and they don't get more. We are equals."
Low proximity	TG	"If player 2 gives back it will be tripled and I will end up with more."
Panel C: Player 2		
Moral good	UG	"Because both amounts were the same and I believe that the sum is a fair proposition!"
Moral bad	TG	"They still earned more money than me so I would not send them back anything"
Self-interest	UG	"It makes sense to accept the offer since while not completely fair, it wasn't that unfair as well. Not accepting means that I get nothing, so the offer I got was much better."
Mutual Benefit/Coop.	TG	"Player 1 has $1 left from the original $6 so I sent $7 back so we both have $8 total"

Notes: Answers to the open-ended reasons question, control condition, reproduced verbatim (including typos), drawn at random within category-by-game cells among responses of typical length (selection rule and further examples in the replication package).

words, a given category would map onto similar behavior wherever it appears; instead, the same category maps onto the behavior the strategic structure dictates: self-interested proposers offer four times what self-interested dictators transfer, because an offer must survive the responder's acceptance decision while a dictator faces no such constraint. A verbal copy of the action could not produce this pattern; a motive filtered through the rules of the game does. Second, categories predict beliefs about the *other* player at a fixed reference action, an object that no rationalization of one's own choice pins down. Third, a second pre-registered measure, elicited *before* the decision and anchored to hypothetical allocations rather than to the choice, delivers the same cross-game and treatment patterns (Appendix F): whatever the categories capture is present before the action is taken. The coherence extends to the person level: the modal pre-decision memory falls in the participant's stated-reason category, and the social proximity of its counterpart tracks the proximity the reasons evoke (Table 38 in Appendix F). The residual caveat is that reasons are elicited after the choice, so category–behavior associations could partly reflect ex-post rationalization of the chosen action; the three features above bound that concern, and we flag it where it bites. A standing language rule follows from the design: throughout the paper, effects of the randomized context manipulations are causal; relationships between categories and behavior are associations, since categories are measured rather than assigned.

Measuring beliefs. First movers in the UG report the probability that their offer is accepted; in the TG, the amount they expect back. We call these *chosen-action beliefs*. Because they condition on endogenously chosen actions, we also elicit *hypothetical beliefs* at a common reference action—the probability that an offer of one third would be accepted, and the expected return to sending one third—which are comparable across participants and treatments. Second movers report both their response to the matched first mover's actual action and their responses to hypothetical actions, including the same one-third reference. In the language of the model, the reasons question measures the attention component $\boldsymbol{\alpha}$

of the representation and the belief questions measure its believed-behavior component $\mathbf{x}$; we elicit hypothetical beliefs at a fixed action precisely because $\mathbf{x}$ is a *function*, and second movers' hypothetical responses trace the same function from the other side.

3.2 Representations and Sender Behavior in Control

The control sample is where the data first meet the model. This subsection establishes four descriptive facts—stated representation categories differ across games, both in what participants emphasize and in the counterpart they evoke; they are associated with actions; they are associated with beliefs; and the action–belief relationship itself varies across categories—and then returns to Section 2 and asks what the facts say about it: the subsection closes with a calibration of the model's parameters, ρ, σ, μ, and the believed behaviors (a, b) and s, category by category.

Representations differ across games. The same population, playing for stakes of comparable magnitude, spontaneously represents the three games in morally different ways. Figure 2 shows the distribution of Player 1 categories in the control condition, by game. In the two fixed-pie games moral reasoning dominates—68.4% of classified DG-KW responses and 76.6% of UG responses—while in the TG the picture changes discretely: Moral falls to 43.8% and Mutual Benefit/Cooperation, negligible in the DG-KW and the UG (0.8% and 3.3%), rises to 37.0%.[16] Only the strategic structure differs across these games, and with it the situations they evoke: the multiplier and the sequential exchange of the TG evoke joint production; the fixed pie of the DG and UG evokes sharing and its ethics.

Social proximity in stated reasons. Stated reasons differ not only in the considerations they emphasize but also in whether and how they refer to the other player. The classification records whether a response refers to the counterpart and, when it does, the counterpart's

[16]Self-interest varies much less across games: 30.8%, 20.1%, and 19.2% in the DG-KW, UG, and TG. One procedural asymmetry qualifies the cross-game comparison: dictator-game participants completed the hypothetical-allocation task of Appendix F before choosing, while UG and TG participants did not.

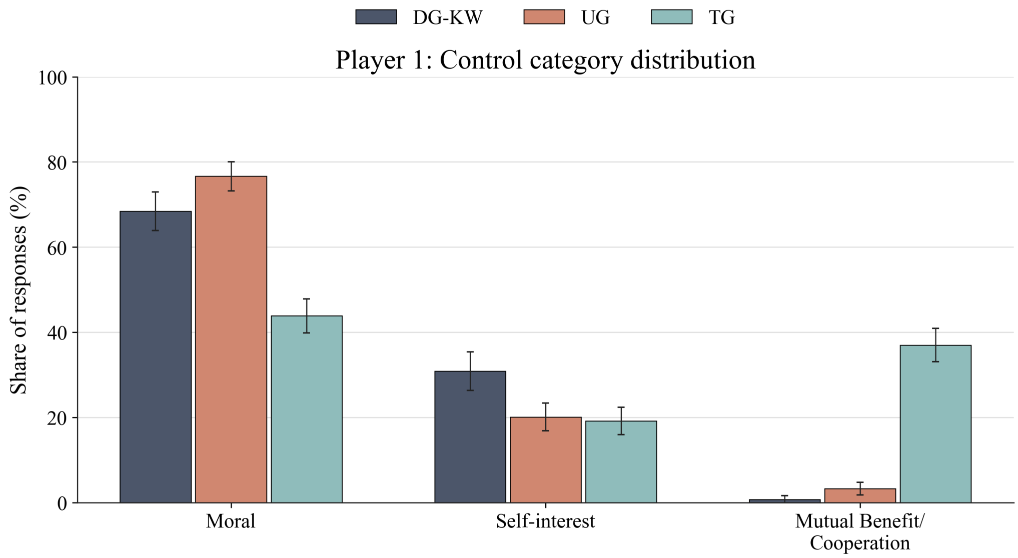

Figure 2: Control-sample representation distributions for Player 1, excluding unclassified observations. Error bars show 95% confidence intervals.

social proximity; Panel A of Table 3 reports the resulting proximity classification within each stated reason category. Self-interested reasons are rarely paired with high social proximity (8.1%), whereas high social proximity appears in 50.6% of Moral reasons. Between these poles sits a discrepancy worth flagging: Mutual Benefit/Cooperation reasons carry high proximity in only 40.6% of cases, a ten-point gap from Moral reasons ($z = 6.51$, $p < 0.001$). Cooperation, as our participants describe it, is not morality extended to a close other: its majority mode is *abstract* cooperation with an anonymous counterpart—joint gains with whoever happens to be on the other side. This distinction does real work below: the market frame, which strips the counterpart of personal identity, crowds out moral representations everywhere yet *raises* cooperative ones (Section 4.2)—possible precisely because cooperation, unlike morality, does not need a concrete person to attach to. Proximity is also associated with behavior. Panel B reports mean actions by proximity and category: within every category, high-proximity answers come with larger transfers (for example, 34.6% versus 23.6% of the budget among self-interested answers). In control, dictators whose response describes a close counterpart transfer 47.0% of the budget, against 29.5% for those whose response describes a distant counterpart or none; the same difference is 2.9 points in the ultimatum game and 10.1 points in the trust game. These patterns are consistent with participants who give different reasons also construing the counterpart differently.

Table 3: **Player 1: Counterpart Proximity within Stated Reason Category**

	Moral	Self-interest	Mutual Benefit/Cooperation
Panel A: Distribution of counterpart proximity (column shares)			
Low social proximity	49.4%	91.9%	59.4%
High social proximity	50.6%	8.1%	40.6%
Panel B: Mean action, share of the budget			
Low social proximity	44.3%	23.6%	57.0%
High social proximity	47.0%	34.6%	63.2%

Notes: Pooled across all games and conditions for Player 1, classified sample. Panel A reports the distribution of counterpart proximity within each stated reason category (columns sum to 100%). Panel B reports the mean action (share of the budget transferred, offered, or sent) in each proximity-by-category cell. High social proximity: the stated reason refers to the counterpart as at least an anonymous peer (anonymous peer, teammate or coworker, friend); low: no mention of the counterpart or an abstract stranger.

Representations and actions. Within each game, stated categories are systematically associated with actions, and these associations differ across strategic environments in the direction anticipated by the model. Figure 3 reports mean actions by category. Where a dictator faces no strategic response, the gap is large: moral dictators transfer 46.2% of the budget and self-interested dictators 10.3%. In the UG, the same category gap is much smaller: moral proposers offer 50.0% and self-interested proposers 41.7%, a pattern consistent with rejection risk constraining low offers. In the TG, cooperative senders send the most (63.5%), moral senders 44.1%, and self-interested senders 29.5%. Thus, the same Self-interest category is associated with near-bottom transfers in the DG and near-fair offers in the UG.

Representations and beliefs. The belief data reveal a sharp contrast between the UG and the TG. Figure 4 reports hypothetical beliefs at the one-third reference action. In the UG, beliefs about acceptance are essentially flat across categories (48.2–48.5%): proposers in different categories report very similar acceptance expectations. These common beliefs

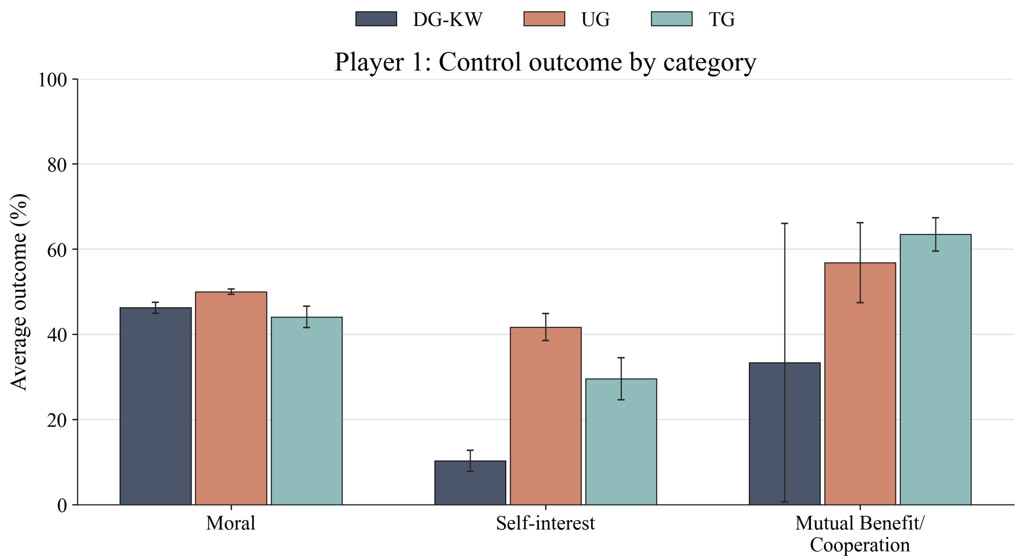

Figure 3: Control-sample outcomes by category for Player 1. Outcomes are the share of the budget transferred (DG-KW), offered (UG), and sent (TG). The DG-KW Mutual Benefit/Cooperation cell contains very few observations (the category is 0.8% of classified DG-KW responses) and should be read accordingly. Error bars show 95% confidence intervals.

may help explain why category differences in UG offers are much smaller than in dictator transfers. In the TG, by contrast, beliefs differ sharply across categories: self-interested senders expect back 22.0% of the tripled amount, moral senders 31.5%, and cooperative senders 35.7%. Participants who describe the game as a cooperative venture also expect more cooperation from their counterpart, whereas those focused on own payoffs expect less. Together, these patterns are consistent with a common perceived acceptance standard in the UG and more category-specific expectations in the TG.

Actions and beliefs jointly. Finally, the relationship between actions and beliefs also varies across categories. Figure 5 shows both patterns in the control data. In the UG, moral proposers' offers are essentially unrelated to their beliefs (slope +0.01), whereas self-interested proposers offer less when they believe a low offer is more likely to be accepted (slope −0.11). In the TG, sending rises with the believed return for every category, and the association is steepest for Self-interest (+0.45, against +0.31 for Moral). Pooling all four conditions sharpens the estimates and rejects equality of slopes across categories in both

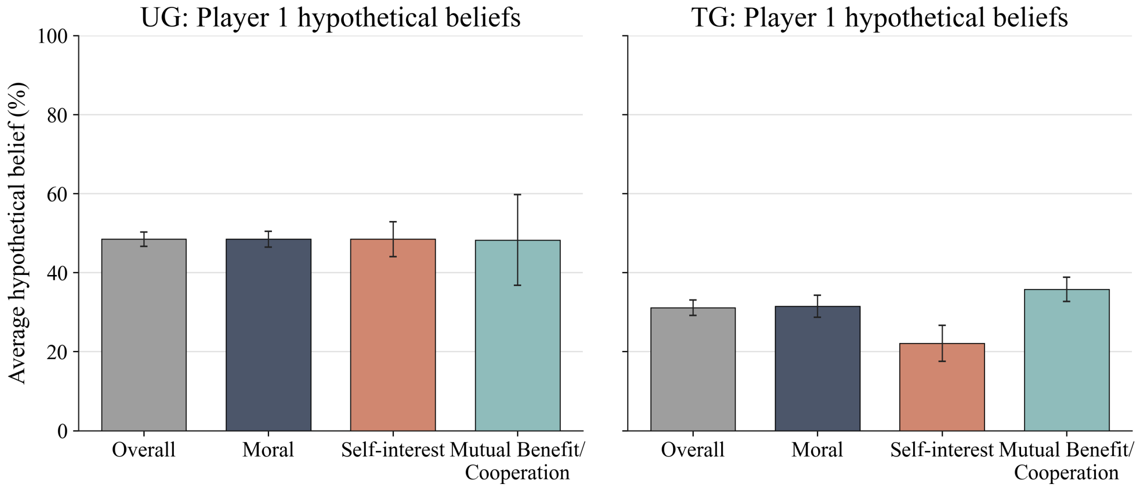

Figure 4: Control-sample hypothetical beliefs for Player 1 at the common one-third reference action, split by UG and TG. Within each game, bars report the overall control-sample average and the category-specific means. Error bars show 95% confidence intervals.

games ($p = 0.002$ in the UG and $p < 0.001$ in the TG; Table 18 in Appendix C). The pooled Mutual Benefit/Cooperation slope in the TG is the smallest of the three (0.13). Finally, the UG slopes are much smaller in magnitude than the TG slopes (-0.09 against $+0.47$ for Self-interest in the pooled specification), consistent with expected returns playing a more direct role in the trust game. These category-specific slopes underlie the pooled belief coefficients in the benchmark regressions of Section 5.1.

Appendix C shows the same three facts for second movers: categories differ across games, predict hypothetical acceptance and returns (with *Moral bad*—condemnation of the first mover—as the strongest predictor of rejection), and interact with the strategic environment as the model predicts.

Calibrating the representations. These moments suffice to put numbers on the model. Behavior identifies attention only up to scale, so the calibrated objects are the ratios $(\sigma/\mu, \rho/\mu)$, category by category: the dictator transfer pins the selfishness ratio (Proposition 1), the trust send together with the measured believed share pins surplus attention (Proposition 3), and the ultimatum first-order condition combined with the reference-action belief pins the believed acceptance schedule (a, b); for Mutual Benefit/Cooperation, whose dictator cell is

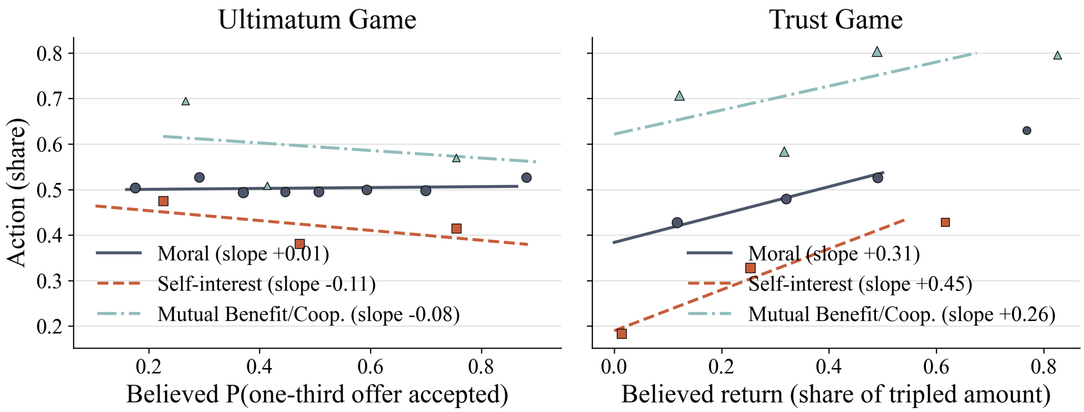

Figure 5: Actions versus hypothetical beliefs by representation category, control sample. Points are quantile-binned means of the Player 1 action against the hypothetical belief at the one-third reference action (marker size increases with bin size); lines are category-specific OLS fits, with slopes in the legends. The UG Mutual Benefit/Cooperation cell is very small ($N = 17$) and should be read accordingly. Slopes, standard errors, and pooled specifications are in Table 18.

negligible, the ultimatum and trust conditions are solved jointly. These moments come from different participants—each participant plays a single game—so the calibration imposes that a category carry the same attention profile wherever it is retrieved: the selfishness ratio estimated on moral dictators is the one applied to moral proposers. That restriction is the model's own claim that attention is a property of the situation rather than of the person, and it is what gives the overidentifying check below its content. Table 4 reports the result: a concrete portrait of each representation. Three features give the calibration content beyond curve-fitting. First, the self-interested representation comes out with $\rho \approx \sigma$ (0.426 against 0.397, relative to μ)—own-earnings attention with no residual surplus motive—which is the mapping of Section 2.1, not an imposed restriction. Second, Moral is norm-dominated ($\sigma/\mu = 0.038$), and for it the inversion returns no valid acceptance schedule: beliefs are unidentified exactly where the model says behavior is insensitive to them. Third, the schedule identified for Self-interest is a genuine probability schedule ($a = 0.324$, $b = 0.481$, $a + b = 0.81 \leq 1$), a restriction nothing in the inversion enforces.

The calibration also fails somewhere informative. The inversion leaves one measurement unused—the belief each proposer attaches to her own chosen offer (the offer itself enters through the first-order condition)—and that is where it fails: measured beliefs at *chosen* offers exceed the calibrated schedule by 11–23 percentage points in every category—proposers are more optimistic about the offer they chose than any coherent schedule through their reference-action beliefs allows—an optimism at chosen actions that returns in the forecast-error analysis of Section 4.3. The calibrated belief sensitivities reproduce the slopes of Figure 5 up to attenuation—estimated over predicted ≈ 0.4 in the ultimatum game and, uniformly across all three categories, 0.15–0.22 in the trust game, where the anchor's common belief channel raises every predicted sensitivity—of the same order as the attenuation the accounting exercise of Section 5 recovers, and consistent with coarse categories—and with return beliefs noisier than acceptance beliefs—measuring the underlying objects with error. Finally, the calibration delivers the coupling between the two components of the representation: assigning each trust-game sender her category's selfishness ratio and her own believed share, $\mathrm{Cov}(\sigma/\mu, s) = -0.008$, and since at interior optima the mean send differs from an independent-marginals counterfactual by exactly $(1+r)\,\mathrm{Cov}(\sigma/\mu, s)$, mean sends sit 2.3 points of the endowment below what independence would deliver—the projection direction, in which self-interested senders also imagine less generous receivers.

These control results establish the empirical content of the representation measure: stated categories vary systematically with games, actions, and beliefs. The next subsection examines the similarity structure of the stories and games, yielding a prediction about where the story treatments should be most consequential. In the decomposition of Section 2.2, a context manipulation moves mean behavior through a retrieval shift, whose size is gated by the similarity measured next and multiplied by the spread of category-typical actions documented above, widest where nothing disciplines the action.

3.3 Treatments and Similarity

The retrieval model of Section 2.2 runs on similarity, and similarity is measurable. Ideally, we want the perceived similarity between our contexts—the two stories, the market and

Table 4: **Calibrated Representations, by Category (Control Sample)**

	Attention		Beliefs			Acceptance at chosen offer	
Category	σ/μ	ρ/μ	a	b	s	implied	measured
Moral	0.038	0.022	–	–	0.315	0.565	0.795
	(0.007)	(0.013)					
Self-interest	0.397	0.426	0.324	0.481	0.220	0.525	0.701
	(0.012)	(0.027)	(0.042)	(0.120)			
Mutual Benefit/Coop.	0.000	0.048	$0.324^{\dagger}$	$0.481^{\dagger}$	0.357	0.598	0.711
	(0.044)	(0.047)					

Notes: Control sample. σ/μ from mean DG-KW transfers (Proposition 1); ρ/μ from the trust-game first-order condition at the measured believed share s (mean hypothetical belief at the one-third reference). For Mutual Benefit/Cooperation, whose dictator-game cell is negligible, $(\sigma/\mu, \rho/\mu)$ solve the ultimatum- and trust-game first-order conditions jointly. (a, b) from the ultimatum-game first-order condition plus the reference-action belief point; for Moral the inversion returns no valid schedule—beliefs are unidentified exactly where the model says behavior is insensitive to them—and † marks the Self-interest schedule imposed on Mutual Benefit/Cooperation (reference-action acceptance beliefs are flat across categories). The last two columns are the overidentifying test: acceptance at the category's mean chosen offer implied by the calibrated schedule—for Moral, the Self-interest schedule, its own being unidentified—versus the measured chosen-action belief; measured beliefs exceed the schedule at chosen offers, the optimism the forecast-error analysis takes up. Bootstrap standard errors in parentheses (participants resampled within game, $B = 1{,}000$).

abstract descriptions—and both the games themselves and broad categories of experience (cooperation problems, conflict problems, moral problems), as well as receiver types (accepting or rejecting, returning or keeping). As a first pass, we measured the perceived *structural* similarity between the two stories and the four games using large language models as raters. A second module takes the measurement to the level the theory specifies: it rates the similarity of every context in the design—the two stories and the Market and Control descriptions of each game—to vignette exemplars of the representation categories themselves (and, for the strategic games, of the receiver types), so that treatment-induced representation shifts can be put against treatment-induced similarity shifts directly. That module is reported at the end of this section; a human-subjects survey replicating both rounds is in progress (designs and materials in the replication package).

Three frontier models—GPT-5.5 Thinking (OpenAI), Claude Opus 4.7 (Anthropic), and Gemini 3 Pro (Google)—each rated, in three separate conversations with independently permuted neutral labels (games presented as A–D, stories as Story 1/Story 2, to rule out position effects and priors about the experimental literature), the similarity of each story to each game—the games' texts being the verbatim Control-condition instructions of Appendix I—on a 0–100 scale, under an instruction to judge the strategic structure of the two situations and to ignore surface thematic content; decomposed the games' structure along five dimensions; and reported, for each game, which story a participant who had just read the game's instructions would be more likely to call to mind as a similar past experience (a 100-point split, with no restriction on what may drive recall). This yields 72 similarity ratings across nine conversations; the findings below hold in every model and every permutation, with small ranges across label permutations.[17]

Table 5: **Perceived Similarity of Stories to Games, LLM Raters**

	Structural similarity (0–100)			Retrieval split
Game	Bonus	Aid	Mean	Bonus / Aid
DG-KW	94.6	94.7	94.6	41 / 59
DG-LT	51.8	50.7	51.2	54 / 46
UG	30.1	29.8	29.9	60 / 40
TG	15.0	13.9	14.4	72 / 28

Notes: Means over nine conversations (three models × three label permutations). The games were presented under neutral labels (Games A–D), their texts the verbatim Control-condition instructions; the stories as Story 1 and Story 2. Structural similarity: 0–100 rating of each story–game pair, under an instruction to judge the strategic structure of the two situations and ignore surface thematic content. Retrieval split: 100 points divided between the two stories according to which the game calls to mind as a similar past experience, with no restriction on what may drive recall.

Table 5 reports three findings. First, the games differ sharply in their structural similarity to the stories, in a fixed order: DG-KW (94.6) > DG-LT (51.2) > UG (29.9) > TG (14.4). Every model identifies the pie structure—expandable versus fixed—as the dimension that isolates the trust game from the other three: the gains from cooperation are what make the

[17] The five dimensions, fixed in the prompt, are: pie structure (fixed-sum, or expandable through one player's action); Player 1's initial endowment (full pie, or partial); Player 2's role (passive recipient, accept/reject responder, or reciprocator with a continuous choice); the source of Player 1's endowment (earned through effort, awarded by selection or luck, produced through interaction, or unspecified); and strategic structure (unilateral allocation, proposer–responder with veto, or sequential interaction with reciprocity). Prompts, the nine transcripts, and the recording workbook are in the replication package.

TG a structurally different situation. Second, *Bonus and Aid are structurally indistinguishable*: their similarity ratings differ by at most 1.1 points for any game. Whatever behavioral differences the two stories produce cannot run through a differently perceived game structure; the stories are the same situation seen from two contexts, which is exactly what the story lever of Section 2.2 requires—it moves F_c while the description stays put—and what a game-specific-norms account cannot use. Third, the *retrieval* direction tells a different story from structural similarity: asked which story each game calls to mind, the models assign Bonus a share that rises monotonically with strategic richness—41% in DG-KW, 54% in DG-LT, 60% in the UG, 72% in the TG—even though structural similarity is flat across the two stories. Surface and contextual cues in the Bonus story (joint production, an ongoing work relationship) increasingly dominate retrieval as the game acquires strategic structure, precisely the wedge between similarity of structure and similarity of context on which the retrieval model runs.

The structural-distance ordering yields a discipline on the story treatments that we use in Section 4.1: if similarity gates retrieval, the boost that a story gives to its category should matter most where the story is structurally close to the game and vanish where it is remote—largest in the dictator games, smaller in the ultimatum game, absent in the trust game. With only four games this is a coarse check, but it is one the data can fail.

From contexts to categories. The second module measures the similarity input at the level the retrieval model specifies. A category, in Section 2.2, is a class of concrete situations, so we represent each Player-1 category by eight vignettes—short descriptions of everyday situations instantiating the category definitions used in the classification, generated blind in a single conversation by a frontier model under constraints that balance domains across categories and ban category-naming words—and each receiver type (accepting or rejecting a proposal, returning or keeping entrusted proceeds) by two vignettes. Fresh rater conversations (a reasoning-tier LLM, three independently permuted label orders, blind to categories and treatments) rated the similarity of each of the ten context texts of the design—the two stories and the Control and Market instructions of each game, verbatim—to each vignette on a 0–100 scale, judging the situation as a whole (what is at stake, the relationship between the parties, the setting, and the structure of the decision); distributed 100 retrieval points across the vignettes per context according to which would come to mind as a similar experience; and, for the strategic games, split 100 points across the receiver-type vignettes. The raters never see the categories: similarity to a category is recovered only afterwards, by mapping each vignette back to the category it instantiates and averaging. Appendix H details the design and robustness checks; the module is a single-model working measurement, with a multi-model grid and the human-subjects survey as the validation path.

Table 6: **Similarity of Contexts to Representation Categories, LLM Raters**

	Similarity (0–100)			Retrieval split (%)		
Context	Moral	Self-int.	Mut. Ben.	Moral	Self-int.	Mut. Ben.
Bonus story	54.9	8.6	9.6	94.0	2.3	3.7
Aid story	57.5	11.1	7.5	93.8	5.5	0.7
DG-KW Control	51.8	10.5	9.1	89.1	5.3	5.6
DG-KW Market	17.6	19.3	19.0	30.8	34.8	34.4
DG-LT Control	45.9	14.3	10.8	79.9	12.5	7.6
DG-LT Market	16.4	19.8	20.0	27.2	36.7	36.1
UG Control	41.1	15.9	13.0	75.4	12.6	12.0
UG Market	16.1	21.8	21.2	22.6	39.9	37.5
TG Control	12.3	20.5	47.3	7.6	17.1	75.2
TG Market	12.9	20.7	48.8	7.8	15.6	76.6

Notes: Similarity: 0–100 rating of each context text against eight vignette exemplars per category, judging the situation as a whole (stakes, relationship, setting, and decision structure), averaged over vignettes and three permuted-label conversations (Claude Opus 4.8 raters with reasoning enabled; single-model working measurement). Retrieval split: 100 points distributed across the 24 vignettes according to which would come to mind as a similar experience, aggregated by category. Context texts are verbatim from the instructions and stories (Appendix I). Design, prompts, and robustness: Appendix H.

Table 6 lines up with the treatment effects of Section 4 on three margins. First, the market frame moves similarity exactly as it moves measured representations in the fixed-pie games: Moral similarity falls by 34.2, 29.5, and 25.0 points in DG-KW, DG-LT, and the UG, while Self-interest rises (+8.8, +5.5, +5.9) *and* Mutual Benefit/Cooperation rises (+9.9,

+9.2, +8.2)—the same replacement of the ethics of sharing by a mixture of own-payoff and joint-gains content as in Figure 8. Second, the two stories are close on Moral similarity (57.5 versus 54.9) but split on the other categories in the directions of Section 4.1: Aid is more similar to the Self-interest exemplars (+2.5), whose need and scarcity content it shares, and Bonus to the Mutual Benefit/Cooperation exemplars (+2.1), whose joint-production content it shares. Third, the control trust game is by far the most cooperation-similar context (47.3, against 9.1 in DG-KW and 13.0 in the UG) and the least moral-similar (12.3)—the similarity counterpart of the cross-game variation in stated categories of Section 3.2. The receiver-type splits (Table 40 in Appendix H) move with the market frame the way first movers' measured beliefs do in the UG: the accepting counterpart's share rises from 42.7 to 52.0 points of 100 while the rejecting share falls symmetrically; the trust-game split barely moves (53.0 to 54.0 toward the returning type).

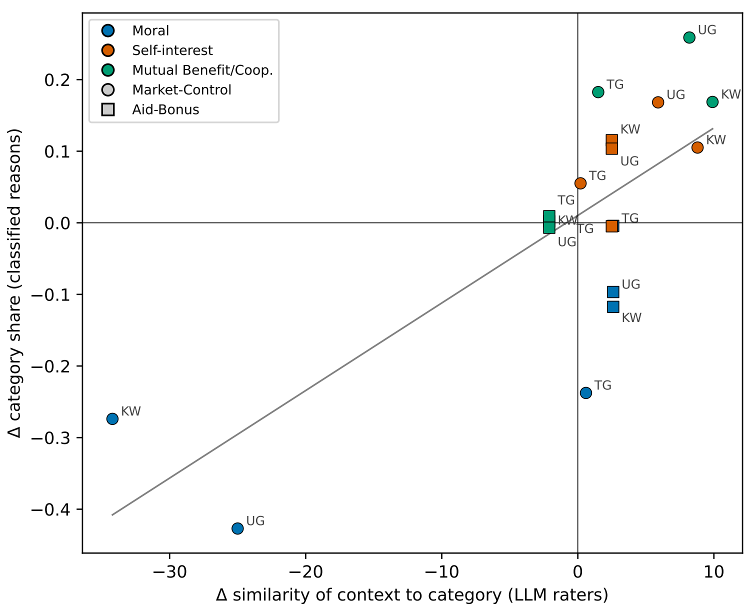

Figure 6: Treatment-induced representation shifts against treatment-induced similarity shifts. Each point is one of the 18 category × game × comparison cells (Market−Control and Aid−Bonus in DG-KW, UG, and TG, three categories each). Color gives the category, the marker the comparison, and the point label the game. Vertical axis: change in the classified category share; horizontal axis: change in the context-to-category similarity of Table 6. The Aid−Bonus cells stack at three similarity values by construction: the story texts do not vary by game, so each category has a single story-induced similarity shift, while the measured representation shifts on the vertical axis vary by game. The line is the OLS fit: slope 0.0122 (se 0.0025), $R^2 = 0.59$.

Figure 6 puts the two measurements together: across the 18 category-by-game-by-comparison cells, the representation shifts measured in the experiment increase in the similarity shifts measured here (slope 0.0122, se 0.0025; $R^2 = 0.59$; Spearman $\rho = 0.57$). The exceptions are as informative as the fit, and both localize the levers. The trust game contributes the off-pattern cells of the Market–Control comparison: its similarity shifts are near zero (at most +1.5 points) while its measured representation and behavior shifts are large—at the level of the instruction text, the control trust game already reads as a joint venture, leaving the market overlay little resemblance to add, so its trust-game effects must run through the availability of commercial experience (F_c) or through beliefs rather than through the described structure. Symmetrically, Aid lowers the classified Moral share without lowering Moral *similarity*: the story lever, as Section 2.2 interprets it, works through what reading the story makes available rather than through what the game text resembles. Both readings are testable in the human-subjects round, and we flag the trust-game cells as the sharpest open item for it.

4 Treatment Effects on Representations and Sender Behavior

Having established how representations organize behavior when context is held fixed, we turn to the randomized context manipulations. For each comparison and game, we report treatment effects on three objects: representation shares, actions, and hypothetical beliefs.[18] Signs implied by the model (via the category-to-weights mapping of Section 2.1, applied to the measured representation shifts) are marked in the figures when the outcome effect is statistically distinguishable from zero and the model's prediction is unambiguous. The two

[18]The figures in this section are drawn on the classified sample—the category panels require it, and each figure uses one sample throughout—while the effect sizes quoted in the text and in the tables use all responses (the residual unclassified category is 1–3.5% of responses). Cell by cell, the two conventions differ by less than one percentage point; both sets are logged in the replication package.

comparisons manipulate different aspects of context. Aid versus Bonus varies the narrative presented before the abstract game, whereas Market versus Control changes the description of the interaction itself. We interpret these treatments through the two levers of Section 2.2: the stories as shifting retrieval frequency (F_c), and the market frame as changing the context of the description ($\boldsymbol{\kappa}_d$).

4.1 Stories: Aid vs Bonus

Reading a story about charitable aid, relative to a story about a workplace bonus, makes representations *less* moral and *more* self-interested in the two sharing games—and behavior follows (Figure 7). Moral falls by 11.7 percentage points in the DG-KW and 9.7 in the UG, and Self-interest rises by 11.5 and 10.4 points; the TG is unaffected on every margin. Dictator transfers fall by 5.6 percentage points of the budget and UG offers by 2.4 points, while TG sends do not move (+0.5, n.s.). Beliefs do not move either (+1.2 points in the UG, −1.6 in the TG, both n.s.), consistent with the representation shift operating through first movers' own weights rather than through their forecasts.

A charitable context does not prime generosity: if anything, a story about need and scarcity licenses self-interest relative to a story about workplace achievement. Context operates not by cueing a behavior ("be generous") but by changing what the situation is *about*—a pattern consistent with stories of hardship directing attention to one's own material stakes. The stories differ in more than setting—the aid story voices the temptation to keep, the bonus story the case for sharing (Appendix I)—so we read the comparison as story content shifting which considerations the game evokes, rather than as isolating hardship per se; the substantive point, that a charitable context fails to trigger generosity, stands either way.

Figure 7 displays a further pattern consistent with the model's interpretation of the story treatment: the effects fade with the structural distance between the stories and the game measured in Section 3.3 (Table 5).

Table 7 places this pattern alongside the structural-similarity ratings, adding DG-LT, the

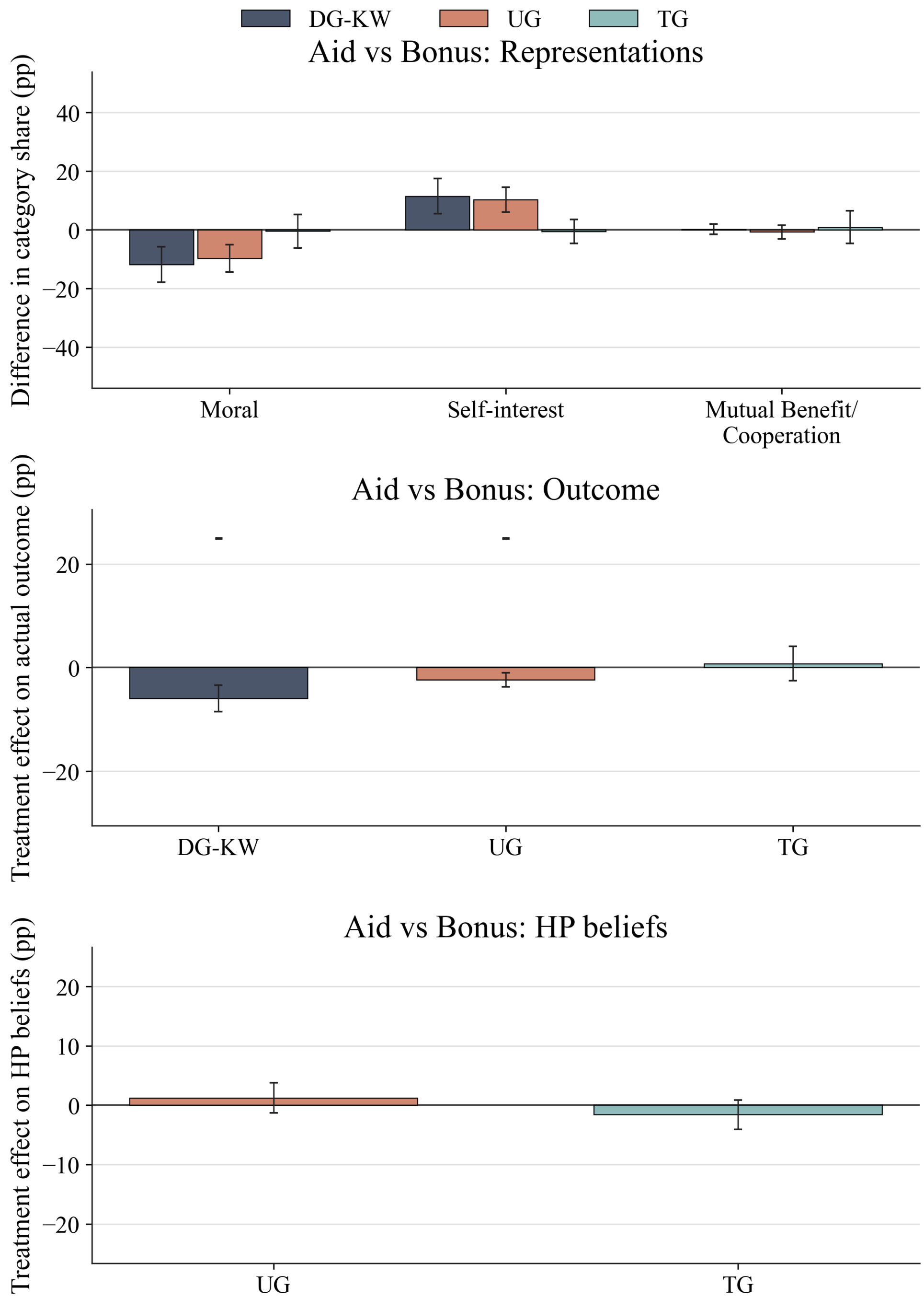

Figure 7: Aid vs Bonus treatment effects for Player 1. The first panel shows treatment-induced shifts in category shares, the second panel shows treatment effects on actual outcomes with model-predicted direction markers, and the third panel shows treatment effects on hypothetical beliefs at the common one-third reference action. A plus or minus sign appears only when the corresponding outcome effect is statistically distinguishable from zero and the model prediction is unambiguous. Error bars show 95% confidence intervals. Each panel shares its y-axis scale with the corresponding panel of Figure 8, so effect sizes are directly comparable across the two treatments.

give-or-take dictator game of Figure 1. DG-LT otherwise appears only in the Introduction and in Section 5.3, and we bring it in here deliberately: the gradient prediction is ordinal, three games give it little bite, and the DG-LT story conditions were fielded and pre-registered precisely to supply the fourth point where the test has power. Story effects on actions decline monotonically as measured similarity falls: −5.6, −4.7, −2.4, and +0.5 (n.s.) percentage points as similarity falls from 94.6 to 51.2 to 29.9 to 14.4. The effect on stated representations shows a similar pattern: the Moral share falls by 9.7 to 14.5 points in the three more structurally similar games and by a negligible 0.4 points in the TG. This ordering is consistent with the prediction that story-induced retrieval shifts matter less when the story and game are structurally remote. With only four games and LLM-based similarity ratings, however, it is an ordinal, exploratory consistency check rather than a separately identified estimate. The negligible TG effects coincide with both a negligible shift in Moral reasoning and the lowest measured similarity to the stories; they do not by themselves establish why trust-game sending is unchanged.

Table 7: **Story Effects Track the Structural Similarity of Stories to Games**

Game	Structural similarity (0–100)	Bonus retrieval share (%)	Aid − Bonus effect (pp) Action	Aid − Bonus effect (pp) Moral share
DG-KW	94.6	41	−5.6	−11.7
DG-LT	51.2	54	−4.7	−14.5
UG	29.9	60	−2.4	−9.7
TG	14.4	72	+0.5	−0.4

Notes: Structural similarity and retrieval shares are means over the nine LLM conversations (Section on Treatments and Similarity); action and Moral-share effects are Aid − Bonus differences (Player 1; Moral share among classified responses). The treatment-effect figures of Section 4 use DG-KW, UG, TG; the DG-LT story conditions enter this table and the dispersion accounting of Section 5.

4.2 Interaction Frame: Market vs Control

Framing the identical interaction as a market transaction does not simply make people selfish: it replaces the ethics of sharing with a mixture of own-payoff reasoning and joint-gains reasoning (Figure 8). Moral representations fall in every game—by 27.4 percentage points in the DG-KW, 42.7 in the UG, and 23.8 in the TG—while *both* Self-interest (+10.5, +16.8, +5.5) and Mutual Benefit/Cooperation (+16.9, +25.9, +18.3) rise.[19]

Behavior tracks the attention weights game by game. Where the pie is fixed and efficiency attention is idle, transfers fall exactly as the moral-to-self-interest shift predicts: dictator giving falls by 10.9 percentage points and UG offers by 21.9 points (95% CI $[-24.2, -19.6]$). In the TG, where efficiency attention is the engine of sending, sends *rise* by 14.5 points (95% CI $[11.3, 17.7]$), in line with the rise of cooperative representations. Beliefs move in the same direction as the cooperative shift: proposers become markedly more optimistic that a one-third offer would be accepted (+17.2 points) and senders more optimistic about repayment (+10.3 points). Second movers show the mirror image: the market frame shifts their representations the same way, yet their hypothetical acceptance and returns *rise* by about 10 points (Section 5).

Treatment effects on means understate what the frame does. Figure 9 plots the full distributions of Player 1 actions. The market frame does not slide a distribution of mostly-fair choices toward selfishness; it dissolves the equal-split anchor. In the UG, the share of proposers offering exactly half collapses from 76.5% to 18.3% while zero offers rise from 1.0% to 32.0%; in the DG-KW, the even-split spike falls from 51.4% to 14.0% and zero transfers rise from 18.9% to 37.2%. In the TG, mass moves the other way, toward the maximum: full sends rise from 16.0% to 27.0%. The equal-payoff anchor predicts this corner: for a

[19] Category shares are computed among classified answers, and the unclassified share itself responds to the frame: it rises from 2.4% of Control answers to 13.0% of Market answers pooled across games (17.8% in the DG-KW). The exclusion is innocuous. Forcing all 309 unclassified Control and Market answers into the three substantive categories and recomputing the shares leaves every category effect in this comparison within 5 percentage points of the baseline, with no sign change. The forced labels come from an example-based classification (Claude Opus 4.8, as in Appendix F) anchored to the 95 unclassified answers whose forced category an earlier classification pass and at least one of two independent human coders assigned identically; leave-one-out accuracy on these examples is 84% ($\kappa = 0.71$), and the result is unchanged substituting a different classifier model or the earlier forced labels throughout. Nor are the unclassified answers random noise: they are the far end of the counterpart-abstraction pattern documented at the end of this section—under the market frame they run shorter than classified answers (15 versus 22 words on average), 92% mention no counterpart (against 46% of classified Market answers), two-thirds read as Self-interest when forced, and their authors' transfers sit slightly below those of classified participants in the same condition.

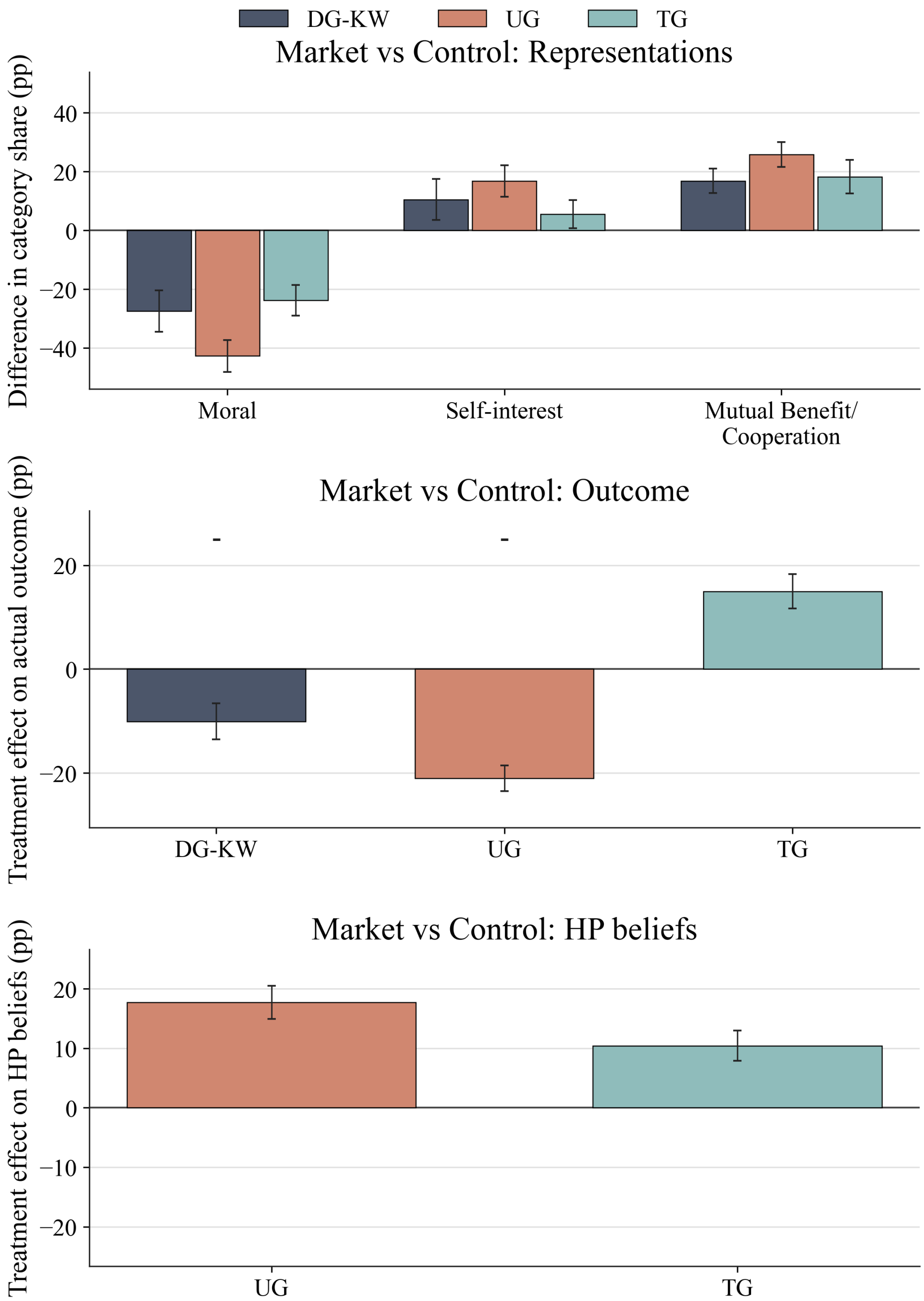

Figure 8: Market vs Control treatment effects for Player 1. Panels as in Figure 7, with identical y-axis scales panel by panel: the market frame moves representations, outcomes, and beliefs by multiples of the story effects.

sender who believes at least half of the output comes back, the full send is the unique payoff-equalizing action—the norm itself points to the corner (Proposition 3)—and the share of senders holding such beliefs rises from 30.5% to 52.0% across the two conditions. A shift in a single generosity or selfishness parameter would translate these distributions; what the data show is mass moving between qualitatively different actions—the norm anchor, zero, the maximum—which is what a change in the mixture of representations, each with its own anchor, produces. Section 5.1 returns to this in variance terms.

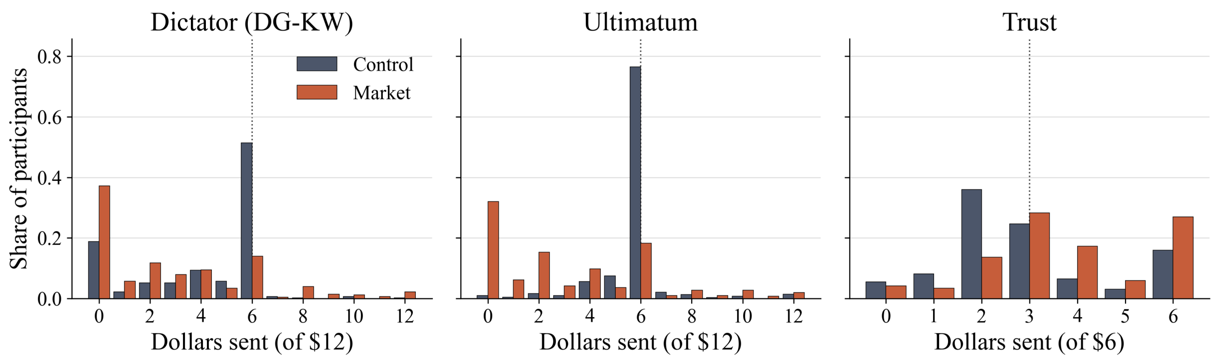

Figure 9: Distributions of Player 1 actions, Control versus Market. Bars are shares of participants at each dollar amount; the dotted line marks the equal split. Kolmogorov–Smirnov tests reject equality of the control and market distributions in every game ($D = 0.32$, 0.58, 0.28 in DG-KW, UG, TG; all $p < 10^{-17}$).

The market frame is deliberately a *bundle* of naturalistic cues rather than a minimal manipulation: it evokes produced—hence earned—endowments, it reframes the dictator's transfer as a price cut from an owned amount, and it gives the broker a concrete value-added role in the TG. Entitlement effects (Hoffman et al., 1994; Cherry et al., 2002) are thus one ingredient of the bundle, exactly as they are in the field settings the frame imitates. The paper's claim is not that any single cue is minimal and payoff-irrelevant. It is that the bundle's effects travel through measurable representations—exactly what the category shifts, the belief shifts, and the accounting exercise of Section 5 test.

Two further features of these results resist explanations based on stable preferences plus standard channels. First, the same manipulation moves behavior in opposite directions across games (giving falls, trusting rises), which no single-dimensional shift in generosity or selfishness can produce, but which a reallocation of attention across three motives produces naturally. Second, beliefs move together with representations, strongly under the market frame and not under the stories. This contrast is consistent with the two treatments affecting different aspects of context: the stories change the narrative presented before an otherwise abstract game, whereas the market frame changes the described interaction itself. The data suggest that the latter treatment also changes how participants describe the counterpart, a possibility we examine next.

Changes in references to counterparts are visible in both measures. Under the market frame, the share of answers that mention no counterpart at all roughly triples in the reasons measure, from 18.5% to 57.4% in the DG-KW, 12.9% to 40.5% in the UG, and 12.2% to 38.7% in the TG; the hypothetical-allocation measure shows the same pattern (28.8% to 75.9%, Appendix F). Together with the rise of Mutual Benefit/Cooperation, the trust-game pattern is consistent with more generalized rather than relational trust: market senders send more while describing the counterpart less often as a particular person. The measures record how participants describe the counterpart; they do not directly measure generalized trust.[20] The market frame may thus make a particular counterpart less salient; where the strategic structure rewards cooperation with abstract counterparties, this abstraction need not lower sending. We close the section with what the treatments do to the accuracy of beliefs; their consequences for surplus follow in Section 5.

4.3 Beliefs and Forecast Errors

Under rational expectations, context could move beliefs only by moving opponents' behavior, and forecast errors would be centered at zero in every treatment. Figure 10 shows they are

[20]Glaeser et al. (2000) find that survey measures of generalized trust predict trustworthiness in the investment game better than they predict sending; Cox (2004) separates trust from unconditional generosity in sending; Ashraf et al. (2006) decompose trust into expectations and kindness; Johnson and Mislin (2011) meta-analyze 162 trust-game replications; Karlan (2005) links game behavior to real repayment; Fehr (2009) reviews.

not. In the UG, proposers systematically underestimate acceptance: signed forecast errors (belief minus realized Player 2 behavior) average roughly −17 percentage points in Control and −16 in the story conditions, and swing to +9 under the market frame—proposers there expect brokers to accept even more readily than brokers actually do. Chosen-action errors condition on endogenously chosen offers, so part of the market swing reflects the shifted offer distribution; reference-action errors, immune to this selection, deliver a similar picture (Figure 17 in Appendix C): substantial pessimism that shrinks sharply under the market frame, with one difference—at the fixed one-third reference the market error stays mildly negative, because the optimism lives at the very low offers market proposers actually choose. At zero offers, market proposers' beliefs exceed realized acceptance by 55 percentage points, while their errors at fair offers remain negative. In the TG, errors are small in Control and Market (+1 to +3 points) and mildly pessimistic in the story conditions (−4 to −5 points).

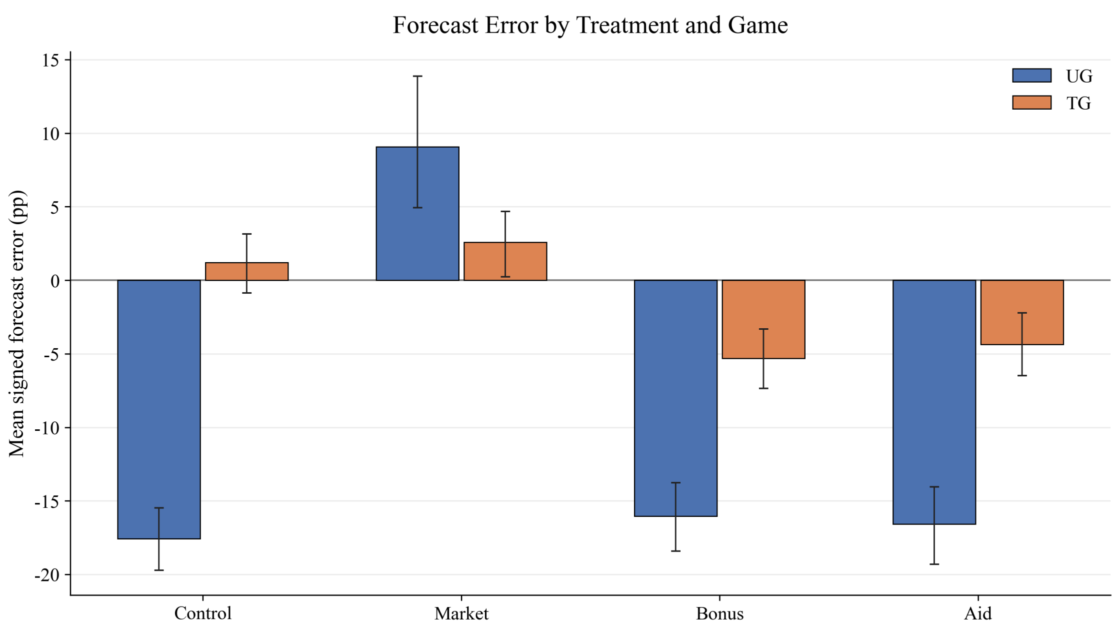

Figure 10: Forecast errors by treatment, chosen-action beliefs. The benchmark is the mean realized Player 2 behavior at the same Player 1 action within the same condition-and-game cell. Positive values indicate optimism and negative values pessimism. Error bars show bootstrap 95% confidence intervals. The counterpart figure for hypothetical beliefs at the one-third reference action is Figure 17 in Appendix C.

Two features tie the errors to representations. First, treatments change forecast errors, not just beliefs: the market frame moves proposers' beliefs upward by more than it moves receivers' behavior, exactly what one expects if beliefs are generated by projecting one's own (treatment-shifted) representation onto the opponent rather than by tracking the opponent's response to the same treatment. Second, within treatments, chosen-action forecast errors covary with representations: in the UG, self-interested and cooperative proposers make errors 13–16 percentage points more optimistic than moral proposers with the same hypothetical beliefs (Table 19 in Appendix C). This individual-level covariation is partly mediated by the offers those proposers choose: for reference-action errors it attenuates and, for Self-interest, reverses sign (Table 20), so we treat it as suggestive. The case against rational expectations rests on the first, treatment-level pattern, which survives the selection-immune measure: rational mechanisms—in which contextual cues shift equilibrium behavior and beliefs adjust correctly—cannot move mean forecast errors at all. The full model of Appendix B makes the pattern a prediction rather than a residual: the sender's imagined receiver and the receiver's actual behavior are retrieved from different memories given different descriptions, and nothing aligns the two retrievals, so a treatment that moves the two descriptions differently moves mean forecast errors.

A third feature pins down what exactly proposers get wrong. Each proposer reports two beliefs—at the reference action and at her chosen offer—which identify a *believed slope* of the acceptance schedule, and second movers' responses identify the actual slope. The actual schedule is steep below the reference action and nearly flat above it: within the market condition, realized acceptance climbs with the offer at a rate of 1.98 (s.e. 0.18) on the below-reference segment. Market proposers' believed slope on that same segment has a median of 0.00: they imagine a broker who takes whatever is profitable, whose acceptance barely depends on the price—and so they cut offers as if acceptance were insensitive precisely where it is most sensitive. In the control and story conditions the distribution of offers sits at or above the reference action, where the actual schedule is flat and the error is one of level, not shape: beliefs undershoot acceptance everywhere. Context thus changes more than the level of beliefs; it changes the imagined counterpart, and with it the believed shape of the response schedule—which answers a natural question about the errors' origin: proposers' forecasts go wrong under the market frame not because second movers respond to context more than proposers think, but because proposers stop believing that second movers respond to the *offer*.

These forecast errors have allocative consequences: they are how the market frame destroys surplus in the UG (Section 5.2). Receivers under the market frame are, if anything, more accepting of a given low offer (Figure 11), but proposers—expecting brokers to accept even more readily than they do—cut offers so aggressively that acceptance collapses and a third of the pie is lost. In the TG, where forecast errors stay small, the same frame raises surplus. We return to these departures in the Conclusions.

5 Explanatory Power of Representations

Sections 3 and 4 establish the two links of the proposed chain separately: representations predict behavior within a context, and randomized context shifts representations. This section combines them. Section 5.1 asks whether measured representation shifts, weighted by how each representation behaves, quantitatively account for treatment effects—on actions and beliefs, for both players—and for the heterogeneity behind the means. Section 5.2 traces the allocative consequences for surplus and inequality. Section 5.3 then returns to the puzzle that opened the paper and reads it with the same instruments.

5.1 Accounting for Treatment Effects and Heterogeneity

Second movers. Before the regressions, we bring in the second movers. They enter through their *hypothetical* responses—acceptance of a given offer, return of a given send—because these are their only clean partial-equilibrium outcomes: realized acceptance and returns respond to endogenously chosen first-mover actions, and even measured representations are elicited after the receiver faces the action, which may itself cue her retrieval (Appendix B). Realized returns also carry the anchor's signature: within *Moral good* and *Mutual Benefit/Cooperation* receivers, returned shares rise with the send along the equal-payoff schedule, while senders' believed schedules stay flat—the wedge behind the trust-game forecast errors (Table 16 in Appendix B). Figure 11 reports treatment effects on hypothetical outcomes at the one-third reference action, together with the treatment-level means of the hypothetical actions. The market frame crowds out receivers' *Moral good* representations in favor of Self-interest and *Moral bad* (Figure 16 in Appendix C), yet both hypothetical acceptance of a one-third offer and trustees' hypothetical returns *rise* by about 10 points—the mirror image of the sender-side pattern, and the reason realized surplus moves as it does below. Control-sample distributions, category–outcome associations, and the underlying regressions are in Appendix C.

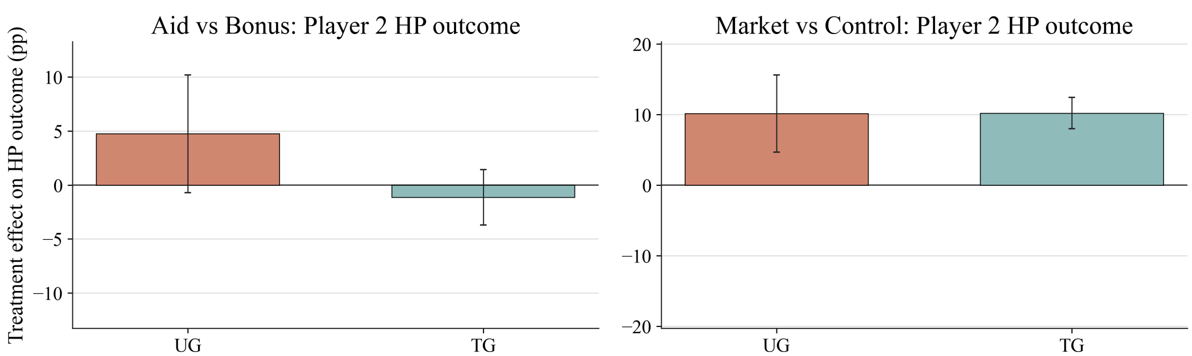

Figure 11: Player 2 hypothetical behavior: treatment effects on hypothetical outcomes at the one-third reference action (acceptance of a one-third offer in the UG, returned share after a one-third send in the TG). Panels report Aid vs Bonus and Market vs Control. A model-predicted direction marker appears only when the effect is statistically distinguishable from zero and the model prediction is unambiguous. Error bars show 95% confidence intervals.

Benchmark regressions. The benchmark regressions estimate the association between representation categories and each outcome—the per-category behavior that, combined with the treatment-induced category shifts, generates the predictions of the accounting exercise below. For each pre-registered comparison (Aid vs Bonus; Market vs Control) we pool the two conditions and, separately by game, regress Player 1 outcomes on category dummies, adding hypothetical beliefs for UG offers, TG sends, and the belief outcomes to capture the belief channel of Section 2. For Player 2, we regress UG acceptance and TG returns on category dummies plus the participant's hypothetical response at the reference action *and* the fitted Player 1 action she faces: realized acceptance and returns respond to the amount sent at least as much as to context (Appendix B), so the amount-sent channel must be inside the benchmark, and the accounting below moves it along with the representation shifts. The belief outcomes in the Player 1 regressions and in the fit exercise below are chosen-action beliefs. Tables 8–11 report the estimates.

The coefficients quantify the control-sample patterns: category gaps are large where representations bite and small where strategic discipline mutes them. Self-interest (relative to Moral) is associated with dictator transfers 30–34 percentage points lower, but with UG offers only 12–16 points lower; Mutual Benefit/Cooperation is associated with TG sends 24–33 points higher. Among second movers, Moral bad is the dominant predictor, associated with acceptance 84–91 percentage points lower and returns 28–29 points lower. Beliefs enter exactly where the model says they should: hypothetical beliefs predict TG sends strongly (coefficients 0.24–0.36) and enter UG offers with the opposite, model-consistent sign (-0.04 to -0.15)—a proposer who believes a low offer would be accepted offers less. These are exactly the belief-channel signs of equations (3) and (5), and their opposite signs across the two games are a joint prediction no belief-free account shares.

Table 8: **Benchmark Regressions: Player 1, Aid vs Bonus, Categories + Beliefs HP**

	(1) KW action	(2) UG action	(3) TG action	(4) UG beliefs	(5) TG beliefs
Self-interest	-0.337***	-0.124***	-0.151***	-0.159***	-0.031**
	(0.010)	(0.009)	(0.024)	(0.016)	(0.015)
Mutual Benefit / Cooperation	0.011	0.059***	0.326***	0.040	0.106***
	(0.033)	(0.016)	(0.017)	(0.028)	(0.010)
No clear justification	-0.245***	-0.032	-0.035	-0.210***	0.015
	(0.036)	(0.024)	(0.077)	(0.041)	(0.047)
Hypothetical beliefs		-0.038**	0.240***	0.345***	0.574***
		(0.015)	(0.042)	(0.026)	(0.026)
Constant	0.457***	0.512***	0.380***	0.629***	0.117***
	(0.005)	(0.008)	(0.018)	(0.014)	(0.011)
Observations	801	1142	947	1142	947
R^2	0.599	0.162	0.407	0.193	0.436

Standard errors in parentheses. Baseline category is Moral. Sample restricted to Bonus and Aid. UG and TG equations include hypothetical beliefs in addition to category dummies; KW action uses category dummies only. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 9: **Benchmark Regressions: Player 2, Aid vs Bonus, Categories + Beliefs HP**

	(1) UG outcome	(2) TG outcome	(3) UG outcome + fitted share sent P1	(4) TG outcome + fitted share sent P1
Moral bad	-0.844***	-0.294***	-0.746***	-0.212***
	(0.024)	(0.020)	(0.029)	(0.019)
Mutual Benefit / Cooperation	0.004	0.004	0.011	0.001
	(0.012)	(0.011)	(0.011)	(0.010)
No clear justification	-0.016	-0.093**	-0.002	-0.067*
	(0.034)	(0.038)	(0.033)	(0.034)
Self-interest	-0.062***	-0.178***	-0.032*	-0.162***
	(0.016)	(0.020)	(0.016)	(0.018)
Hypothetical acceptance	0.027**		0.031***	
	(0.011)		(0.011)	
Hypothetical share sent P2		0.516***		0.504***
		(0.025)		(0.023)
Predicted share sent P1			1.259***	0.558***
			(0.220)	(0.040)
Constant	0.964***	0.236***	0.350***	-0.089***
	(0.010)	(0.011)	(0.108)	(0.025)
Observations	1142	923	1142	923
R^2	0.529	0.497	0.542	0.585

Standard errors in parentheses. Baseline category is Moral good. Sample restricted to Bonus and Aid. Columns (3) and (4) add fitted share sent P1, computed from the comparison-specific Player 1 action regression in the same specification by averaging fitted values within each observed Player 1 donation level. UG equations include hypothetical acceptance and TG equations include hypothetical share sent P2; columns (3) and (4) additionally include fitted share sent P1. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: **Benchmark Regressions: Player 1, Market vs Control, Categories + Beliefs HP**

	(1) KW action	(2) UG action	(3) TG action	(4) UG beliefs	(5) TG beliefs
Self-interest	-0.299***	-0.164***	-0.083***	-0.093***	0.001
	(0.016)	(0.016)	(0.022)	(0.015)	(0.013)
Mutual Benefit / Cooperation	-0.082***	-0.162***	0.244***	-0.066***	0.080***
	(0.027)	(0.020)	(0.019)	(0.018)	(0.011)
No clear justification	-0.211***	-0.197***	0.074*	-0.148***	0.067***
	(0.025)	(0.022)	(0.041)	(0.021)	(0.023)
Hypothetical beliefs		-0.148***	0.364***	0.281***	0.674***
		(0.027)	(0.038)	(0.025)	(0.022)
Constant	0.418***	0.543***	0.368***	0.629***	0.100***
	(0.010)	(0.017)	(0.020)	(0.016)	(0.011)
Observations	803	1137	1011	1137	1011
R^2	0.324	0.179	0.330	0.129	0.542

Standard errors in parentheses. Baseline category is Moral. Sample restricted to Control and Market. UG and TG equations include hypothetical beliefs in addition to category dummies; KW action uses category dummies only. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11: **Benchmark Regressions: Player 2, Market vs Control, Categories + Beliefs HP**

	(1) UG outcome	(2) TG outcome	(3) UG outcome + fitted share sent P1	(4) TG outcome + fitted share sent P1
Moral bad	-0.906***	-0.280***	-0.724***	-0.230***
	(0.033)	(0.022)	(0.035)	(0.023)
Mutual Benefit / Cooperation	-0.057**	-0.011	-0.022	-0.007
	(0.024)	(0.010)	(0.023)	(0.010)
No clear justification	-0.243***	-0.024	-0.157***	-0.025
	(0.055)	(0.021)	(0.053)	(0.021)
Self-interest	-0.302***	-0.148***	-0.161***	-0.152***
	(0.024)	(0.015)	(0.026)	(0.015)
Hypothetical acceptance	0.038*		0.071***	
	(0.020)		(0.019)	
Hypothetical share sent P2		0.529***		0.504***
		(0.026)		(0.025)
Predicted share sent P1			1.738***	0.445***
			(0.160)	(0.054)
Constant	0.951***	0.226***	0.206***	-0.039
	(0.019)	(0.012)	(0.071)	(0.035)
Observations	1137	1016	1137	964
R^2	0.432	0.481	0.486	0.515

Standard errors in parentheses. Baseline category is Moral good. Sample restricted to Control and Market. Columns (3) and (4) add fitted share sent P1, computed from the comparison-specific Player 1 action regression in the same specification by averaging fitted values within each observed Player 1 donation level. UG equations include hypothetical acceptance and TG equations include hypothetical share sent P2; columns (3) and (4) additionally include fitted share sent P1. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Predicted versus actual treatment effects. We then combine each benchmark regression with the treatment-induced shifts in its regressors—category shares and, where included, hypothetical beliefs (for Player 2, also the fitted Player 1 action)—to *predict* the treatment effect on each outcome, and compare the prediction with the actual effect. This is a demanding exercise: the benchmark coefficients are identified mostly from within-condition heterogeneity, and the prediction uses only the movement of measured representations and beliefs across conditions.

Figure 12 plots predicted against actual effects for Player 1 actions, Player 1 beliefs, Player 2 outcomes, and Player 2 hypothetical actions; Table 12 reports the calibration regressions. Three findings stand out. First, predicted and actual effects agree in sign in 17 of the 18 cells, including every action cell—and including the reversal between UG offers (down under Market) and TG sends (up under Market). The single sign miss is a belief cell: the model predicts essentially no effect of the market frame on proposers' chosen-action acceptance beliefs (+0.003), while the actual effect is negative (−0.062).[21] Second, the fit is tight: regressing actual on predicted effects across all 18 outcome-by-comparison cells yields a slope of 1.24 and $R^2 = 0.87$ (Player 1: slope 1.77, $R^2 = 0.95$; Player 2: slope 1.10, $R^2 = 0.92$). We read slope and R^2 as descriptive fit statistics: the cells are estimated from overlapping samples, so the conventional standard errors in Table 12 are indicative at best.[22] Third, predicted magnitudes fall short of actual ones (predicted-to-actual ratios of roughly 0.5 for the large first-mover effects). Coarse categories measuring the underlying attention weights with classical error would produce exactly this attenuation, and the pre-choice measure of Appendix F supports that reading; measurement error of the rationalization type would

[21] Treatment effects on beliefs in Section 4 refer to hypothetical beliefs at the reference action; the belief cells in the fit exercise are chosen-action beliefs, so magnitudes are not comparable across the two.

[22] Categories alone, without the belief channel, already deliver $R^2 = 0.80$ with slope 1.41 (Player 1 alone: slope 2.77), so the accounting power does not hinge on beliefs; beliefs mainly improve the fit for the trust game. See Table 21 in Appendix C. For Player 2, the sender-action channel matters: predicting receiver outcomes from categories and hypothetical responses alone, without the fitted Player 1 action, lowers the predicted-to-actual ratio for UG acceptance under the market comparison from 0.88 to 0.62—a measure of how much of the receiver-side treatment effects travels through treatment-shifted sender behavior rather than through receivers' own representations.

push the other way, however, so we read the ratios as descriptive rather than as bounds. A specification check follows from the model itself: belief sensitivities differ by category (Table 1), so the additive benchmark is a simplification. Re-estimating the benchmarks with category-by-belief interactions recovers the interactions with the model's signs—negative for Self-interest in the UG (-0.17, $p = 0.02$), positive in the TG ($+0.31$, $p = 0.009$)—but leaves the predicted effects essentially unchanged (Table 22 in Appendix C): the missing interaction was not the reason for the distance between predicted and actual effects.

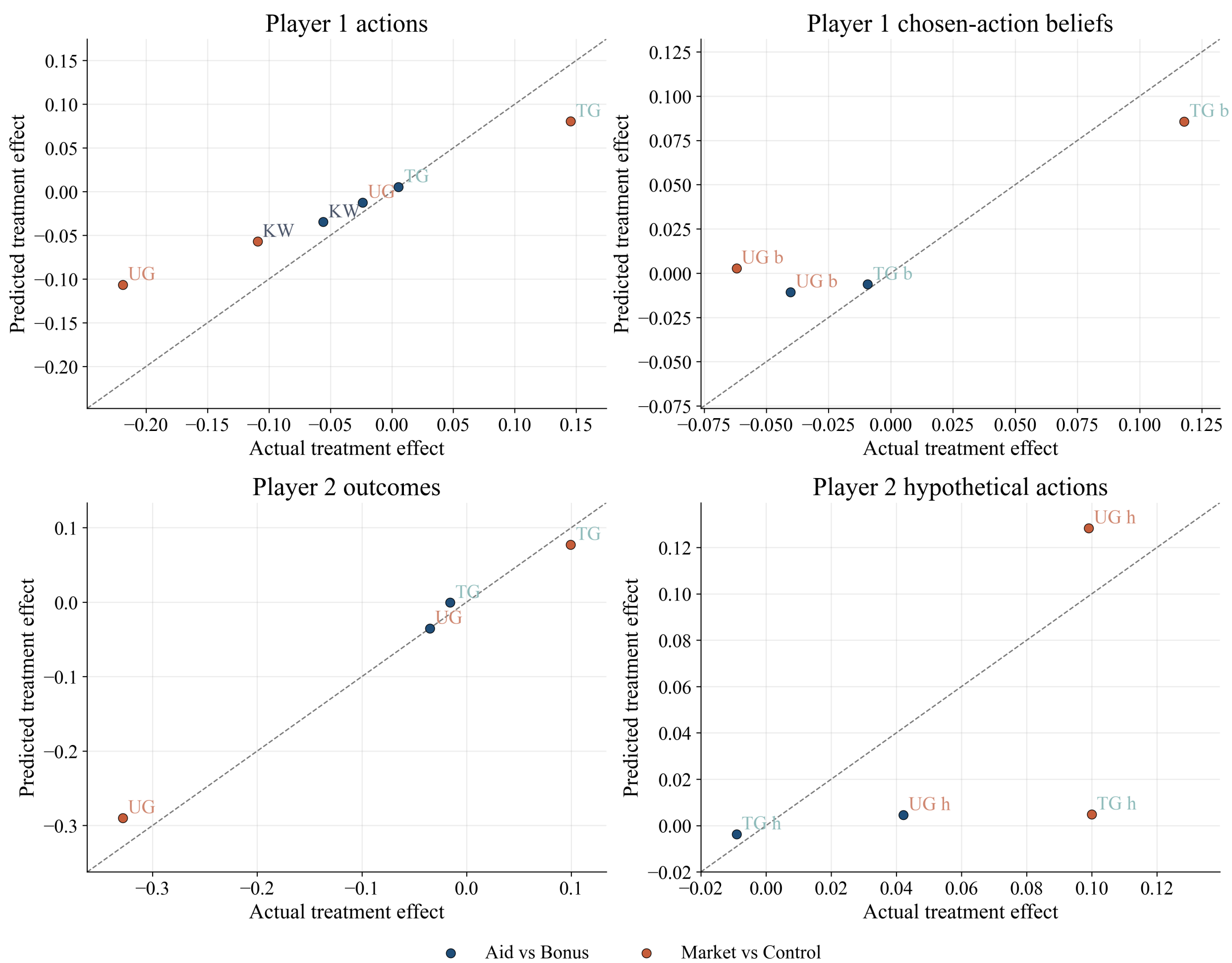

Figure 12: Predicted versus actual treatment effects, Categories + Beliefs specification (the Player 2 panels add the fitted Player 1 action): Player 1 actions (top left), Player 1 chosen-action beliefs (top right), Player 2 outcomes (bottom left), and Player 2 hypothetical actions (bottom right). Each point is an outcome-by-comparison cell; point labels give the game, with the b and h suffixes marking the belief and hypothetical-action cells; the dashed 45-degree line marks exact fit.

The economic content of this exercise is the paper's central claim: the same handful

Table 12: **Instability calibration regressions, Categories + Beliefs HP**

	All	Player 1	Player 2
Intercept	-0.005	-0.016*	0.010
	(0.010)	(0.008)	(0.016)
Predicted	1.238***	1.771***	1.100***
	(0.119)	(0.140)	(0.137)
Observations	18	10	8
R^2	0.871	0.952	0.915

Notes: Unit of observation is an outcome-by-comparison cell from the fitted-values exercise. Each column reports the regression of the actual treatment effect on the predicted treatment effect. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

of measured representations that distinguish who gives, offers, trusts, and rejects *within* a context also account for how behavior moves *across* contexts. The same categories therefore organize both variation across participants within a given context and behavioral change when the context changes.

From mean effects to heterogeneity. The model's first implication in Section 2.2 is about dispersion, not means: within a condition, behavior is heterogeneous because representations are, and a treatment that shifts the category mixture should reshape the whole outcome distribution, not just its center. Two statistics tie heterogeneity to representations directly. The first is the *between-category share* of outcome variance: within a game-and-condition cell with category shares q_c, category mean actions $\bar{t}_c$, and overall mean $\bar{t}$,

$$\text{BCS} \;=\; \frac{\sum_c q_c \left(\bar{t}_c - \bar{t}\right)^2}{\text{Var}(t)}, \tag{7}$$

the fraction of dispersion that lies between representations rather than within them.[23] The second takes the accounting to dispersion: holding each category-conditional action distribution fixed and reweighting it by the other condition's measured category shares yields a counterfactual distribution, whose standard deviation is compared with the actual one—the dispersion analogue of the mean decomposition $\Delta E[t^g] = \sum_c \Delta q_c \, \bar{t}_c^g$ of Section 2.2. Table 13 reports the key cells (the full decomposition, including DG-LT, is Table 23 in Appendix D). In control, the between-category share of action variance is 0.66 in the DG-KW against 0.11 in the UG and 0.20 in the TG: where nothing disciplines the dictator, which representation a participant holds is most of the story, while rejection risk compresses offers toward the common standard so strongly that categories have little variance left to explain—the variance-domain counterpart of the mean compression of Section 3.2, and of Proposition 2. The treatments move the sorting itself: under the market frame the DG-KW between-share collapses from 0.66 to 0.14 while overall dispersion widens (the standard deviation rises from 0.20 to 0.26). The frame does not merely reshuffle participants across categories that go on sorting behavior as before; it also moves behavior within categories—the sender-side counterpart of the within-category margin that the receiver decomposition of Appendix B makes explicit.

Figure 13 takes the accounting to dispersion. Reweighting the category-conditional action distributions by each condition's measured category shares—holding within-category behavior fixed—predicts the change in the standard deviation of behavior with the correct sign in all eight game-by-comparison cells (the six DG-KW/UG/TG cells in the figure, plus the two DG-LT cells in Table 24 in Appendix D). Magnitudes are close in the story comparisons of the two dictator games and understated where dispersion moves most (the UG story

[23] Equation (7) uses the coarsest cut of the representation—the stated category alone. The belief component adds little dispersion on top: on the subsample with non-missing hypothetical beliefs, refining the cells to category interacted with belief terciles raises the between-cell share by at most 0.06 across the eight ultimatum- and trust-game condition cells, under any of three tercile conventions, relative to categories alone on the same subsample (for example, from 0.28 to 0.31 in the control trust game; replication package). This is consistent with the control patterns of Section 3.2: acceptance beliefs are nearly flat across categories in the ultimatum game, and in the trust game beliefs differ by category, so the category dummies already absorb much of that variation.

Table 13: **Heterogeneity Tied to Representations: Variance Decomposition**

Game	Condition	N	SD of action	Between-category share
DG-KW	Control	399	0.204	0.663
	Market	329	0.263	0.136
UG	Control	582	0.111	0.109
TG	Control	584	0.282	0.202

Notes: Player 1, classified responses only. "Between-category share" is the share of the within-condition variance of the action lying between the three representation categories (the R^2 of the action on category dummies). The full decomposition for all games, conditions, and DG-LT is reported in the appendix table.

conditions, and the DG-KW and UG under the market frame); the small actual changes in the remaining cells are matched only loosely. The mean panel shows the same attenuation as Figure 12 (predicted-to-actual ratios of 0.28–0.46 for the large market effects). The pattern is one message read twice: composition measured from text carries the direction in which every treatment reshapes both the level and the spread of behavior, and the market frame's residual is the within-category movement that composition alone cannot carry.

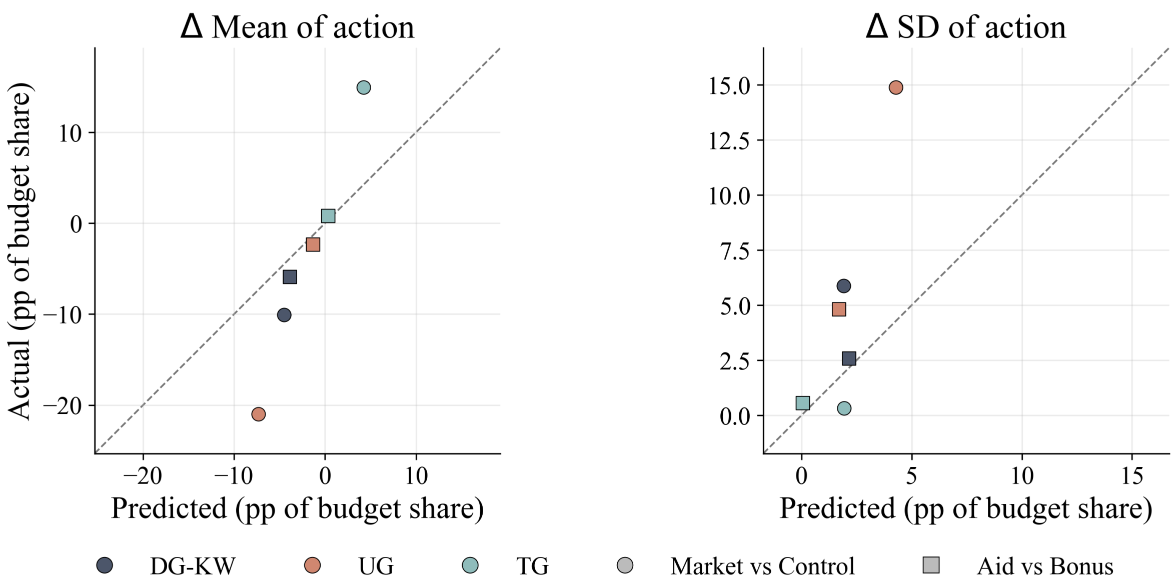

Figure 13: Predicted versus actual treatment effects on the mean (left) and standard deviation (right) of Player 1 actions. Predictions reweight the category-conditional action distributions by each condition's measured category shares, holding within-category behavior fixed; the dashed line marks exact fit. Circles: Market vs Control; squares: Aid vs Bonus; colors denote games. The two DG-LT cells appear in Table 24 in Appendix D.

A preregistered decomposition. The pre-registered analysis plan specified a complementary, regression-free split of every treatment effect into the two channels the model distinguishes: a symmetrized (Shapley/Oaxaca–Blinder) decomposition of the difference in mean actions into a component due to the *distribution of representations* and a component due to *behavior conditional on the representation*. We implement it with representation cells defined by both measured components of the representation—the stated category interacted with terciles of the hypothetical belief—in the UG and TG, and by categories alone in the DG, where no beliefs were elicited.[24] Table 14 reports the results, with conditional means evaluated three ways: symmetrized, held at the baseline condition, and held at the four-condition pooled sample. The representation component carries 39–65% of the treatment effect in the dictator and trust cells and about half of the story effect in the UG. The exception is informative rather than embarrassing: under the market frame in the UG the representation component is 16%, because there the frame moves behavior *within* representation cells—the same within-category margin that dominates the variance accounting above and that composition alone cannot carry. (The trust-game story cell's total effect is a negligible +0.8 points, so its split is uninformative.)

5.2 Surplus and Inequality

The market frame has opposite allocative consequences in the two strategic games: it destroys surplus in the UG and creates it in the TG. Table 15 reports realized surplus by treatment (the DG is omitted because its total is fixed by design). In the UG, lower offers meet receivers who—despite representing the game less morally—still reject them often

[24] The pre-registrations word the decomposition over representation categories alone (over social-proximity levels in the dictator game); the belief terciles add the representation's second measured component. Wording-literal variants deliver the same picture: with category-only cells on the same samples, the representation components are 11% (Market) and 47% (Aid) in the ultimatum game, and dictator-game cells defined by social proximity give 36% (Market) and 50% (Aid). The largest difference is itself informative: in the trust game's market cell, categories alone carry 35% against 59% with the belief terciles—the belief component of the representation does real work exactly in the game where beliefs differ by category (Section 3.2). No representation cell is empty in either condition of any comparison, so the empty-cell rule in the table notes never binds (replication package).

Table 14: **Decomposition of Treatment Effects into Representation and Behavior Components**

Comparison	Game	Δ mean	Symmetrized (preregistered) Repr.	Behav.	Repr. %	Representation comp. Baseline ref.	Pooled ref.	N
Market vs Control	DG-KW	-0.101	-0.039	-0.061	39%	-0.060	-0.048	728
		(0.018)	(0.016)	(0.020)		(0.027)	(0.012)	
Market vs Control	UG	-0.220	-0.036	-0.184	16%	-0.003	-0.088	1027
		(0.013)	(0.015)	(0.020)		(0.024)	(0.009)	
Market vs Control	TG	+0.133	+0.078	+0.054	59%	+0.075	+0.089	970
		(0.019)	(0.012)	(0.018)		(0.015)	(0.012)	
Aid vs Bonus	DG-KW	-0.059	-0.039	-0.021	65%	-0.039	-0.037	790
		(0.013)	(0.010)	(0.008)		(0.011)	(0.010)	
Aid vs Bonus	UG	-0.024	-0.012	-0.011	52%	-0.009	-0.017	1120
		(0.007)	(0.003)	(0.006)		(0.003)	(0.004)	
Aid vs Bonus	TG	+0.008	+0.011	-0.003	—	+0.011	+0.009	937
		(0.020)	(0.014)	(0.016)		(0.014)	(0.012)	

Notes: Pre-registered symmetrized (Shapley/Oaxaca–Blinder) decomposition of the difference in mean Player 1 actions between conditions into a component due to the distribution of representations and a component due to behavior conditional on the representation. Representation cells are the stated category interacted with terciles of the hypothetical belief (ultimatum and trust games, with terciles computed within game on the two conditions compared); the dictator game has no belief elicitation, so its cells are categories only. The last two columns recompute the representation component holding conditional means fixed at the baseline condition of each comparison (Baseline ref.: Control or Bonus) and at the four-condition pooled sample (Pooled ref.), with the two-condition tercile edges applied to the pooled sample; each reference scheme decomposes the mean difference exactly. Cells empty in one condition contribute their observed mean to the representation component. Bootstrap standard errors in parentheses ($B = 1{,}000$ participant-level draws within game $\times$ condition strata; terciles and the full decomposition recomputed in each draw). Representation shares are point estimates of the symmetrized split; the share is not reported for the trust-game story comparison, where the total effect is +0.008 (bootstrap SE 0.020) and the share is undefined or degenerate in 44% of bootstrap draws. Classified sample with non-missing beliefs where applicable.

enough that acceptance falls from 94.7% to 61.8%, and expected surplus per pair falls from \$11.36 to \$7.42, roughly a third of the pie. In the TG, sends rise and realized surplus increases from \$11.81 to \$13.55 (a 15% expansion; the maximum feasible surplus is \$18). The aid story has a small negative effect on UG surplus and none in the TG. The same contextual manipulation thus destroys surplus in a division problem and creates it in a production problem—a distinction invisible to any account in which "market framing" uniformly promotes or degrades prosociality. The forecast errors of Section 4.3 are the mechanism: under the market frame, proposers expect brokers to accept low prices more readily than brokers actually do.

Table 15: **Treatment Effects on Realized Surplus**

	Control	Market	Market − Control	Bonus	Aid	Aid − Bonus
UG: acceptance rate	0.947	0.618	-0.328***	0.947	0.912	-0.035**
UG: expected surplus ($)	11.36	7.42	-3.94***	11.36	10.94	-0.42**
TG: realized surplus ($)	11.81	13.55	1.74***	12.13	12.20	0.07

Notes: In the Ultimatum Game, realized surplus per matched pair is $12 if the receiver accepts and $0 otherwise; the acceptance rate is computed over receivers responding to their matched sender's offer. In the Trust Game, realized surplus is $6 + 12s$ dollars, where s is the sender's share sent; the receiver's return does not change total surplus. The Dictator Game is omitted because total surplus is fixed at $12 by design. Differences are tested with Welch two-sample t-tests. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.3 Circling Back: The Two Dictator Games

Return to Figure 1. Under the representations account, DG-KW and DG-LT differ not in what participants can do but in what the situation evokes: a windfall to be shared versus an allocation already in place, where taking from another's pile is theft. The evidence bears this out. Applying the pre-registered representation measures to the two games (Appendix F) shows that they evoke different moral situations: taking-type categories that are virtually absent in DG-KW appear in DG-LT (Theft: 0.5% vs 4.0% of responses). A symmetrized decomposition of the roughly $1 gap in mean allocations in the hypothetical-allocation sample attributes essentially all of it to behavior conditional on the representation and next to none to differences in the *distribution* of representations, whether cells are defined by social proximity or by moral category (Table 32): as with the market frame in the ultimatum game (Section 5.1), the change in the evoked situation moves behavior *within* representation cells.

The treatments put the gap on a common scale. The KW–LT difference in mean transfers—6.4 points of the budget—is of the same order as the Aid–Bonus effect on dictator giving (5.6 points) and smaller than the market effect (10.9 points): two payoff-equivalent descriptions of the *same* dictator game move behavior about as much as reading a different story, and less than embedding the game in a market.[25] The within-subject designs sharpen the point at the individual level (Appendix E): when the same participant plays both games, 46% change their choice, and choice switching is tightly linked to representation switching—80% of those whose category changes across the two games change their choice, against 36% of those whose category is stable. When the frame switches from abstract to market with the game fixed, 73–82% change their choice and over half change their representation category. To first order, behavioral instability tracks representational instability—across frames, across stories, and within the person.

6 Conclusions

Strategic behavior is shaped by context; on the account this paper develops and tests, representations are the channel. Identical rules evoke different situations depending on strategic structure and on payoff-irrelevant cues; the evoked situation directs attention across own earnings, total surplus, and the sharing norm, and supplies the counterpart one imagines facing; attention and beliefs together shape actions. On this account, within-context heterogeneity and cross-context instability are the same phenomenon: people differ in the situations a game evokes for them, and contexts differ in the situations they evoke on average.

Could stable preferences plus correct beliefs explain our treatment effects, with context operating through equilibrium expectations—for instance, the market frame signaling that others will behave selfishly, and best responses adjusting? The forecast-error evidence of Section 4.3 says no. In the UG, beliefs overshoot behavior under the market frame and undershoot it elsewhere, while TG errors stay small; treatments move chosen-action errors by more than 25 percentage points, and the selection-immune reference-action errors move in the same direction, if by less. Beliefs look like outputs of the representation—a proposer who sees a market expects market-like acceptance, a sender who sees a joint venture expects

[25] With 95% confidence intervals, in percentage points of the budget: LT − KW +6.4 [+3.8, +8.9] (bootstrap, 5,000 draws); Aid − Bonus −5.6 [−8.2, −3.0] and Market − Control −10.9 [−14.1, −7.7] (Welch intervals).

reciprocation—rather than like calibrated forecasts of a treatment-shifted opponent. This also cautions against interpreting elicited beliefs in games as measures of information: two participants facing the same distribution of opponents can hold different beliefs because they represent the game differently.

Experimenter demand is the other cross-cutting candidate. Three features of the design resist it. The same manipulation moves behavior in opposite directions across games, so a compliant participant would need to divine that the experimenter "wants" lower ultimatum offers but higher trust-game sends from the same market frame. The stories never mention the game, and their effects fall with the measured structural distance between story and game rather than following the stories' surface moral content—the charitable story *reduces* giving. And the open-ended answers, written after an unincentivized prompt, predict incentivized beliefs about the other player and hypothetical responses at fixed reference actions, which telling the experimenter what she wants to hear does not produce. Formal bounding exercises are available (de Quidt et al., 2018); the sign-reversal pattern alone places tight limits on the demand channel.

Two substantive lessons follow. First, social context need not mechanically produce generous behavior. Relative to a workplace-bonus story, the aid story shifted reported reasoning toward self-interest and reduced dictator transfers and ultimatum offers. Because the stories differ jointly in focal characters, entitlement, need, and the explicit case for sharing, this comparison does not isolate the effects of charity or hardship per se. It shows instead that a narrative with charitable content need not increase sharing relative to an alternative narrative that foregrounds contribution and reciprocity. Interventions that rely on context to encourage prosociality may therefore have unintended effects when they change what the situation appears to be about. Second, market framing crowds out moral reasoning but can foster cooperation. The market frame crowded out moral representations in every game, echoing longstanding concerns about markets and morals (Falk and Szech, 2013; Gneezy and Rustichini, 2000); but in the trust game the very same frame raised trust, repayment expectations, and surplus (Section 5), consistent with the ethics of sharing being replaced by the logic of mutually beneficial exchange. Whether a market cue degrades or improves outcomes depends on whether the underlying interaction is a division or a joint production problem.

The representation lens also speaks to game theory beyond our three games. If representations rather than game forms are the objects players best-respond to, then learning to play a game—the prisoner's dilemma, a public goods game—is learning a *shared* representation, and stability is representational: occasional deviations can be destabilizing not because they change payoffs but because they change what the situation appears to be. Repeated play under a stable context may work precisely by fading the naturalistic frame and creating a context of its own. And the salience that makes focal points focal (Schelling, 1960) is, in this language, a property of the representations a description evokes rather than of the game form—a reading consistent with the predictable effects of visual salience on strategic choice documented by Li and Camerer (2022).

The measurement strategy developed here—eliciting open-ended reasoning at scale and classifying it into representation categories, now paired with direct measurement of the beliefs inside the representation—is portable: to bargaining and labor-market experiments where entitlements and frames are known to matter (Hoffman et al., 1994; Kahneman et al., 1986), to field settings where the same transaction is embedded in different relationships, and to the study of how experience reshapes the categories themselves (Bordalo et al., 2020, 2026a). What determines which representation a context evokes for whom—and how policy can move representations deliberately—strikes us as the central open question.

References

James Andreoni and B. Douglas Bernheim. Social image and the 50–50 norm: A theoretical and experimental analysis of audience effects. *Econometrica*, 77(5):1607–1636, 2009.

Nava Ashraf, Iris Bohnet, and Nikita Piankov. Decomposing trust and trustworthiness. *Experimental Economics*, 9(3):193–208, 2006.

Nicholas Bardsley. Dictator game giving: Altruism or artefact? *Experimental Economics*, 11(2):122–133, 2008.

Björn Bartling, Roberto A. Weber, and Lan Yao. Do markets erode social responsibility? *Quarterly Journal of Economics*, 130(1):219–266, 2015.

Michal Bauer, Christopher Blattman, Julie Chytilová, Joseph Henrich, Edward Miguel, and Tamar Mitts. Can war foster cooperation? *Journal of Economic Perspectives*, 30(3): 249–274, 2016.

Joyce Berg, John Dickhaut, and Kevin McCabe. Trust, reciprocity, and social history. *Games and Economic Behavior*, 10(1):122–142, 1995.

Cristina Bicchieri. *The Grammar of Society: The Nature and Dynamics of Social Norms*. Cambridge University Press, New York, 2006.

Iris Bohnet and Bruno S. Frey. Social distance and other-regarding behavior in dictator games: Comment. *American Economic Review*, 89(1):335–339, 1999.

Gary E. Bolton and Axel Ockenfels. ERC: A theory of equity, reciprocity, and competition. *American Economic Review*, 90(1):166–193, 2000.

Pedro Bordalo, Nicola Gennaioli, and Andrei Shleifer. Salience theory of choice under risk. *Quarterly Journal of Economics*, 127(3):1243–1285, 2012.

Pedro Bordalo, Katherine Coffman, Nicola Gennaioli, and Andrei Shleifer. Stereotypes. *Quarterly Journal of Economics*, 131(4):1753–1794, 2016.

Pedro Bordalo, Nicola Gennaioli, and Andrei Shleifer. Memory, attention, and choice. *Quarterly Journal of Economics*, 135(3):1399–1442, 2020.

Pedro Bordalo, John J. Conlon, Nicola Gennaioli, Spencer Y. Kwon, and Andrei Shleifer. How people use statistics. *Review of Economic Studies*, 93(1):250–285, 2026a.

Pedro Bordalo, Nicola Gennaioli, Giacomo Lanzani, and Andrei Shleifer. A cognitive theory of reasoning and choice. Working paper, 2026b.

Samuel Bowles and Sandra Polanía-Reyes. Economic incentives and social preferences: Substitutes or complements? *Journal of Economic Literature*, 50(2):368–425, 2012.

Colin F. Camerer. *Behavioral Game Theory: Experiments in Strategic Interaction*. Princeton University Press, Princeton, NJ, 2003.

Gary Charness and Uri Gneezy. What's in a name? anonymity and social distance in dictator and ultimatum games. *Journal of Economic Behavior and Organization*, 68(1): 29–35, 2008.

Gary Charness and Matthew Rabin. Understanding social preferences with simple tests. *Quarterly Journal of Economics*, 117(3):817–869, 2002.

Todd L. Cherry, Peter Frykblom, and Jason F. Shogren. Hardnose the dictator. *American Economic Review*, 92(4):1218–1221, 2002.

Miguel A. Costa-Gomes and Georg Weizsäcker. Stated beliefs and play in normal-form games. *Review of Economic Studies*, 75(3):729–762, 2008.

James C. Cox. How to identify trust and reciprocity. *Games and Economic Behavior*, 46 (2):260–281, 2004.

Jason Dana, Roberto A. Weber, and Jason Xi Kuang. Exploiting moral wiggle room: Experiments demonstrating an illusory preference for fairness. *Economic Theory*, 33(1):67–80, 2007.

Jonathan de Quidt, Johannes Haushofer, and Christopher Roth. Measuring and bounding experimenter demand. *American Economic Review*, 108(11):3266–3302, 2018.

Martin Dufwenberg, Simon Gächter, and Heike Hennig-Schmidt. The framing of games and the psychology of play. *Games and Economic Behavior*, 73(2):459–478, 2011.

Catherine C. Eckel and Philip J. Grossman. Altruism in anonymous dictator games. *Games and Economic Behavior*, 16(2):181–191, 1996.

Tore Ellingsen, Magnus Johannesson, Johanna Mollerstrom, and Sara Munkhammar. Social framing effects: Preferences or beliefs? *Games and Economic Behavior*, 76(1):117–130, 2012.

Armin Falk and Nora Szech. Morals and markets. *Science*, 340(6133):707–711, 2013.

Ernst Fehr. On the economics and biology of trust. *Journal of the European Economic Association*, 7(2–3):235–266, 2009.

Ernst Fehr and Gary Charness. Social preferences: Fundamental characteristics and economic consequences. *Journal of Economic Literature*, 63(2):440–514, 2025.

Ernst Fehr and Simon Gächter. Fairness and retaliation: The economics of reciprocity. *Journal of Economic Perspectives*, 14(3):159–181, 2000.

Ernst Fehr and Klaus M. Schmidt. A theory of fairness, competition, and cooperation. *Quarterly Journal of Economics*, 114(3):817–868, 1999.

Ernst Fehr and Ivo Schurtenberger. Normative foundations of human cooperation. *Nature Human Behaviour*, 2:458–468, 2018.

Beatrice Ferrario and Stefanie Stantcheva. Eliciting people's first-order concerns: Text analysis of open-ended survey questions. *AEA Papers and Proceedings*, 112:163–169, 2022.

Robert Forsythe, Joel L. Horowitz, N. E. Savin, and Martin Sefton. Fairness in simple bargaining experiments. *Games and Economic Behavior*, 6(3):347–369, 1994.

Itzhak Gilboa and David Schmeidler. Case-based decision theory. *Quarterly Journal of Economics*, 110(3):605–639, 1995.

Edward L. Glaeser, David I. Laibson, José A. Scheinkman, and Christine L. Soutter. Measuring trust. *Quarterly Journal of Economics*, 115(3):811–846, 2000.

Uri Gneezy and Aldo Rustichini. A fine is a price. *Journal of Legal Studies*, 29(1):1–17, 2000.

Jacob K. Goeree, Margaret A. McConnell, Tiffany Mitchell, Tracey Tromp, and Leeat Yariv. The 1/d law of giving. *American Economic Journal: Microeconomics*, 2(1):183–203, 2010.

Werner Güth, Rolf Schmittberger, and Bernd Schwarze. An experimental analysis of ultimatum bargaining. *Journal of Economic Behavior and Organization*, 3(4):367–388, 1982.

Ingar K. Haaland, Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart. Measuring what is top of mind. NBER Working Paper 32421, 2024.

Joseph Henrich, Robert Boyd, Samuel Bowles, Colin Camerer, Ernst Fehr, Herbert Gintis, and Richard McElreath. In search of homo economicus: Behavioral experiments in 15 small-scale societies. *American Economic Review*, 91(2):73–78, 2001.

Elizabeth Hoffman, Kevin McCabe, Keith Shachat, and Vernon Smith. Preferences, property rights, and anonymity in bargaining games. *Games and Economic Behavior*, 7(3):346–380, 1994.

Elizabeth Hoffman, Kevin McCabe, and Vernon L. Smith. Social distance and other-regarding behavior in dictator games. *American Economic Review*, 86(3):653–660, 1996.

Philippe Jehiel. Analogy-based expectation equilibrium. *Journal of Economic Theory*, 123 (2):81–104, 2005.

Noel D. Johnson and Alexandra A. Mislin. Trust games: A meta-analysis. *Journal of Economic Psychology*, 32(5):865–889, 2011.

Daniel Kahneman, Jack L. Knetsch, and Richard Thaler. Fairness as a constraint on profit seeking: Entitlements in the market. *American Economic Review*, 76(4):728–741, 1986.

Dean S. Karlan. Using experimental economics to measure social capital and predict financial decisions. *American Economic Review*, 95(5):1688–1699, 2005.

Erik O. Kimbrough and Alexander Vostroknutov. Norms make preferences social. *Journal of the European Economic Association*, 14(3):608–638, 2016.

Erin L. Krupka and Roberto A. Weber. Identifying social norms using coordination games: Why does dictator game sharing vary? *Journal of the European Economic Association*, 11(3):495–524, 2013.

Edward P. Lazear, Ulrike Malmendier, and Roberto A. Weber. Sorting in experiments with application to social preferences. *American Economic Journal: Applied Economics*, 4(1): 136–163, 2012.

Stephen Leider, Markus M. Möbius, Tanya Rosenblat, and Quoc-Anh Do. Directed altruism and enforced reciprocity in social networks. *Quarterly Journal of Economics*, 124(4): 1815–1851, 2009.

Steven D. Levitt and John A. List. What do laboratory experiments measuring social preferences reveal about the real world? *Journal of Economic Perspectives*, 21(2):153–174, 2007.

Xiaomin Li and Colin F. Camerer. Predictable effects of visual salience in experimental decisions and games. *Quarterly Journal of Economics*, 137(3):1849–1900, 2022.

Varda Liberman, Steven M. Samuels, and Lee Ross. The name of the game: Predictive power of reputations versus situational labels in determining prisoner's dilemma game moves. *Personality and Social Psychology Bulletin*, 30(9):1175–1185, 2004.

John A. List. The behavioralist meets the market: Measuring social preferences and reputation effects in actual transactions. *Journal of Political Economy*, 114(1):1–37, 2006.

John A. List. On the interpretation of giving in dictator games. *Journal of Political Economy*, 115(3):482–493, 2007.

Pedro Rey-Biel. Equilibrium play and best response to (stated) beliefs in normal form games. *Games and Economic Behavior*, 65(2):572–585, 2009.

Alvin E. Roth, Vesna Prasnikar, Masahiro Okuno-Fujiwara, and Shmuel Zamir. Bargaining and market behavior in Jerusalem, Ljubljana, Pittsburgh, and Tokyo: An experimental study. *American Economic Review*, 81(5):1068–1095, 1991.

Thomas C. Schelling. *The Strategy of Conflict*. Harvard University Press, Cambridge, MA, 1960.

Deborah A. Small and George Loewenstein. Helping a victim or helping the victim: Altruism and identifiability. *Journal of Risk and Uncertainty*, 26(1):5–16, 2003.

Amos Tversky. Features of similarity. *Psychological Review*, 84(4):327–352, 1977.

Maarten J. Voors, Eleonora E. M. Nillesen, Philip Verwimp, Erwin H. Bulte, Robert Lensink, and Daan P. van Soest. Violent conflict and behavior: A field experiment in burundi. *American Economic Review*, 102(2):941–964, 2012.

A Proofs of Propositions

Throughout, attention weights are strictly positive, $\rho, \sigma, \mu > 0$, and in the ultimatum game $a, b > 0$. Each objective below is strictly concave and differentiable on $[0, 1]$, so the constrained maximizer is unique and equals the unconstrained solution projected onto $[0, 1]$ (the censoring in the propositions); comparative statics are stated at interior optima, where the projection is the identity.

Proof of Proposition 1. With $W(t) = 1$ and $\pi_R = t$, the objective is $u(t) = \rho - \sigma t - \frac{\mu}{2}\left(t - \frac{1}{2}\right)^2$, which is strictly concave ($u'' = -\mu < 0$). The first-order condition $-\sigma - \mu\left(t - \frac{1}{2}\right) = 0$ gives $t^* = \frac{1}{2} - \sigma/\mu$. Since $\sigma/\mu > 0$, $t^* < \frac{1}{2}$, so the upper bound never binds and $t^{DG} = \max\{0, t^*\} < \frac{1}{2}$. The solution is strictly decreasing in σ/μ where interior and constant at zero for $\sigma/\mu \geq \frac{1}{2}$; ρ enters the objective only through the additive constant $\rho W(t) = \rho$, so it leaves the maximizer unchanged. □

Proof of Proposition 2. The objective $v(t) = (a + bt)(\rho - \sigma t) - \frac{\mu}{2}\left(t - \frac{1}{2}\right)^2$ has $v'(t) = b\rho - a\sigma - 2b\sigma t - \mu\left(t - \frac{1}{2}\right)$ and $v''(t) = -(2b\sigma + \mu) < 0$: v is strictly concave. The first-order condition gives $t\,(2b\sigma + \mu) = b\rho - a\sigma + \frac{\mu}{2}$, and subtracting $\frac{1}{2}$ from the resulting t yields (3). The censored solution lies in $(0, 1)$ if and only if $\left|b\rho - (a+b)\,\sigma\right| < b\sigma + \mu/2$, half the curvature $2b\sigma + \mu$.

(i) If $t^{DG} = 0$, interiority gives $t^{UG} > 0 = t^{DG}$. If $t^{DG} = \frac{1}{2} - \sigma/\mu > 0$, putting the difference over the common denominator $\mu\,(2b\sigma + \mu)$ gives

$$t^{UG} - t^{DG} = \frac{b\rho - (a+b)\,\sigma}{2b\sigma + \mu} + \frac{\sigma}{\mu} = \frac{\mu b\rho + 2b\sigma^2 + \mu\sigma\,(1 - a - b)}{\mu\,(2b\sigma + \mu)},$$

and $a + b \leq 1$ makes every term in the numerator non-negative, with $\mu b\rho > 0$; hence $t^{UG} > t^{DG}$.

(ii) Differentiating (3),

$$\frac{\partial t^{UG}}{\partial \rho} = \frac{b}{2b\sigma + \mu} > 0, \qquad \frac{\partial t^{UG}}{\partial \sigma} = -\frac{(a+b)\,\mu + 2b^2\rho}{(2b\sigma+\mu)^2} < 0, \qquad \frac{\partial t^{UG}}{\partial \mu} = -\frac{t^{UG} - \frac{1}{2}}{2b\sigma + \mu},$$

where the middle expression follows from the quotient rule and is negative unconditionally and the last expression is positive when $t^{UG} < \frac{1}{2}$ and negative when $t^{UG} > \frac{1}{2}$: the norm weight pulls the offer toward the equal split.

(iii) Only the numerator of the second term in (3) depends on a, so $\partial t^{UG}/\partial a = -\sigma/(2b\sigma + \mu) < 0$. The sensitivity $\sigma/(2b\sigma+\mu)$ has derivative $\mu/(2b\sigma+\mu)^2 > 0$ in σ and $-\sigma/(2b\sigma+\mu)^2 < 0$ in μ. □

Proof of Proposition 3. The objective $w(t) = \rho\,(1+rt) - \sigma(1+r)(1-s)\,t - \frac{\mu}{2}\,(t - t^e(s))^2$ has $w'(t) = \rho r - \sigma(1+r)(1-s) - \mu\,(t - t^e(s))$ and $w''(t) = -\mu < 0$, so the first-order condition gives (5). The censored solution lies in $(0,1)$ if and only if $-\mu\, t^e(s) < \rho r - \sigma(1+r)(1-s) < \mu\big(1 - t^e(s)\big)$. At an interior optimum,

$$\frac{\partial t^{TG}}{\partial \rho} = \frac{r}{\mu} > 0, \qquad \frac{\partial t^{TG}}{\partial s} = \frac{\mathrm{d}t^e}{\mathrm{d}s} + \frac{\sigma(1+r)}{\mu} > 0, \qquad \frac{\partial t^{TG}}{\partial \sigma} = -\frac{(1+r)(1-s)}{\mu} < 0,$$

where $\mathrm{d}t^e/\mathrm{d}s = 2(1+r)\big[1 + (1+r)(1-2s)\big]^{-2} > 0$ by direct differentiation, and $\partial t^{TG}/\partial \mu = -\big[\rho r - \sigma(1+r)(1-s)\big]/\mu^2 = -\big(t^{TG} - t^e(s)\big)/\mu$: the norm weight pulls the send toward the equal-payoff send. As $\mu \to 0$ the objective converges to the affine function $\rho + \big[\rho r - \sigma(1+r)(1-s)\big]\,t$, whose maximizer is $t = 1$ if the slope is positive and $t = 0$ if it is negative. For a purely material sender, $\rho = \sigma$, the objective reduces to $\sigma\,\pi_S$ with $\pi_S = 1 + t\,\big[s(1+r) - 1\big]$, so the slope is $\sigma\big[s(1+r) - 1\big]$: she sends everything if $s(1+r) > 1$ and nothing if $s(1+r) < 1$. The anchor's own censoring is immediate: $t^e(s) \geq 1$ if and only if $(1+r)(1-2s) \leq 0$, i.e., $s \geq \frac{1}{2}$. Finally, if $t^{DG} = 0$, interiority gives $t^{TG} > 0 = t^{DG}$; if $t^{DG} = \frac{1}{2} - \sigma/\mu$, then

$$t^{TG} - t^{DG} \;=\; t^e(s) - \tfrac{1}{2} + \frac{\rho r - \sigma(1+r)(1-s) + \sigma}{\mu} \;=\; t^e(s) - \tfrac{1}{2} + \frac{r\,\big[\rho - \sigma(1-s)\big] + \sigma s}{\mu},$$

which is non-negative if and only if $r\left[\rho-\sigma(1-s)\right]+\sigma s \geq \mu\left(\frac{1}{2}-t^e(s)\right)$. Since $t^e(s) \geq \frac{1}{2}$ exactly when $s \geq r/\left(2(1+r)\right)$, the condition $\rho \geq \sigma(1-s)$ suffices there: the numerator is then a sum of two non-negative terms and the right side is non-positive. $\square$

Proof of Proposition 4. Welfare statements follow from the chain rule $\partial V_g/\partial\theta = W_g' \cdot \partial t^g/\partial\theta$, where the pie's sensitivity to the action is $W_{UG}' = p'(t) = b$ (expected pie $p(t)\cdot 1$) and $W_{TG}' = r$, combined with the action derivatives in the proofs of Propositions 2 and 3: for (i), $\partial V_{UG}/\partial\rho = b\cdot b/(2b\sigma+\mu)$ and $\partial V_{TG}/\partial\rho = r\cdot r/\mu$; for (ii), $\partial V_{UG}/\partial\sigma = b\cdot\partial t^{UG}/\partial\sigma < 0$ and $\partial V_{TG}/\partial\sigma = r\cdot\partial t^{TG}/\partial\sigma < 0$; for (iii), $\partial V_g/\partial\mu = W_g'\cdot\partial t^g/\partial\mu$, and both games have $\partial t^g/\partial\mu$ proportional to $-(t^g-t^e)$, with $t^e=\frac{1}{2}$ in the ultimatum game; for (iv), $\partial V_{UG}/\partial a = 1+b\cdot\partial t^{UG}/\partial a = 1-b\sigma/(2b\sigma+\mu)$ and $\partial V_{TG}/\partial s = r\cdot\left[\mathrm{d}t^e/\mathrm{d}s+\sigma(1+r)/\mu\right]$. The counterpart's expected payoff: in the ultimatum game $\partial\left[p(t)t\right]/\partial\sigma = \left(bt+p(t)\right)\partial t^{UG}/\partial\sigma<0$; in the trust game $\partial\left[(1-s_a)(1+r)t\right]/\partial\sigma = (1-s_a)(1+r)\,\partial t^{TG}/\partial\sigma < 0$. The sender's own expected payoff: $\partial\left[p(t)(1-t)\right]/\partial\sigma = \left[b(1-t^{UG})-p(t^{UG})\right]\partial t^{UG}/\partial\sigma$, and $b(1-t)-p(t) = b(1-2t)-a$, so with $\partial t^{UG}/\partial\sigma<0$ the sender's own expected payoff falls precisely when $b(1-2t^{UG})>a$; a witness: $a=1/20$, $b=3/5$, $\rho=1/10$, $\sigma=1$, $\mu=1$ gives the interior offer $t^{UG}=0.232$, $b(1-2t^{UG})=0.32>0.05=a$, and $\partial\left[p(t)(1-t)\right]/\partial\sigma=-0.04<0$. $\square$

Proof of Remark 1. For an actual acceptance schedule $\hat{p}$ with $\hat{p}'>0$, the chain rule gives $\partial\hat{V}_{UG}/\partial\theta = \hat{p}'(t^{UG})\cdot\partial t^{UG}/\partial\theta$ for $\theta\in\{\rho,\sigma,\mu\}$, the same signs as in (i)–(iii), while $\partial\hat{V}_{UG}/\partial a = \hat{p}'(t^{UG})\cdot\partial t^{UG}/\partial a = -\hat{p}'\,\sigma/(2b\sigma+\mu)<0$, reversing (iv); in the trust game $\hat{V}_{TG} = 1+r\,t^{TG}$ coincides with V_{TG}, so $\partial\hat{V}_{TG}/\partial s = r\left[\mathrm{d}t^e/\mathrm{d}s+\sigma(1+r)/\mu\right]>0$ is unchanged. $\square$

Proof of Proposition 5. Conditional on trade, $D_{DG} = 1-2t^{DG} = 2\sigma/\mu$ and $D_{UG} = 1-2t^{UG} = 2\left[(a+b)\sigma-b\rho\right]/(2b\sigma+\mu)$, whence $\partial D_{UG}/\partial\sigma = 2\left[(a+b)\mu+2b^2\rho\right]/(2b\sigma+\mu)^2>0$, $\partial D_{UG}/\partial\rho = -2b/(2b\sigma+\mu)<0$, and $\partial D_{UG}/\partial\mu = -D_{UG}/(2b\sigma+\mu)$ by direct differentiation. In the trust game $D_{TG} = (1-t)+s_a(1+r)t-(1-s_a)(1+r)t = 1-t\left[1+(1+r)(1-2s_a)\right]$, affine in t with slope $-k$, $k\equiv 1+(1+r)(1-2s_a)$, positive iff $s_a<(2+r)/\left(2(1+r)\right)$; since $\partial t^{TG}/\partial\mu = -\left(t^{TG}-t^e(s)\right)/\mu$, $\partial D_{TG}/\partial\mu = -\left(D_{TG}-D_{TG}(t^e(s))\right)/\mu$ with $D_{TG}(t^e(s)) =$

$1 - k\,t^e(s)$; at $s = s_a$ the definition of the anchor gives $k = 1/t^e(s_a)$, so $D_{TG}(t^e(s_a)) = 0$: the gap at the anchor vanishes exactly under correct beliefs. □

B Endogenizing Second-Mover Behavior

The main text treats the acceptance schedule $p(t) = a + bt$ and the returned share s as primitives of the sender's representation. This appendix endogenizes them: both schedules arise from the receiver's own moral psychology, so what the sender holds as a belief is a representation of the *receiver's motives*—of how the sender perceives the receiver's representation. It then closes the model by making the receiver a full player, with her own retrieval problem cued by the very action she faces.

Remark 2 (Endogenous acceptance schedule). The linear acceptance schedule can be founded in a moral psychology similar to the one we introduced above for senders. Endow the receiver with attention weights $\hat{\boldsymbol{\alpha}} = (\hat{\rho}, \hat{\sigma}, \hat{\mu})$. Assume that the idealistic principle is applied to the sender's action: the receiver suffers from the offer's inequity and feels this cost whether or not the offer is accepted. Moreover, assume that rejection delivers a psychological benefit of *protest*, $\psi \geq 0$, which is drawn from a distribution F and weighted by moral attention. In this case, the inequity-aversion term cancels from the accept–reject margin and the receiver accepts t if and only if $\hat{\rho} - \hat{\sigma}(1-t) \geq \hat{\mu}\,\psi$, so the acceptance probability is $F\big[(\hat{\rho} - \hat{\sigma} + \hat{\sigma}t)/\hat{\mu}\big]$. For ψ uniform on $[0, \bar{\Psi}]$ this is exactly $p(t) = a + bt$ with

$$a = \frac{\hat{\rho} - \hat{\sigma}}{\hat{\mu}\bar{\Psi}}, \qquad b = \frac{\hat{\sigma}}{\hat{\mu}\bar{\Psi}}.$$

The believed acceptance schedule (a, b) is therefore a representation of the *receiver's motives*: imagining a punitive receiver (high $\hat{\mu}\bar{\Psi}$—our *Moral bad* category) scales acceptance down across the board, while imagining a self-interested receiver (high $\hat{\sigma}$) makes acceptance sensitive to money. Note also that $p(1) = a + b = \hat{\rho}/(\hat{\mu}\bar{\Psi})$, so the condition $a + b \leq 1$ in Proposition 2 has a natural reading: the imagined receiver's protest weight is large enough that even a full-split offer is not certain to be accepted.

Remark 3 (Endogenous return schedule). The believed return can be founded in the same moral psychology as the acceptance schedule of Remark 2. Endow the imagined receiver with attention weights $\hat{\boldsymbol{\alpha}} = (\hat{\rho}, \hat{\sigma}, \hat{\mu})$. Holding the output $(1+r)t$, the receiver is a dictator over it: total wealth $1 + rt$ is fixed no matter how much she returns, so $\hat{\rho}$ is idle and she trades own earnings against a norm of returning half of a target pie X, choosing the returned amount y to maximize $\hat{\rho}(1+rt) - \hat{\sigma}(1-t+y) - \frac{\hat{\mu}}{2}\bigl(y/X - \frac{1}{2}\bigr)^2$. The natural targets are the amount sent ($X = t$), the output ($X = (1+r)t$), and total wealth ($X = 1+rt$), and for every target the optimal returned share of the target is the same expression, $y^*/X = \frac{1}{2} - (\hat{\sigma}/\hat{\mu})X$. Under the first two targets the believed returned share of the output is affine and *decreasing* in the send—with the amount-sent target, $s(t) = \bigl[\frac{1}{2} - (\hat{\sigma}/\hat{\mu})t\bigr]/(1+r)$; with the output target, $s(t) = \frac{1}{2} - (\hat{\sigma}/\hat{\mu})(1+r)t$. Thus, the foundation has a testable signature: believed return shares should decline in the send, which the two elicited belief points per sender (reference and chosen action) can check. The total-wealth target is ill behaved—at small sends the norm target exceeds the pot itself, forcing full return below a threshold and a kinked schedule—so the amount-sent and output targets are the tractable specifications. As in Remark 2, the believed schedule is a representation of the receiver's motives, here through the imagined selfishness ratio $\hat{\sigma}/\hat{\mu}$.

Replacing the constant share with the founded, declining schedule—or any affine one—leaves the trust-game analysis intact:

Proposition 6 (Trust send with a declining return schedule)**.** *Under any affine believed schedule $s(t) = s_0 - s_1 t$ with $0 \le s_0 < 1$ and $s_1 \ge 0$, expected payoffs equalize at the unique positive root t^e of $2(1+r)s_1x^2 + \bigl[1 + (1+r)(1-2s_0)\bigr]x - 1 = 0$, which reduces to $t^e(s_0)$ at $s_1 = 0$. The sender's objective remains strictly concave, and the optimal send is*

$$t^{TG} = \frac{\mu\, t^e + \rho r - \sigma(1+r)(1-s_0)}{\mu + 2\sigma(1+r)s_1}, \tag{8}$$

censored to $[0,1]$*, which nests equation* (5) *at* $s_1 = 0$*; the send is interior if and only if* $-\mu\, t^e < \rho r - \sigma(1+r)(1-s_0) < \mu(1-t^e) + 2\sigma(1+r)s_1$*. At an interior optimum: (i) the send is increasing in* ρ*, decreasing in* σ*, and moved toward the equal-payoff send* t^e *by* μ*, as in Proposition 3; (ii) the send is increasing in the believed level* s_0*, with sensitivity* $\big[\mu\,\partial t^e/\partial s_0 + \sigma(1+r)\big]/\big(\mu + 2\sigma(1+r)s_1\big)$*—damped by the believed slope exactly as the ultimatum sensitivity* $\sigma/(2b\sigma+\mu)$ *is damped by* b*, with constant* s *the undamped limit; (iii) the send is decreasing in the believed slope,* $\partial t^{TG}/\partial s_1 = \big[\mu\,\partial t^e/\partial s_1 - 2\sigma(1+r)\,t^{TG}\big]/\big(\mu + 2\sigma(1+r)s_1\big) < 0$*: greater imagined receiver selfishness lowers both the anchor and the material pull of sending.*

Proof of Proposition 6. Write $\beta \equiv 1 + (1+r)(1-2s_0)$ and $q(x) \equiv 2(1+r)s_1x^2 + \beta x - 1$. Since $q(0) = -1 < 0$ and $q(x) \to \infty$, q has a unique positive root t^e, and the quadratic's slope at that root is positive: $q'(t^e) = 4(1+r)s_1t^e + \beta > 0$. The objective $w(t) = \rho\,(1+rt) - \sigma(1+r)\,\big(1 - s_0 + s_1 t\big)\,t - \frac{\mu}{2}\,(t-t^e)^2$ has $w'(t) = \rho r - \sigma(1+r)(1-s_0) - 2\sigma(1+r)s_1 t - \mu\,(t-t^e)$ and $w''(t) = -\big(\mu + 2\sigma(1+r)s_1\big) < 0$ for every $s_1 \geq 0$: w is strictly concave. The first-order condition gives $t\,\big(\mu + 2\sigma(1+r)s_1\big) = \mu\,t^e + \rho r - \sigma(1+r)(1-s_0)$, which is (8) and reduces to (5) at $s_1 = 0$. Write $D \equiv \mu + 2\sigma(1+r)s_1$ and $N \equiv \mu\,t^e + \rho r - \sigma(1+r)(1-s_0)$, so that $t^{TG} = N/D$; the censored solution lies in $(0,1)$ if and only if $0 < N < D$, i.e., $-\mu\,t^e < \rho r - \sigma(1+r)(1-s_0) < \mu(1-t^e) + 2\sigma(1+r)s_1$. The comparative statics below hold at an interior optimum.

(i) $\partial t^{TG}/\partial\rho = r/D > 0$; $\partial t^{TG}/\partial\sigma = -(1+r)\big[(1-s_0)D + 2s_1N\big]/D^2 < 0$ since $N > 0$; and $\partial t^{TG}/\partial\mu = \big(t^eD - N\big)/D^2 = -\big(t^{TG} - t^e\big)/D$, so the norm weight pulls the send toward the equal-payoff send.

(ii) Implicit differentiation of $q(t^e) = 0$ gives $\partial t^e/\partial s_0 = 2(1+r)\,t^e/q'(t^e) > 0$, so $\partial t^{TG}/\partial s_0 = \big[\mu\,\partial t^e/\partial s_0 + \sigma(1+r)\big]/D > 0$, and $D \geq \mu$ with equality exactly at $s_1 = 0$ gives the damping.

(iii) Implicit differentiation gives $\partial t^e/\partial s_1 = -2(1+r)\,(t^e)^2/q'(t^e) < 0$, so $\partial t^{TG}/\partial s_1 = \big[\mu\,\partial t^e/\partial s_1 \cdot D - 2\sigma(1+r)N\big]/D^2 = \big[\mu\,\partial t^e/\partial s_1 - 2\sigma(1+r)\,t^{TG}\big]/D < 0$: both terms are negative. $\square$

The receiver as a full player. In the main text the receiver exists only as the sender's belief. The same two building blocks—moral psychology given a representation (Remarks 2 and 3) and retrieval given a description (equation (6))—turn her into a full player. The receiver's description differs from the sender's in one crucial way: it contains the sender's action. She faces $\mathbf{d}_R = (\mathbf{r}_d, \boldsymbol{\kappa}_d, t)$—not an abstract game, but a game in which she has just been offered t or entrusted $(1+r)t$. Retrieval then assigns category c a probability $q_c(t)$ (a logit, under the extreme-value shocks), *with the sender's action as a contextual cue*, and conditional on the category, behavior follows the second-mover margins derived above—reading the hatted weights of Remarks 2 and 3 as the receiver's own retrieved attention profile: an acceptance schedule $p_c(t) = a_c + b_c t$ from the protest psychology, and a returned share from the dictator-over-the-output problem. For the return margin the equal-payoff anchor matters: the receiver moves last, so her norm anchors at the *realized* equalizing return—the share $s^e(t) = \frac{2+r}{2(1+r)} - \frac{1}{2(1+r)t}$ of the output, the inverse of the sender's anchor ($t^e(s^e(t)) = t$), rising in the send and reaching zero at $t = 1/(2+r)$—and trading the anchor against own earnings gives $s_c(t) = \max\{0,\ s^e(t) - (\hat{\sigma}/\hat{\mu})_c(1+r)t\}$: near-equalizing and *rising* schedules for low-selfishness categories, low and flat ones as $(\hat{\sigma}/\hat{\mu})_c$ grows. (Anchoring instead at half of a target pie, as in the belief foundation of Remark 3, gives affine *declining* schedules—the two anchors are separable in the data, below.) Aggregate second-mover behavior is the mixture

$$p(t) = \sum_c q_c(t)\, p_c(t), \qquad s(t) = \sum_c q_c(t)\, s_c(t). \tag{9}$$

The sender's action now works through two channels,

$$p'(t) \;=\; \underbrace{\sum_c q_c(t)\, p_c'(t)}_{\text{margin within categories}} \;+\; \underbrace{\sum_c q_c'(t)\, p_c(t)}_{\text{composition across categories}} \quad ,$$

and if low offers retrieve condemnation—an insulting offer resembles situations of exploitation, so $q_{\text{Moral bad}}$ falls with t and the released probability mass flows to categories with higher acceptance—the composition term is positive and the aggregate schedule is steeper than the within-category margins alone, and can be steeper than *every* category schedule. Three consequences follow. First, measured acceptance can respond to offers more strongly than any fixed protest psychology implies: part of the response is a change in what the offer makes the receiver think the situation *is*. Second, the mixture is measurable: receiver categories should covary with the action faced—condemnation concentrated at low offers, cooperation at generous sends—which the second-mover representation data can check directly. Third, treatments reach the receiver through the same two levers as the sender (F_c and $\boldsymbol{\kappa}_d$), but nothing forces the two sides to move equally—the asymmetry between first- and second-mover treatment effects documented in Sections 4 and 5.

The mixture in the data. The second consequence can be checked directly, and the data bear it out. In the control ultimatum game, the receiver's stated category covaries strongly with the offer she faces: *Moral bad* is the modal category at zero offers (67% of responses) and virtually disappears at fair ones (2%), while *Moral good* rises to 62% at fair offers; linear-probability slopes of the category indicators on the offer faced are +0.86 (Moral good) and −0.79 (Moral bad), both $p < 0.001$. Decomposing the aggregate acceptance slope as in equation (9) splits it into the within-category margin and the composition term: of the control-sample aggregate slope of 0.87, the within-category margin contributes 0.35 and composition 0.52—six tenths of the aggregate response to the offer is the offer changing what the receiver thinks the situation *is*, not movement along any fixed schedule. In the trust game the same decomposition attributes 14% to composition: returns respond to the send mostly within categories. Full tables are in the replication package.

The return margin: equalization against the norm-target family. The within-category trust-game schedules separate the two anchors directly, and they take the equal-payoff side. Table 16 reports the category schedules—Panel A's acceptance schedules are the within-category margins of the decomposition above; Panel B carries the anchor test: returned shares *rise* with the send among *Moral good* receivers (slope +0.28, s.e. 0.03, $N = 303$) and *Mutual Benefit/Cooperation* receivers (+0.28, s.e. 0.05, $N = 150$)—rejecting every member of the weakly declining norm-target family: returning half of a target amount can never produce a rising schedule, whatever the target—while *Self-interest* receivers sit near the selfish benchmark (slope -0.05), the type-dependence the weights predict; *Moral bad* receivers face mostly minimal sends, where the equalizing and selfish returns coincide at zero. Against the fitted one-parameter norm-target schedules, the *zero-parameter* equalization curve $s^e(t)$ is the best fit for both low-selfishness categories from the reference send upward, with one systematic deviation: below $t = 1/(2 + r)$, where equalization asks for nothing, receivers still return a positive share—a moral floor. Senders' *believed* schedules show the opposite pattern: the two elicited belief points per sender imply individual believed slopes with medians of exactly 0.00 in every category, rejecting the rise that equalization predicts and matching the constant-share specification of the main text. Senders thus attribute to the receiver the norm-target psychology of Remark 3 while facing receivers who equalize—the trust-game forecast-error wedge of Section 4.3, located in the imagined psychology itself.

This closes the model on the forecast-error side. The sender's imagined receiver—the schedule (a, b) or the share s—is retrieved from the *sender's* memory given the sender's description; actual second-mover behavior is retrieved from the *receiver's* memory given a description that includes the action. Nothing aligns the two retrievals, so systematic, context-dependent forecast errors are an equilibrium object, and a treatment that moves the two descriptions differently moves *mean* forecast errors—the treatment-level result of Section 4.3. Learning under a stable context is then convergence of the sender's retrieved receiver toward the receiver's retrieved self.

Table 16: **Second-Mover Schedules by Category (Control Sample)**

Category	Intercept	(SE)	Slope in t	(SE)	N
Panel A: UG acceptance, $p_c(t) = a_c + b_c\,t$					
Moral good	0.755	(0.117)	0.455	(0.223)	332
Moral bad	-0.064	(0.091)	1.106	(0.405)	29
Self-interest	0.898	(0.096)	0.143	(0.190)	81
Mutual Benefit / Cooperation	0.959	(0.042)	0.068	(0.072)	150
Panel B: TG returned share, $s_c(t)$					
Moral good	0.227	(0.022)	0.282	(0.033)	303
Moral bad	-0.062	(0.030)	0.452	(0.168)	42
Self-interest	0.122	(0.052)	-0.049	(0.072)	58
Mutual Benefit / Cooperation	0.185	(0.032)	0.278	(0.053)	150

Notes: Control sample. OLS of the second mover's realized response (UG: acceptance of the offer faced; TG: share of the tripled amount returned) on the first mover's action t (share of the budget), by stated category. Robust (HC1) standard errors. The offers and sends faced are endogenous to the matched first mover; hypothetical responses at the one-third reference action give the treatment-clean counterpart in Figure 11.

C Second Movers: Control Sample and Treatment Effects

This appendix reports the Player 2 control-sample results, the treatment-induced shifts in Player 2 category shares, the forecast-error regressions referenced in Section 4.3, the belief-sensitivity regressions behind Figure 5, and the Player 2 social-proximity cross-tabulation (the Player 1 counterpart is in Section 3.2); treatment effects on Player 2 hypothetical behavior appear in Figure 11 in Section 5.

Table 17: **Player 2: Counterpart Proximity within Stated Reason Category**

Counterpart proximity	Moral good	Moral bad	Self-interest	Mutual Benefit/Cooperation
Low social proximity	48.6%	82.2%	91.0%	19.1%
High social proximity	51.4%	17.8%	9.0%	80.9%

Notes: Pooled across all games and treatments for Player 2. Columns report the distribution of counterpart proximity within each stated reason category and sum to 100%.

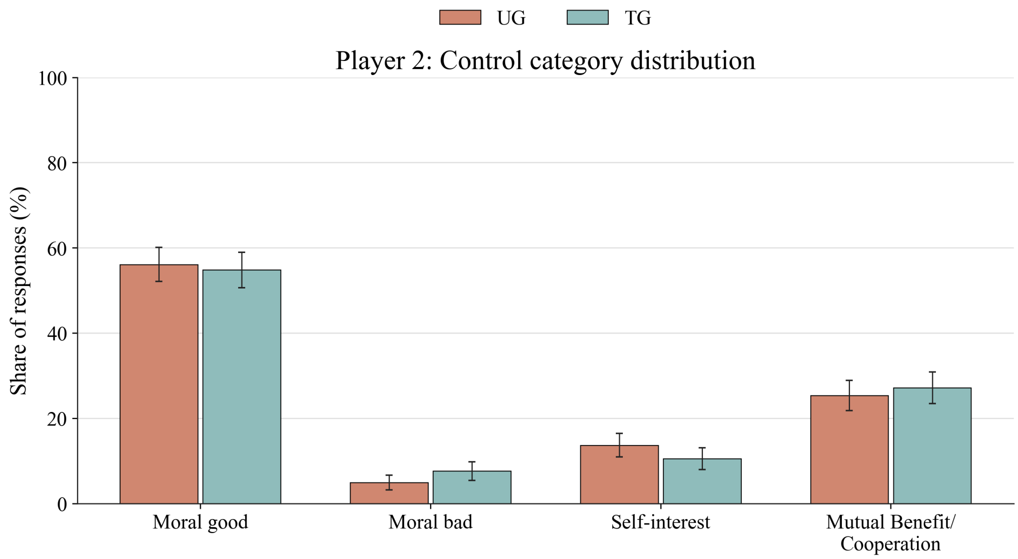

Figure 14: Control-sample representation distributions for Player 2, excluding unclassified observations. Error bars show 95% confidence intervals.

Table 18: **Belief Sensitivity of Actions, by Representation Category**

		Control		All conditions (condition FE)	
Game	Category	Slope	(SE)	Slope	(SE)
UG	Moral	0.010	(0.020)	-0.003	(0.016)
	Self-interest	-0.108	(0.096)	-0.093	(0.048)
	Mutual Benefit/Coop.	-0.083	(0.319)	-0.128	(0.089)
TG	Moral	0.306	(0.095)	0.319	(0.045)
	Self-interest	0.451	(0.144)	0.465	(0.086)
	Mutual Benefit/Coop.	0.264	(0.098)	0.128	(0.045)

Notes: OLS slopes of the Player 1 action (share) on the hypothetical belief at the one-third reference action, estimated separately by representation category; robust (HC1) standard errors. "All conditions" pools the four conditions with condition fixed effects. Slope-equality F-tests across categories: UG control $p = 0.464$, UG pooled $p = 0.002$, TG control $p = 0.560$, TG pooled $p = 0.000$. Model predictions: negative slopes in the UG, positive in the TG for every category (the equal-payoff norm anchor moves with the believed return); magnitude largest for Self-interest, whose material channel adds to the common anchor channel. The UG Mutual Benefit/Cooperation cell is very small and should be read accordingly.

Table 19: **Benchmark Regressions: Player 1, Forecast Error, Categories + Beliefs HP**

	(1) UG	(2) TG
Self-interest	0.129***	-0.017
	(0.019)	(0.016)
Mutual Benefit / Cooperation	0.155***	0.019*
	(0.026)	(0.011)
No clear justification	0.131***	0.038
	(0.032)	(0.033)
Hypothetical beliefs	0.469***	0.517***
	(0.032)	(0.026)
Constant	-0.400***	-0.189***
	(0.018)	(0.012)
Observations	2279	1887
R^2	0.139	0.183

Forecast Error is defined as Player 1 belief minus realized Player 2 behavior. Baseline category is Moral. Sample restricted to pooled Control, Market, Bonus, and Aid. Regressions include category dummies and hypothetical beliefs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 20: **Benchmark Regressions: Player 1, Hypothetical FE, Categories + Beliefs HP**

	(1) UG	(2) TG
Self-interest	-0.061**	-0.037***
	(0.025)	(0.013)
Mutual Benefit / Cooperation	-0.017	-0.030***
	(0.035)	(0.010)
No clear justification	-0.066	-0.015
	(0.043)	(0.028)
Hypothetical beliefs	0.933***	0.996***
	(0.043)	(0.023)
Constant	-0.627***	-0.317***
	(0.024)	(0.010)
Observations	2279	1887
R^2	0.176	0.513

Hypothetical FE is defined as Player 1 hypothetical belief minus the corresponding hypothetical Player 2 response. Baseline category is Moral. Sample restricted to pooled Control, Market, Bonus, and Aid. Regressions include category dummies and hypothetical beliefs. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

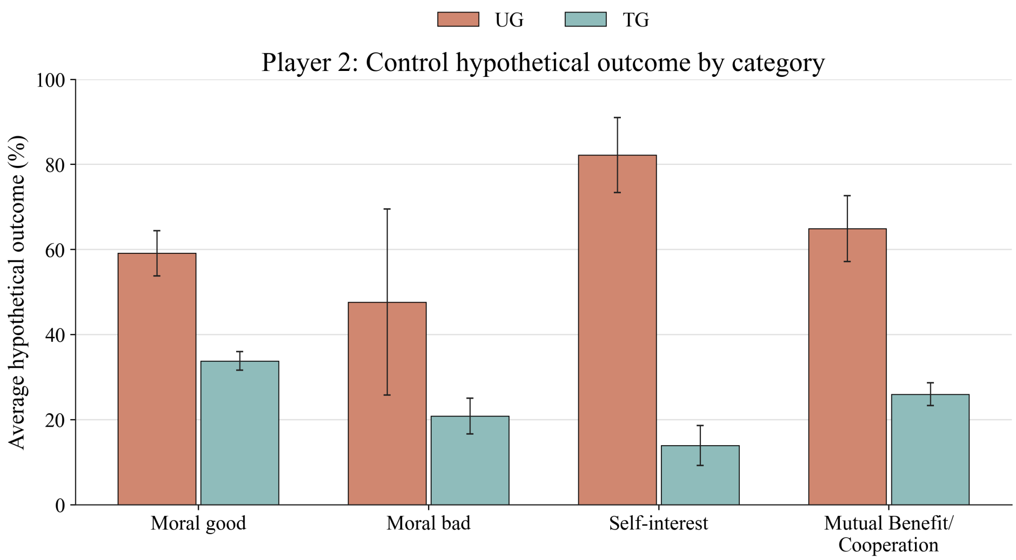

Figure 15: Control-sample hypothetical outcomes by category for Player 2. UG is measured using hypothetical acceptance and TG using hypothetical share sent. Error bars show 95% confidence intervals.

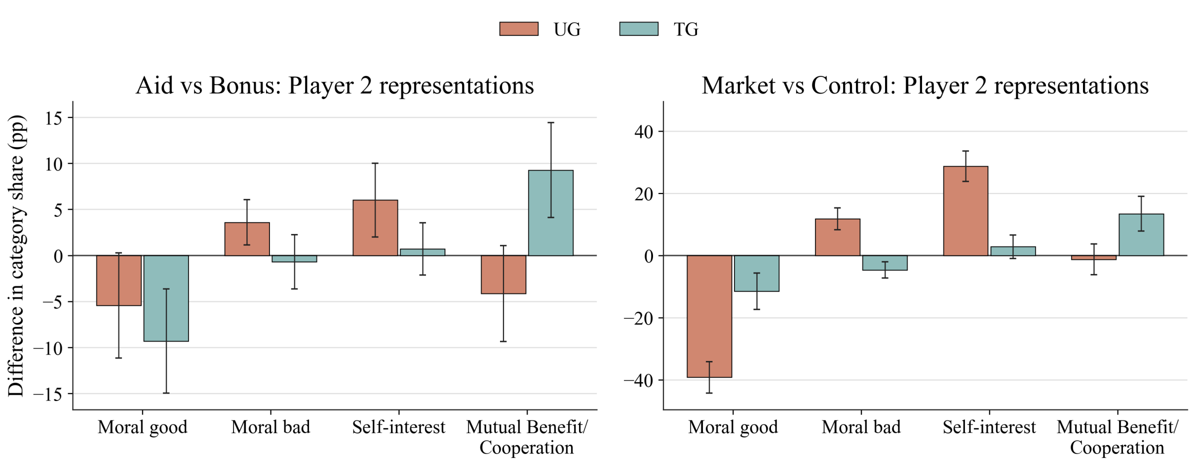

Figure 16: Treatment-induced shifts in Player 2 category shares. Panels report Aid vs Bonus and Market vs Control. Error bars show 95% confidence intervals.

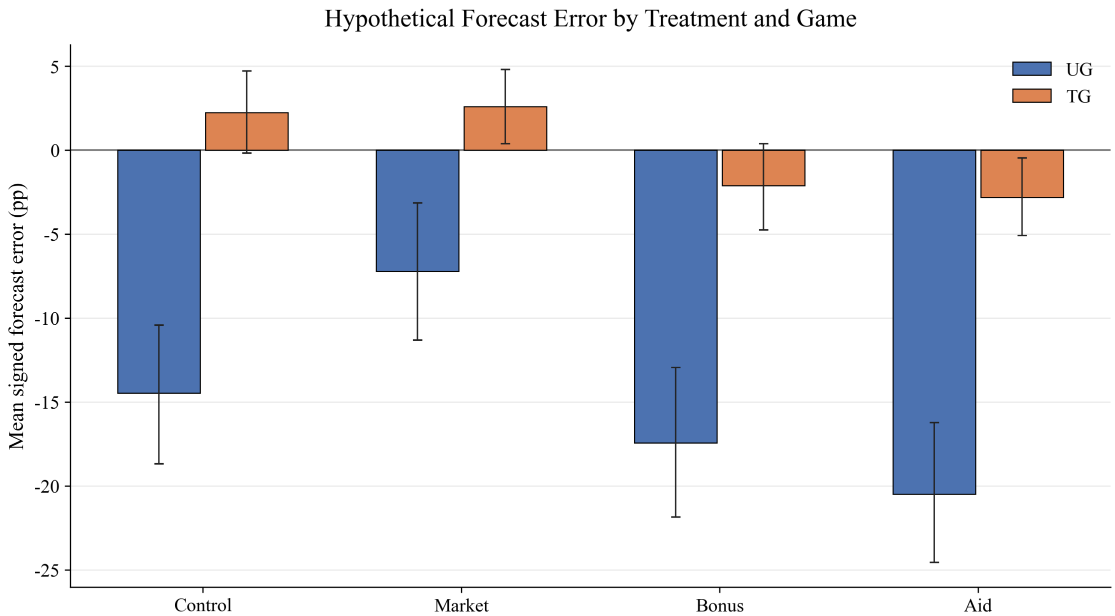

Figure 17: Forecast errors by treatment, hypothetical beliefs at the common one-third reference action, using the corresponding mean hypothetical Player 2 response as the benchmark. Positive values indicate optimism and negative values pessimism. Error bars show bootstrap 95% confidence intervals. Main-text counterpart: Figure 10.

Table 21: **Instability calibration regressions, Categories Only**

	All	Player 1	Player 2
Intercept	0.007	0.021**	0.015
	(0.013)	(0.009)	(0.019)
Predicted	1.413***	2.772***	1.218***
	(0.177)	(0.223)	(0.179)
Observations	18	10	8
R^2	0.799	0.951	0.886

Notes: Unit of observation is an outcome-by-comparison cell from the fitted-values exercise. Each column reports the regression of the actual treatment effect on the predicted treatment effect. Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 22: **Predicted Treatment Effects on Actions: Additive versus Interacted Specification**

Comparison	Game	Actual	Predicted		N
			Additive	Interacted	
Aid vs Bonus	DG-KW	-0.059	-0.039	-0.039	790
Aid vs Bonus	UG	-0.024	-0.013	-0.014	1120
Aid vs Bonus	TG	+0.008	+0.006	+0.004	937
Market vs Control	DG-KW	-0.101	-0.045	-0.045	728
Market vs Control	UG	-0.220	-0.095	-0.096	1027
Market vs Control	TG	+0.133	+0.083	+0.085	970

Notes: Player 1 actions, classified sample with non-missing hypothetical beliefs (UG, TG). Additive: action on category dummies and the hypothetical belief (the benchmark specification); Interacted: action on categories, the belief, and their interactions, so predicted effects include the treatment-induced movement of the category-belief cross-moments. Predicted effects are $\hat{\beta}'(\bar{x}_T - \bar{x}_C)$ on the pooled two-condition fit. DG-KW has no belief elicitation, so both specifications reduce to categories only.

D Heterogeneity and Dispersion: Full Decompositions

This appendix reports the full versions of the two dispersion exhibits of Section 5.1: the between/within-category variance decomposition for every game and condition, including DG-LT (Table 23), and the predicted-versus-actual dispersion accounting for all eight game-by-comparison cells (Table 24).

Table 23: **Heterogeneity Tied to Representations: Variance Decomposition**

Game	Condition	N	SD of action	Between-category share
DG-KW	Control	399	0.204	0.663
	Market	329	0.263	0.136
	Bonus	390	0.171	0.613
	Aid	400	0.197	0.584
DG-LT	Control	389	0.175	0.492
	Market	347	0.178	0.109
	Bonus	387	0.166	0.407
	Aid	387	0.194	0.481
UG	Control	582	0.111	0.109
	Market	501	0.260	0.061
	Bonus	587	0.090	0.097
	Aid	588	0.138	0.226
TG	Control	584	0.282	0.202
	Market	563	0.285	0.250
	Bonus	589	0.289	0.296
	Aid	588	0.295	0.314

Notes: Player 1, classified responses only. "Between-category share" is the share of the within-condition variance of the action lying between the three representation categories (the R^2 of the action on category dummies).

Table 24: **Predicted versus Actual Treatment Effects on Dispersion**

		Δ Mean (pp)		Δ SD	
Comparison	Game	Predicted	Actual	Predicted	Actual
Aid vs Bonus	DG-KW	-3.9	-5.9	+0.022	+0.026
	DG-LT	-4.0	-5.0	+0.023	+0.028
	UG	-1.3	-2.4	+0.017	+0.048
	TG	+0.3	+0.8	+0.001	+0.005
Market vs Control	DG-KW	-4.5	-10.1	+0.019	+0.059
	DG-LT	-9.9	-21.5	+0.010	+0.003
	UG	-7.3	-21.0	+0.043	+0.149
	TG	+4.2	+15.0	+0.019	+0.003

Notes: Player 1, classified responses only. Predicted values reweight the category-conditional action distributions (means and variances pooled across the two conditions of each comparison) by each condition's category shares; the predicted effect is the mixture implication of the representation shift alone. Actual values are the raw between-condition differences.

E Within-Subject Evidence from the Dictator Game

The dictator-game pre-registration (AsPredicted #261370) included three within-subject designs of 600 participants each (599 complete pairs in the DG-KW market design), all without stories: (i) game fixed at DG-KW, frame varying between abstract and market; (ii) game fixed at DG-LT, frame varying between abstract and market; (iii) frame fixed at abstract, game varying between DG-KW and DG-LT. Order was randomized; order effects are small—within condition, mean shares sent differ by at most 2.5 percentage points between first and second decisions—and order enters all specifications below.

Table 25 summarizes switching behavior. When the same participant plays DG-KW and DG-LT under the abstract frame, 46.2% change their choice (share sent) and 22.7% change their representation category; when the frame switches between abstract and market with the game fixed, 73.1% (DG-KW) and 82.3% (DG-LT) change their choice and over half change their category. In all three designs, choice switching is strongly predicted by category switching: participants whose representation category changes are 14–44 percentage points more likely to change their choice than participants whose category is stable (linear-probability estimates: 0.44, robust SE 0.04, in the DG-KW/DG-LT design; 0.19, SE 0.04, and 0.14, SE 0.03, in the two market designs; all $p < 0.001$). Choice switching counts any change in the share sent, so it includes small adjustments; the conditional contrast between category switchers and stayers is the informative statistic.

Tables 26–29 report within-subject regressions with participant fixed effects. Holding the participant fixed, moving from DG-LT to DG-KW lowers the share sent by 3.3–3.6 percentage points, and moving from the abstract to the market frame lowers it substantially; representation categories predict behavior within participant, with self-interested representations associated with significantly lower transfers.

Table 25: **Within-Subject Switching in the Dictator Game**

Comparison	N pairs	Share switching			P(choice switch | category)	
		Choice	Category	Social proximity	Switched	Stayed
DG-KW vs DG-LT (abstract frame)	600	46.2%	22.7%	40.5%	80.1%	36.2%
Market vs Control (DG-KW)	599	73.1%	51.6%	66.4%	82.2%	63.4%
Market vs Control (DG-LT)	600	82.3%	55.0%	62.0%	88.5%	74.8%

Notes: Each participant makes two dictator-game decisions. In the first row, both decisions use the abstract (control) frame and the game varies (KW vs LT); in the second and third rows, the game is fixed and the frame varies (abstract vs market). "Choice" switching is any change in the share sent; "Category" switching is a change in the reason-based representation category (Moral, Self-interest, Mutual Benefit/Cooperation, No clear justification); "Social proximity" switching is a change in the social-proximity class of the representation. The last two columns report the probability of a choice switch conditional on the representation category switching or staying fixed.

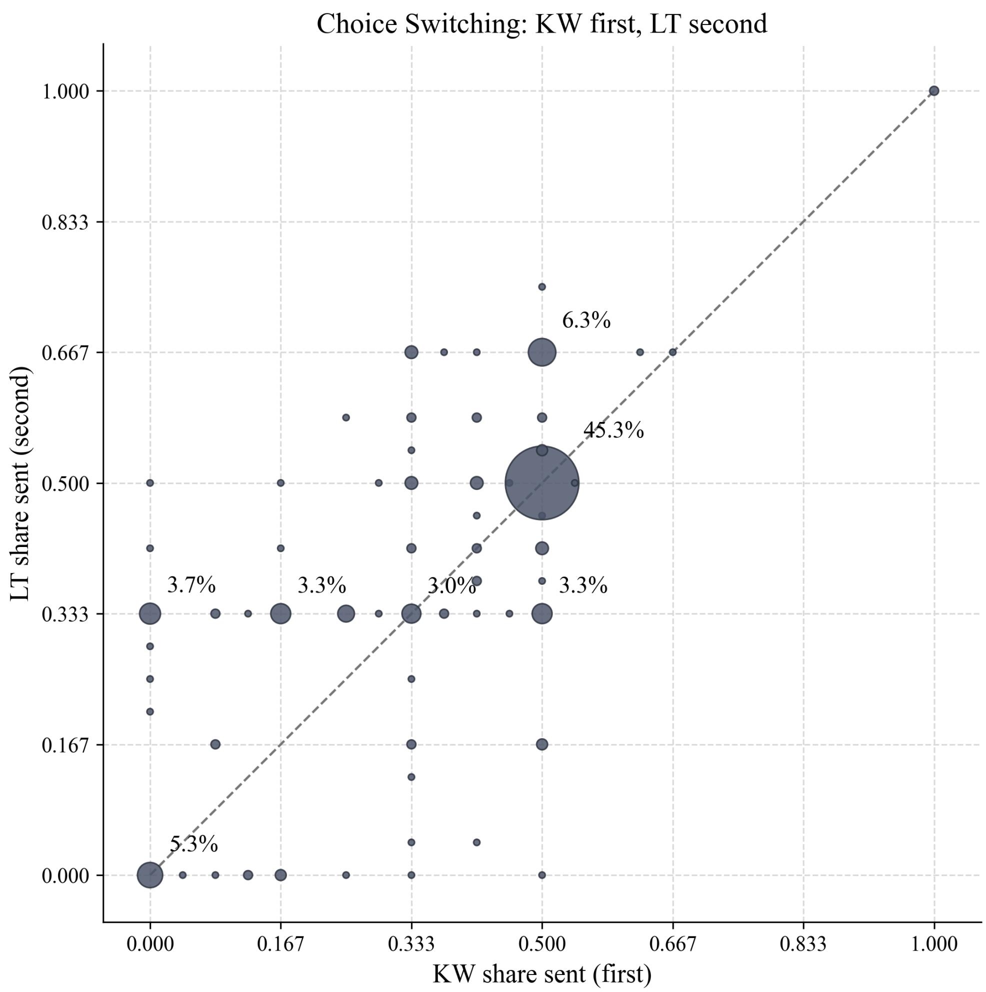

Figure 18: Within-subject choice switching between DG-KW (first decision) and DG-LT (second decision), control frame. Point size is proportional to the share of participants at each pair of choices; the dashed line marks unchanged choices. The mass below the diagonal at high KW shares and above it at low KW shares shows regression toward the LT default.

Table 26: **Within-Subject Regressions**: Control sample, LT vs. KW

	(1)	(2)	(3)	(4)
	Share Sent P1			
KW	-0.036***	-0.035***	-0.033***	-0.033***
	(0.011)	(0.011)	(0.010)	(0.010)
KW x LTFirst	-0.026	-0.026	-0.033**	-0.033**
	(0.016)	(0.016)	(0.015)	(0.015)
HighSP		0.011		-0.004
		(0.015)		(0.015)
Mutual Benefit / Cooperation			-0.070	-0.070
			(0.072)	(0.072)
Self-interest			-0.091***	-0.093***
			(0.026)	(0.026)
No clear justification			-0.053	-0.054
			(0.039)	(0.040)
Participant fixed effects	Yes	Yes	Yes	Yes
Observations	1200	1200	1200	1200
Participants	600	600	600	600
R^2	0.867	0.868	0.877	0.877

Outcome is DG share sent by Player 1, defined as $1 - \texttt{allocation}/12$. Models use the within-control LT/KW sample only. All specifications include participant fixed effects and cluster standard errors by participant. Baseline text category is Moral. HighSP equals 1 if `sp_num¿1`. LTFirst equals 1 when LT was shown before KW; its main effect is collinear with the participant fixed effects and therefore not identified. Estimated without fixed effects (standard errors clustered by participant), the LTFirst level effect is +0.019 (SE 0.015). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 27: **Within-Subject Regressions**: Market vs. Control within subject

	(1)	(2)	(3)	(4)
	Share Sent P1			
Market	-0.160***	-0.139***	-0.135***	-0.126***
	(0.017)	(0.018)	(0.018)	(0.019)
Market x ControlFirst	0.016	0.017	0.017	0.017
	(0.025)	(0.025)	(0.024)	(0.024)
HighSP		0.113***		0.062***
		(0.021)		(0.021)
Mutual Benefit / Cooperation			-0.076**	-0.073**
			(0.033)	(0.033)
Self-interest			-0.197***	-0.177***
			(0.022)	(0.023)
No clear justification			-0.084**	-0.068**
			(0.034)	(0.034)
Participant fixed effects	Yes	Yes	Yes	Yes
Observations	2398	2398	2398	2398
Participants	1199	1199	1199	1199
R^2	0.597	0.614	0.642	0.647

Outcome is DG share sent by Player 1, defined as $1 - \texttt{allocation}/12$. Models use the within-subject Market vs. Control sample pooled across the fixed-game designs. All specifications include participant fixed effects and cluster standard errors by participant. Baseline text category is Moral. HighSP equals 1 if `sp_num`¿1. ControlFirst equals 1 when Control was shown before Market; its main effect is collinear with the participant fixed effects and therefore not identified. Estimated without fixed effects (standard errors clustered by participant), the ControlFirst level effect is +0.009 (SE 0.013). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 28: **Within-Subject Category Regressions**: Control sample, LT vs. KW

	Mutual Benefit / Cooperation	Self-interest	No clear justification
	Outcome category relative to Moral		
KW	1.633	0.250*	-0.711**
	(1.105)	(0.142)	(0.329)
LTFirst	1.391	0.217	-0.912**
	(1.128)	(0.206)	(0.388)
KW x LTFirst	-2.379*	-0.480**	0.753
	(1.410)	(0.202)	(0.489)
Observations	1200	1200	1200
Participants	600	600	600
Pseudo R^2	0.009	0.009	0.009

Outcome is the main reason category. Entries are multinomial-logit coefficients relative to the omitted category Moral. The sample is the within-control LT/KW design only. Regressors are KW, LTFirst, and their interaction. Standard errors are clustered by participant. LTFirst equals 1 when LT was shown before KW. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 29: **Within-Subject Category Regressions**: Market vs. Control within subject

	Mutual Benefit / Cooperation	Self-interest	No clear justification
	Outcome category relative to Moral		
Market	3.150***	0.720***	1.553***
	(0.333)	(0.123)	(0.184)
ControlFirst	-1.322**	-0.237*	-0.661**
	(0.664)	(0.131)	(0.262)
Market x ControlFirst	0.258	-0.094	0.351
	(0.679)	(0.167)	(0.279)
KW design	-0.278	-0.106	-0.072
	(0.172)	(0.108)	(0.152)
Observations	2398	2398	2398
Participants	1199	1199	1199
Pseudo R^2	0.070	0.070	0.070

Outcome is the main reason category. Entries are multinomial-logit coefficients relative to the omitted category Moral. The sample is the within-subject Market vs. Control design pooled across the fixed-game designs. Regressors are Market, ControlFirst, their interaction, and a KW-design indicator. Standard errors are clustered by participant. ControlFirst equals 1 when Control was shown before Market. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

F Representations from Hypothetical Allocations

The dictator-game pre-registration specified a second measure of representations, elicited before the transfer decision: participants were shown hypothetical final allocations and asked to describe a real-world situation they find similar to each. The answers were classified with the same LLM-plus-validation procedure into social-proximity categories and into fine-grained moral categories (Need, Egalitarian, Generous, Neutral, Entitlement, Greedy, Theft). Because this measure is anchored to *allocations* rather than to the participant's chosen action, it is less susceptible to ex-post rationalization of the choice. This appendix shows that it delivers the same conclusions as the reasons-based measure, both for the KW–LT comparison of the Introduction and Section 5.3 and for the market treatment.

F.1 DG-KW versus DG-LT

Table 30: Distribution of social-proximity categories (hypothetical-allocation measure) and mean allocation kept by the dictator, by game. Control condition.

	Share of responses		**Mean Allocation**	
Social proximity	KW	LT	KW	LT
No mention of recipient	26.50%	31.00%	8.74	7.39
Abstract stranger	28.00%	28.50%	9.31	7.10
Anonymous peer	17.50%	19.00%	6.00	6.24
Teammate / coworker	7.50%	5.50%	6.60	7.09
Friend	20.50%	16.00%	6.83	6.14

Table 31: Allocation kept by the dictator and allocation-specific outcomes, by social proximity (hypothetical-allocation measure): OLS estimates, control condition. High SP indicates a socially close counterpart. Standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$.

	Mean Allocation	=12	>8	=8	=6	=4
High SP	-1.703^{***}	-0.228^{***}	-0.299^{***}	-0.028	0.357^{***}	0.006
	(0.220)	(0.034)	(0.039)	(0.034)	(0.047)	(0.019)
Constant	8.101^{***}	0.246^{***}	0.351^{***}	0.145^{***}	0.364^{***}	0.035^{***}
	(0.144)	(0.022)	(0.026)	(0.022)	(0.031)	(0.013)
Observations	400	400	400	400	400	400
R^2	0.131	0.101	0.126	0.002	0.125	0.000

Table 32: Symmetrized (Shapley/Oaxaca–Blinder) decomposition of mean differences between DG-KW and DG-LT into a representation component (differences in the distribution of social proximity) and a behavior component (differences in allocations conditional on social proximity), Control condition, hypothetical-allocation sample. The decomposition is a descriptive accounting; no sampling uncertainty is attached.

Variable	Observed Diff	Representation Component	Behavior Component
KW − LT			
Mean Allocation	0.998	−0.067	1.064
Allocation = 4	−0.065	0.002	−0.067
Allocation = 6	−0.035	0.008	−0.043
Allocation = 8	−0.075	0.000	−0.075
Allocation > 8	0.205	−0.010	0.215
Allocation = 12	0.105	−0.011	0.116

Notes: The construction is the symmetrized decomposition of Section 5.1, with representation cells given by the five social-proximity levels of Table 30 ($N = 200$ per game); no cell is empty in either game. Cells defined by the seven moral categories instead split the mean-allocation gap into −0.066 (representation) and 1.064 (behavior).

Table 33: Distribution of moral categories (hypothetical-allocation measure) and mean allocation kept by the dictator, by game. Control condition.

	Share of responses		Mean Allocation	
Moral category	KW	LT	KW	LT
Need	2.50%	1.00%	11.40	6.00
Egalitarian	32.00%	27.50%	6.02	5.91
Generous	33.00%	37.50%	7.43	6.03
Neutral	8.50%	11.00%	8.00	7.48
Entitlement	10.50%	11.00%	9.90	8.48
Greedy	13.00%	8.00%	10.96	8.75
Theft	0.50%	4.00%	12.00	11.75

F.2 Market versus Control

The hypothetical-allocation measure also replicates the market treatment effects on representations. Under the market frame, the recipient largely disappears from participants' representations, and moral categories collapse toward Neutral.

Person-level consistency under the paper's category scheme. The classification above predates the three-category scheme of the reasons measure. To compare the two mea-

Table 34: Allocation kept by the dictator and allocation-specific outcomes, by moral category (hypothetical-allocation measure): OLS estimates, control condition. Baseline category is Neutral. Standard errors in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.10$.

	Mean Allocation	=12	>8	=8	=6	=4
Generous	-1.021^{***}	-0.144^{***}	-0.103^{*}	-0.169^{***}	0.278^{***}	-0.009
	(0.315)	(0.053)	(0.060)	(0.058)	(0.073)	(0.034)
Entitlement	1.469^{***}	0.146^{**}	0.188^{**}	0.113	-0.166^{*}	-0.051
	(0.385)	(0.064)	(0.073)	(0.071)	(0.089)	(0.042)
Egalitarian	-1.739^{***}	-0.179^{***}	-0.231^{***}	-0.248^{***}	0.609^{***}	0.008
	(0.321)	(0.054)	(0.061)	(0.059)	(0.074)	(0.035)
Greedy	2.414^{***}	0.321^{***}	0.484^{***}	-0.163^{**}	-0.187^{**}	-0.051
	(0.387)	(0.065)	(0.074)	(0.072)	(0.090)	(0.042)
Theft	4.073^{***}	0.709^{***}	0.769^{***}	-0.282^{**}	-0.282^{*}	-0.051
	(0.644)	(0.107)	(0.122)	(0.119)	(0.149)	(0.070)
Need	2.152^{***}	0.392^{***}	0.484^{***}	-0.282^{**}	0.004	-0.051
	(0.714)	(0.119)	(0.136)	(0.132)	(0.165)	(0.078)
Neutral (Constant)	7.705^{***}	0.179^{***}	0.231^{***}	0.282^{***}	0.282^{***}	0.051^{*}
	(0.279)	(0.046)	(0.053)	(0.051)	(0.064)	(0.030)
Observations	400	400	400	400	400	400
R^2	0.452	0.342	0.377	0.116	0.362	0.014

Table 35: Distribution of social proximity within each condition (column shares, hypothetical-allocation measure). Dictator game, Market and Control conditions. Columns sum to 100%. p-values from two-sided tests of equal proportions with continuity correction.

Social proximity	Market	Control	p-value
No mention of recipient	75.88%	28.75%	< 0.001
Abstract stranger	10.75%	28.25%	< 0.001
Anonymous peer	5.00%	18.25%	< 0.001
Teammate / coworker	2.13%	6.50%	< 0.001
Friend	6.25%	18.25%	< 0.001

sures within person, we re-classified the 1,200 retained memory texts (one per participant and frame, Control and Market) into the Moral, Self-interest, and Mutual Benefit/Cooperation scheme, using the written classification rules verbatim except for the two preamble sentences, which describe the texts as recalled situations rather than justifications of a game decision. Because the original classifier model was no longer available, the re-classification uses a different frontier model (Claude Opus 4.8); on a stratified sample of 800 already-classified reasons answers, the two classifiers agree in 83.3% of cases ($\kappa = 0.78$)—above

Table 36: Distribution of moral categories within each condition (column shares, hypothetical-allocation measure). Dictator game, Market and Control conditions. Columns sum to 100%. p-values from two-sided tests of equal proportions with continuity correction.

Moral category	Market	Control	p-value
Egalitarian	7.25%	29.75%	< 0.001
Entitlement	22.50%	10.75%	< 0.001
Generous	11.13%	35.25%	< 0.001
Greedy	7.75%	10.50%	0.14
Need	2.50%	1.75%	0.54
Neutral	48.25%	9.75%	< 0.001
Theft	0.63%	2.25%	0.029

Table 37: Symmetrized (Shapley/Oaxaca–Blinder) decomposition of mean differences between Control and Market into a representation component (differences in the distribution of social proximity) and a behavior component (differences in allocations conditional on social proximity), dictator games pooled, hypothetical-allocation sample. The decomposition is a descriptive accounting; no sampling uncertainty is attached.

Variable	**Observed Diff**	**Representation Component**	**Behavior Component**
Control − Market			
Mean Allocation	−1.996	−0.494	−1.502
Allocation = 4	0.025	−0.002	0.027
Allocation = 6	0.416	0.097	0.319
Allocation = 8	0.028	−0.012	0.040
Allocation > 8	−0.461	−0.083	−0.378
Allocation = 12	−0.150	−0.047	−0.103

Notes: The construction is the symmetrized decomposition of Section 5.1, with representation cells given by the five social-proximity levels of Table 35 ($N = 400$ Control, $N = 800$ Market); no cell is empty in either condition. Cells defined by the seven moral categories instead split the mean-allocation gap into −0.617 (representation) and −1.379 (behavior).

the human-coder benchmark of Appendix G—with disagreements concentrated at the same Mutual Benefit boundary, so the model swap doubles as an out-of-family robustness check of the classification rules. Table 38 reports the person-level cross-tabulation: within every stated-reason category, the modal pre-decision memory falls in the same category (agreement 63.4% pooled, $\kappa = 0.39$, Cramér's $V = 0.38$, $p < 0.001$; 80.5% in Control). Social proximity coheres the same way: 38.7% of participants whose reasons are Moral had recalled a socially

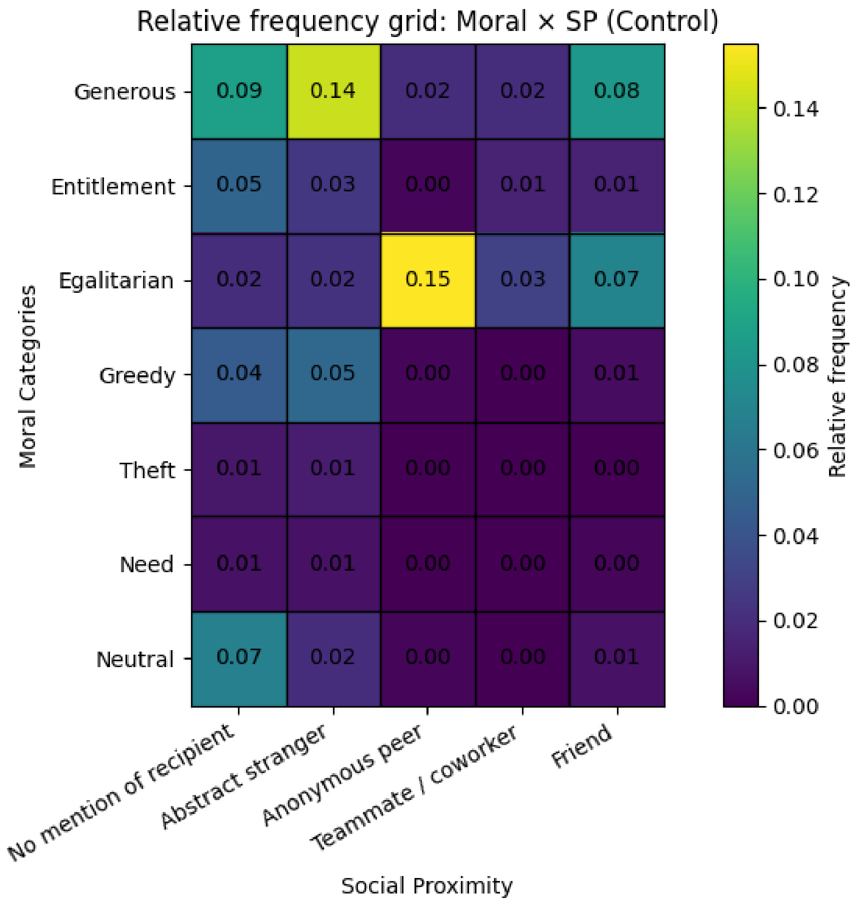

Figure 19: Joint distribution of social proximity and moral categories (hypothetical-allocation measure), Control.

close counterpart, against 11.1% of those with Self-interested reasons ($p < 0.001$), and Control allocations decline across memory categories as they do across reason categories (Moral memories transfer 45.4% of the budget, Self-interested memories 14.6%). The market frame moves the pre-decision measure the way it moves the reasons: Moral memories fall from 68.0% to 19.2% of responses, Mutual Benefit/Cooperation rises from 3.2% to 22.9%, and the residual rises from 11.8% to 43.4%. The retained memory is the one whose hypothetical allocation is closest to the participant's actual allocation, so memory–behavior comparisons are partly mechanical; the category and proximity content of the text is not, and it is elicited before the decision.

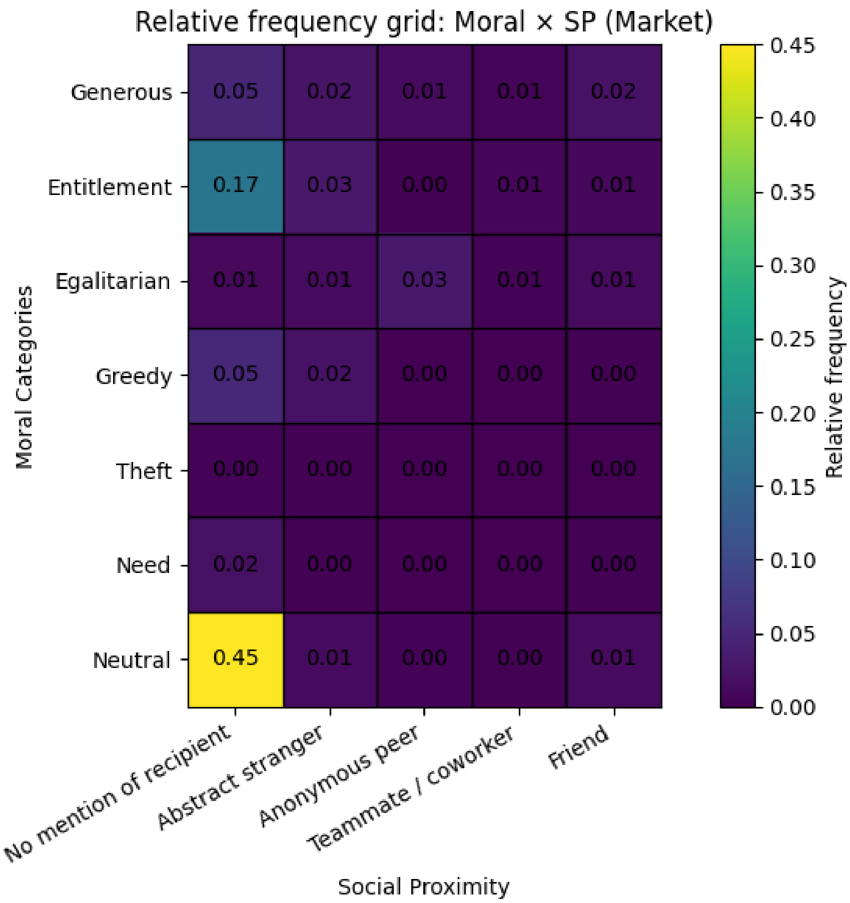

Figure 20: Joint distribution of social proximity and moral categories (hypothetical-allocation measure), Market.

Table 38: Person-level consistency of the two representation measures: distribution of the hypothetical-allocation (memory) category within each stated-reason category, dictator games, Control and Market conditions. The memory measure is elicited before the transfer decision and classified with the same category scheme as the reasons; rows sum to 100%.

	Memory (pre-decision) category			
Reasons category	Moral	MBC	Self-interest	N
Moral	76.8%	13.0%	10.3%	409
MBC	24.7%	54.1%	21.2%	85
Self-interest	27.9%	27.5%	44.5%	247

Notes: $N = 741$ participant-frame observations with substantive categories on both measures. Agreement 63.4%, Cohen's $\kappa = 0.39$, Cramér's $V = 0.38$ (χ^2 $p < 0.001$). The memory retained for each participant is the one attached to the hypothetical allocation closest to the actual allocation, so memory–behavior comparisons are partly mechanical; the category and social-proximity content of the memory text is not.

G Validation of the LLM Classification

Following the pre-registered procedure, the classification of the reasons question—produced by GPT-4.1 from a prompt containing the written classification rules—was validated against human coders. For each of the three classification batches—Player 1 in the story conditions, Player 1 in Market and Control, and Player 2 in all four conditions—two disjoint validation subsamples were drawn, stratified on the LLM-assigned category, and coded by hand by two research assistants, each responsible for one subsample throughout and blind to the condition and to the game, using the same classification rules given to the LLM; in total, human coders classified 3,990 responses, and no response was coded in more than one subsample. Table 39 reports agreement rates and Cohen's κ by batch, subsample, and condition. Agreement is high for Player 1's four-category scheme (three substantive categories plus the residual; 79–80% pooled across subsamples, with κ between 0.63 and 0.69) and lower for Player 2's finer five-category scheme (71%, $\kappa = 0.57$), where disagreements concentrate at the *Moral good/Mutual Benefit* boundary. The LLM labels in the validation subsamples coincide with the final classification used in the analysis files.

Table 39: **Validation of the LLM Classification Against Human Coders**

	Subsample 1			Subsample 2		
	N	Agreement	κ	N	Agreement	κ
Player 1, story conditions (Aid/Bonus)						
Bonus	300	0.84	0.67	300	0.82	0.62
Aid	300	0.81	0.65	300	0.76	0.56
All conditions	600	0.82	0.66	600	0.79	0.59
Player 1, Market/Control						
Control	300	0.91	0.83	300	0.76	0.58
Market	300	0.79	0.71	298	0.70	0.59
All conditions	600	0.85	0.78	598	0.73	0.61
Player 2, all conditions						
Control	200	0.70	0.54	200	0.73	0.55
Market	199	0.66	0.56	200	0.71	0.61
Bonus	197	0.68	0.48	199	0.67	0.42
Aid	197	0.72	0.58	200	0.81	0.69
All conditions	793	0.69	0.55	799	0.73	0.59
Overall	$N = 3,990$, agreement $= 0.76$, $\kappa = 0.64$					

Notes: Human coders classified free-text answers to the reasons question using the same classification rules given to the LLM (GPT-4.1). Each batch comprises two disjoint validation subsamples drawn stratified on the LLM-assigned category (for Player 1, 100 responses per game–condition cell across DG-KW, UG, and TG; for Player 2, roughly 100 per game–condition cell across UG and TG in all four conditions); each subsample was coded by one of two research assistants, blind to the condition and to the game. Agreement is the share of responses assigned to the same category; κ is Cohen's kappa over the full category set (four categories for Player 1, five for Player 2), including the residual "No clear justification".

H The Context–Category Similarity Module

This appendix documents the vignette module of Section 3.3: the measurement of the similarity between every context text in the design and the representation categories and receiver types.

Design. The retrieval model of Section 2.2 runs on the similarity between a described situation and *categories of experience*, and a category is a class of concrete situations rather than an abstract label. The module therefore represents each Player-1 category by eight vignettes—short, third-person descriptions of everyday situations instantiating the category—and measures the similarity of each of the ten context texts (the Bonus and Aid stories; the Control and Market instructions of DG-KW, DG-LT, UG, and TG, verbatim from Appendix I) to each vignette. Receiver types are represented by two vignettes each: counterparts who accept or reject one-sided terms (UG) and counterparts who return or keep entrusted proceeds (TG).

Generation. The 32 vignettes were generated in a single conversation by a frontier model distinct from the raters (Claude Fable 5), from category definitions reconciled verbatim with the written classification rules given to the classifier (Section 3.1 and Appendix G); following those rules, the Self-interest class includes both pure own-use and strategic self-protection content (four vignettes each). The generator prompt imposes: 60–90 words per sender vignette (40–60 per counterpart vignette); one named protagonist and one identified counterpart; a concrete resource at stake; no category-naming words ("fair", "selfish", "generous", "greedy", "cooperate", "moral", "trust", "betray", and derivatives); no mention of games, experiments, or studies; no amounts of 6, 12, or 18 dollars and no tripling of any amount; and domain balance—each category's eight vignettes span the same eight domains (workplace; family or friends; neighbors or community; strangers in a one-off encounter; small business or commerce; online or anonymous; education; recreation), so that category is not confounded with setting. The set was reviewed and frozen before any rating; the only post-generation edit was one word in one vignette ("craft fairs" to "craft markets", removing a banned stem used in its festival sense). The full texts, the generator prompt and transcript, and the freeze record are in the replication package.

Rating protocol. Raters are fresh conversations of a reasoning-tier LLM (Claude Opus 4.8 with reasoning enabled), blind to the design: vignettes appear as V1–V24 and contexts as TEXT A–J in three independently permuted orders, one conversation per permutation, with no category or treatment information. For each context the rater (i) rates the similarity of the context situation to each vignette on a 0–100 scale, considering the situation as a whole—what is at stake, the relationship between the parties, the setting, and the structure of the decision; (ii) distributes exactly 100 retrieval points across the 24 vignettes according to which would come to mind as a similar experience; and (iii) for the four strategic contexts, distributes exactly 100 points across the four counterpart vignettes according to which counterpart it would picture on the other side. Transcripts and the recording workbook are in the replication package.

Receiver-type splits. Table 40 reports the output of the third rating task, aggregated by type: the ultimatum-game split moves toward the accepting counterpart under the market frame, the way first movers' measured acceptance beliefs move (Section 3.3), while the trust-game split barely moves—part of the same trust-game pattern flagged there.

Robustness. Three checks accompany the exhibits of Section 3.3. First, cell means vary across the three permuted conversations (median range 12.6 points over the 30 context–category cells, maximum 20.8), but the three-conversation means are stable: an independent second run of the full protocol—identical except for the one-word edit above—reproduces the context–category table almost exactly (Pearson $r = 0.994$ across the 30 cells; mean absolute difference 1.9 points, maximum 5.1). Second, a second rater from a different model tier

Table 40: **Similarity of Strategic Contexts to Receiver Types, LLM Raters**

	Ultimatum game		Trust game	
Condition	Accepting	Rejecting	Returning	Keeping
Control	42.7	57.3	53.0	47.0
Market	52.0	48.0	54.0	46.0

Notes: For each strategic context, raters distributed 100 points across the game's four counterpart vignettes—two per receiver type: counterparts who accept or reject one-sided terms (ultimatum game) and counterparts who return or keep entrusted proceeds (trust game)—according to which counterpart they would picture on the other side. Cells aggregate the points by type and average over the three permuted conversations of the headline rater (protocol of Table 6); each game's pair sums to 100. The accepting and returning columns are the splits quoted in Section 3.3.

(Claude Fable 5) re-ran the protocol; all three permuted conversations completed. The third required persistence: the model's safety filter declined the benign opening message—a false positive, and a stochastic one, refusing and later accepting the identical text—eleven times across eight candidate label orders (drawn by pre-committed successor seeds, all recorded in the runner) before one conversation ran to completion. Over the 30 context–category cells its means track the headline rater's closely (Pearson $r = 0.982$; mean absolute difference 4.9 points), and it reproduces the key contrasts, including the near-zero trust-game market shifts. Third, both raters belong to a single model family, so the module remains a working exhibit: a multi-model grid (the convention of Table 5) and a pre-registered human-subjects survey replicating both similarity rounds are the validation path.

I Instructions and Stories

This appendix reproduces the instructions as seen by participants and the two stories; screenshots are available upon request.

I.1 Dictator Game, Abstract Frame (KW)

In this task, you are matched with another Prolific participant. You will never know the identity of the person with whom you are paired, and this person will never know your identity. You and your match are provisionally allocated a certain amount of dollars, adding to a total of **$12**. You will have the chance to decide **whether and how to change this original allocation**.

The allocation chosen will be implemented for **1 participant out of 10**. **This means that your decisions affect a bonus payment you and the other participant may receive**. **It is thus in your best interest to choose the allocations that you truly prefer**.

[new page]

You were provisionally allocated **$12**. The other participant was provisionally allocated **$0**.

You have the option to make a transfer to the other participant. You can transfer any amount between $0 and $12.

We now present three hypothetical final allocations and we ask you to describe a real-world situation that you find similar to each. The situation can involve two people or groups. It can be an experience you or somebody you know lived, or it could be fictional.

Then, you may choose the amount of your transfer.

I.2 Dictator Game, Abstract Frame (LT)

In this task, you are matched with another Prolific participant. You will never know the identity of the person with whom you are paired, and this person will never know your identity. You and your match are provisionally allocated a certain amount of dollars, adding to a total of **$12**. You will have the chance to decide **whether and how to change this original allocation**.

The allocation chosen will be implemented for **1 participant out of 10**. **This means that your decisions affect a bonus payment you and the other participant may receive**. **It is thus in your best interest to choose the allocations that you truly prefer**.

[new page]

You were provisionally allocated **$8**. The other participant was provisionally allocated **$4**.

You can transfer any amount between −$4 and $8. If you choose a negative dollar transfer to the other participant, this means that you take that amount *from the other participant*.

We now present four hypothetical final allocations and we ask you to describe a real-world situation that you find similar to each. The situation can involve two people or groups. It can be an experience you or somebody you know lived, or it could be fictional.

Then, you may choose the amount of your transfer.

I.3 Dictator Game, Market Frame (KW)

In this task, you are matched with another Prolific participant. You will never know the identity of the person with whom you are paired, and this person will never know your identity. You take the role of **data provider (producer)**, and the other participant takes the role of **broker (intermediary)**. As a data provider, you produce survey data through online questionnaires like the one you are in. A computer will randomly determine an initial price for your data. You will then have the opportunity to **adjust this price** before it is sent to the broker.

The broker will sell your data to the research team for a fixed price of \$12. You will receive the final price you set. Your decision affects a bonus payment. Specifically, 1 in 10 participant pairs will be randomly selected, and for those selected:

- The **data provider (you)** receives a bonus payment equal to the final price you set.
- The **broker** receives \$12 from the research team for the sold data and pays the provider the set price, keeping the difference.

[new page]

The computer selected a **starting price of \$12** for the data.

You may change the price by any amount between −\$12 and \$0. A negative change means that you decrease the price.

I.4 Dictator Game, Market Frame (LT)

In this task, you are matched with another Prolific participant. You will never know the identity of the person with whom you are paired, and this person will never know your identity. You take the role of **data provider (producer)**, and the other participant takes the role of **broker (intermediary)**. As a data provider, you produce survey data through online questionnaires like the one you are in. A computer will randomly determine an initial price for your data. You will then have the opportunity to **adjust this price** before it is sent to the broker.

The broker will sell your data to the research team for a fixed price of \$12. You will receive the final price you set. Your decision affects a bonus payment. Specifically, 1 in 10 participant pairs will be randomly selected, and for those selected:

- The **data provider (you)** receives a bonus payment equal to the final price you set.
- The **broker** receives $12 from the research team for the sold data and pays the provider the set price, keeping the difference.

[new page]

The computer selected a **starting price of $8** for the data.

You may change the price by any amount between −$8 and +$4. A positive change means that you increase the price, while a negative change means that you decrease the price.

I.5 Ultimatum Game, Abstract Frame

In this task, you are matched with another Prolific participant. You will never know the identity of the person with whom you are paired, and this person will never know your identity. One of you will be assigned to be Player 1, and the other will be assigned to be Player 2.

Player 1 is provisionally allocated $12. **Player 1** will have the chance to decide **whether and how to change this original allocation**.

After seeing the proposed allocation, **Player 2** will decide **whether to accept or reject it**. If Player 2 **accepts**, the proposed **allocation will be implemented**. If Player 2 **rejects**, **both players will receive $0**.

The final allocation will be implemented for 1 participant out of 10. This means that your decision affects a bonus payment you and the other participant may receive.

[new page]

Please choose how much of the $12 you would like to transfer to Player 2.

If the transfer is accepted, it will be implemented.

I.6 Ultimatum Game, Market Frame

In this task, you are matched with another Prolific participant. You will never know the identity of the person with whom you are paired, and this person will never know your identity. One of you will be assigned to be **data provider (producer)**, and the other will be assigned to be **broker (intermediary)**.

The data provider produces survey data through online questionnaires like the one you are in. A computer determines an **initial price** for your data. If you are assigned the role of **data provider**, you will have the opportunity to **adjust this initial price** before it is sent to the broker.

The broker can sell the data to the research team for a fixed price of **$12**.

After seeing the proposed price set by the data provider, the **broker will decide whether to accept or reject it**.

- If the broker **accepts**, the data will be sold for $12.
- The **data provider** receives a bonus equal to the final price set.
- The **broker** receives $12 from the research team, pays the provider the set price, and keeps the difference.
- If the broker **rejects**, the sale does not take place and **both participants receive $0**.

The final outcome will be implemented for 1 participant pair out of 10. This means that your decision affects a bonus payment you and the other participant may receive.

[new page]

The computer selected a **starting price of $12.**

You may change the price by any amount between −$12 and $0. A negative change means that you decrease the price.

If the price is accepted by the broker, the sale will be implemented.

I.7 Trust Game, Abstract Frame

In this task, you are matched with another Prolific participant. You will never know the identity of the person with whom you are paired, and this person will never know your identity. One of you will be assigned to be **Player 1**, and the other will be assigned to be **Player 2**.

Player 1 will receive $6, and Player 2 will receive $0. **Player 1** will decide **how much of the $6 to send** to Player 2. **Any amount sent will be multiplied by 3** before being received by Player 2.

After receiving this amount, **Player 2** will decide **how much of it to send back to Player 1**. The amount sent back will not be changed.

The final allocation will be implemented for 1 participant out of 10. This means that your decision affects a bonus payment you and the other participant may receive.

[new page]

Please choose how much of the $6 you would like to send to Player 2. The remaining amount will be yours.

Any amount you send will be multiplied by 3 before being received by Player 2.

Player 2 will then decide how much of the amount received to send back to you.

I.8 Trust Game, Market Frame

In this task, you are matched with another Prolific participant. You will never know the identity of the person with whom you are paired, and this person will never know your identity. One of you will be assigned to be the data provider (producer), and the other will be assigned to be the broker (intermediary).

The data provider produces survey data through online questionnaires like the one you are in. The data provider has produced 6 units of data, each valued at $1 by the research team, for a total of $6.

If you are assigned the role of data provider, you will decide how many units to sell to the research team for $1 per unit, and how many units to transfer to the broker.

The broker has access to a specialized buyer who is willing to pay $3 per unit, so the broker's revenue is 3 times the number of units you send.

If you are assigned the role of broker, you sell the units received to the high-value buyer, and your revenue is three times the units of data sent to you. After the sale, as a **broker you will decide whether and how much of the revenue to transfer back to the data provider.**

The final outcome will be implemented for 1 participant pair out of 10. This means that your decision affects a bonus payment you and the other participant may receive.

[new page]

Please choose how much of the $6 worth of data you would like to send to the broker. The remaining amount will be sold to the research team and you will cash in its value.

Any amount you send will be multiplied by 3 before being received by the broker and sold to the research team.

The broker will then decide how much of the revenue received to send back to you.

I.9 Stories

Bonus.

After weeks of intense preparation, Steve was thrilled to learn that his team was awarded a performance bonus for exceeding their sales target. The recognition felt satisfying—a tangible reward for the long hours he'd poured into work and the many nights he'd stayed late fine-tuning his pitch. The director called Steve to discuss the bonus and said: "You guys did great. Since you are the team leader, the full bonus is allocated to you. However, you may choose to give a portion to your teammate Max if you wish. The final decision is yours."

Now Steve faced a choice. He could keep the full bonus, as initially allocated. Or he could give some to Max, knowing that Max made a significant contribution to the team's success. What should he do?

Aid.

James was going through a rough financial patch. He struggled to make ends meet and had no savings left. One morning, he received an unexpected email from a government welfare agency he had applied to months ago. "Congratulations! You have been selected for financial assistance. You will be receiving the full aid amount to help with your expenses."

As James went to the agency to collect, he found Mark, a neighbor. He learned that Mark also applied but wasn't selected. Only James received the funds.

Now James faced a choice. Should he keep all the money for himself? James was tempted because he really needed the money, and Mark was just an acquaintance. Or should he give some of it to Mark, given that he was struggling? What should he do?